

AGENTIC LEARNING

GAME-THEORETIC AND AI FOUNDATIONS



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Birkhäuser

Agentic Learning

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Sample Chapters 1–3

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About this sample

This sample contains the table of contents of the complete book, the list of notation, and Chapters 1–3 in full. The manuscript comprises fifteen chapters in five parts and a mathematical appendix, about 570 pages with 170 figures, 113 exercises, and 616 references.

Page numbers, and cross-references from the sample chapters to later parts of the book, are those of the complete manuscript. In the table of contents, entries in colour are included in this sample. The reference list at the end covers the works cited in Chapters 1–3.

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Notation

Symbols are listed in the order in which the book introduces them. Three conventions hold throughout. Calligraphic letters denote sets and spaces $(\mathcal{S}, \mathcal{A}, \mathcal{O})$; sans-serif letters denote the components of an agent (m, T, M, H) ; and a star denotes a true or optimal object (θ^*, V^*) . A subscript t always indexes time, and $-i$ always means “all agents other than i ”. Symbols used in a single chapter are declared where they appear.

Sets, numbers, and operations

$\mathbb{R}, \mathbb{R}_{\geq 0}, \mathbb{N}, \mathbb{Z}$	real numbers, non-negative reals, natural numbers, integers
$ \cdot , \ \cdot\ $	absolute value; norm (Euclidean unless subscripted)
$\langle \cdot, \cdot \rangle$	inner product
$\mathbf{1}$	vector of ones; $\mathbb{1}\{\cdot\}$ the indicator of an event
$:=$	equal by definition
dx	differential, as in $\int f(x) dx$
$\arg \max, \arg \min$	maximizers, minimizers of a function
$\text{supp}, \text{dom}, \text{dist}, \text{Proj}$	support, domain, distance to a set, projection
$\text{diag}, \text{tr}, \text{rank}$	diagonal matrix, trace, rank
softmax	$\text{softmax}(v)_i = e^{v_i} / \sum_j e^{v_j}$
$D_\psi(x, y)$	Bregman divergence generated by ψ

Probability

$\Delta(X)$	probability measures on a measurable set X
$\mathbb{E}, \mathbb{P}, \text{Var}, \text{Cov}$	expectation, probability, variance, covariance
$ $	conditioning bar, as in $P(s' s, a)$
$N(m, \sigma^2)$	Gaussian law with mean m and variance σ^2
i.i.d.	independent and identically distributed as
$D_{\text{KL}}(p \ q)$	Kullback–Leibler divergence; $\ p - q\ _{\text{TV}}$ total variation distance
$H(\cdot), I(\cdot; \cdot)$	entropy, mutual information
$\mathcal{F}, \mathcal{G}$	σ -algebras (what an agent can distinguish)
$\text{do}(x)$	causal intervention: x is set rather than observed
$\text{pa}(X)$	parents of X in a causal graph
ρ	a coherent risk measure (Definition 2.41); $\text{CVaR}_\alpha, \text{VaR}_\alpha$

Decision processes

$\mathcal{S}, \mathcal{A}, \mathcal{O}$	state, action, and observation spaces
s_t, a_t, o_t, r_t	state, action, observation, reward at time t
$P(s' s, a), r(s, a)$	transition kernel, reward function
ρ_0	initial distribution over states
γ, T	discount factor, horizon
$h_t \in \mathcal{H}_t$	history, the information available at time t
π	policy; π_w a policy with parameters $w \in \mathbb{R}^d$
$V^\pi, Q^\pi, V^\star, Q^\star$	value and action-value functions; optimal values
$b_t \in \Delta(\mathcal{S})$	belief state
$\mathcal{T}$	Bellman operator
$\mathcal{M}$	a decision process (MDPS, POMDPS)
$\mathcal{D}$	a dataset, or a distribution over tasks
$\text{Reg}(T)$	regret after T rounds

Uncertainty and models

$\theta \in \Theta$	unknown parameter or type; $\theta^\star$ its true value
$\mu \in \Delta(\Theta)$	prior over parameters; μ_t the posterior at time t
p_ϕ	parametric model (generative model, world model, LLM) with parameters ϕ
$\mathcal{X}, \mathcal{Y}$	input and output (or observation) spaces of a model
$\mathcal{P}$	a set of probability laws; $\mathcal{M}$ a model or manifold, as declared

Games

$\mathcal{N} = \{1, \dots, N\}$	set of agents; $-i$ denotes all agents other than i
$\mathcal{G}$	a game
$\sigma_i \in \Sigma_i$	strategy of agent i ; $\sigma = (\sigma_i, \sigma_{-i})$ a profile
u_i	utility of agent i
$\text{BR}_i(\sigma_{-i})$	best-response correspondence
$\mathcal{I}$	information structure
$\mathcal{K}$	knowledge structure or knowledge graph, as declared
$\mathcal{E}$	a set of equilibria, or an event, as declared

Agentic learning

Ag, E	an agent; an environment
$z_t = (\pi_t, m_t, T_t, M_t, H_t)$	internal state: policy, memory, tools, world model, vocabulary
m_t	memory; T_t the tools available
M_t	the agent's world model; $\mathfrak{M}$ the class of admissible models
H_t	hypothesis space, the agent's vocabulary
L_t	learning rule in force at time t
Rev	revision operator
e_{t+1}	evaluation signal
$\mathcal{W}$	space of cognitive states (Chapter 5)
W	a window of rounds (Chapter 7)
$\mathcal{L}$	a loss or Lagrangian, as declared

Agents, Agentic AI, and Agentic Learning

This chapter introduces the three notions in its title and the conceptual framework that connects them to the rest of the book. An agent is a goal-directed system that closes a loop between what it observes and what it does. Agentic AI is the engineering of such systems around large generative models equipped with memory, tools, and the authority to act. Agentic learning is the process by which an agent improves its future decisions from experience it generates itself, and it may revise not only its behavior but its learning rule, its model of the world, and the very vocabulary in which that model is written. We explain why the outcomes of such processes are best analyzed through solution concepts, organize revisions into behavioral, cognitive, and epistemic levels, and show how these levels structure the book.

In this chapter

- Explain what an agent is in control, decision and game theory, and artificial intelligence, and give a definition that covers all three.
- Describe the anatomy of an agent and interpret autonomy, reactivity, proactiveness, and social ability formally.
- Explain how a large language model becomes an agent, what agentic AI systems consist of, and what is new about them.
- Define agentic learning, distinguish it from supervised learning, online learning, and reinforcement learning, and identify its characteristic difficulty: data generated by the learner's own decisions.
- Explain what a solution concept is for a learning system, through the triad of information, belief, and policy.
- Introduce the behavioral, cognitive, and epistemic levels of agentic learning and the corresponding distinction between known and unknown unknowns.

Consider a defender responsible for a large enterprise network. Every second it receives flow records, authentication logs, and alerts from intrusion detectors. It must decide which connections to allow, which hosts to isolate, whether to query a threat-intelligence service, and when to escalate to a human analyst. Its decisions change what it will see next: an isolated host generates no further traffic, a blocked connection reveals nothing about what it would have carried,

and a query to a tool returns information that exists only because the query was made. The attackers it faces adapt to its behavior, and some of their techniques are unlike anything in its training data.

Twenty years ago such a defender would have been a set of fixed rules. Today it may be a team of software agents built around large language models, each with its own memory, tools, and responsibilities, deciding for itself what to do next. This book is about how such systems should *learn*: how they should use the consequences of their own actions to become better defenders, better traffic managers, or better collaborators, and what can be proved about where that learning leads. We return to the defender throughout this chapter, and treat it in full in Chapter 13.

The purpose of this chapter is conceptual. It explains what we mean by an agent (Section 1.1), by agentic AI (Section 1.2), and by agentic learning (Section 1.3); why the long-run behavior of agentic learning is best described by solution concepts (Section 1.4); how the revisions an agent may make are organized into three levels (Section 1.5); and how the book is organized around those levels (Section 1.6). Formal treatments of the agent, the learning loop, and the solution concepts are developed in later chapters, and we point to them as we go.

1.1 Agents

1.1.1 Three traditions

The word *agent* has been used in at least three intellectual traditions, and the book draws on all of them.

Control and cybernetics. In the tradition that runs from Wiener (1948) to modern feedback control, the central object is a *controller*: a device that measures the output of a system and adjusts its input so as to keep the system near a desired behavior. The controller is not usually called an agent, but it has the essential structure of one. It closes a loop, it acts on the basis of partial and noisy measurements, and it is designed against a performance criterion. This tradition contributes a mature theory of feedback, stability, estimation, and optimal control, and, crucially for us, an understanding of how *information* constrains what a controller can achieve.

Decision and game theory. In economics, statistics, and operations research, an agent is a *decision maker* characterized by preferences over outcomes, beliefs about uncertain events, and a set of feasible choices (Savage, 1954; von Neumann and Morgenstern, 1944). Rationality means choosing the action that is best according to one's preferences given one's beliefs. When several decision makers interact, each must reason about the others, and game theory supplies the solution concepts, such as Nash equilibrium, that describe the outcome (Nash, 1950). This tradition contributes a precise language for goals, beliefs, information, and strategic interdependence.

Table 1.1. Three traditions that study agents. Control supplies the loop and its information constraints; decision and game theory supply the goals and the solution concepts; artificial intelligence supplies the architectures and algorithms that make an agent run.

Tradition	The agent is	Uncertainty enters through	Landmarks this book uses
Control and cybernetics	a feedback law regulating a plant to minimize cost over time	disturbances and noise; the <i>information structure</i> : who measures what, and when	dynamic programming; the Kalman filter and separation; decentralized control (Chapters 2 and 3)
Decision and game theory	a decision maker with preferences and beliefs, facing others who also choose	subjective probability over states, over others' types, and over their strategies	expected utility; Nash and Bayesian Nash equilibrium; self-confirming equilibrium (Chapters 2, 6 and 11)
Artificial intelligence	a system that perceives, remembers, plans, calls tools, and acts on a performance measure	partial observability, model error, and shift between training and deployment	reinforcement learning (Chapter 3); pretrained generative models and language agents (Chapters 4 and 5)

Artificial intelligence. In artificial intelligence an agent is anything that perceives its environment through sensors and acts upon it through actuators, and a rational agent selects actions expected to maximize a performance measure (Russell and Norvig, 2021). Research on multi-agent systems emphasized autonomy and interaction (Shoham and Leyton-Brodie, 2009; Wooldridge, 2009; Wooldridge and Jennings, 1995), and reinforcement learning made learning from interaction the center of attention (Sutton and Barto, 2018). This tradition contributes the engineering of agents: representations, algorithms, and architectures that make agents work at scale.

Table 1.1 sets the three views side by side. They differ in what they take as given and in which landmark results they contribute, but they agree on structure: an agent closes a loop between observation and action, it does so under uncertainty, and it does so in pursuit of a goal. Figure 1.1 shows how their contributions combine.

1.1.2 Environments and agents

We now give a definition general enough to cover all three traditions. Throughout the book $\Delta(X)$ denotes the set of probability measures on a measurable space X , and time is discrete.

Definition 1.1 (Environment). An *environment* is a tuple $E = (\mathcal{S}, \mathcal{A}, \mathcal{O}, P, O, \rho_0)$, where $\mathcal{S}$ is a state space, $\mathcal{A}$ an action space, $\mathcal{O}$ an observation space, $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ a transition kernel, $O : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{O} \times \mathbb{R})$ an emission kernel

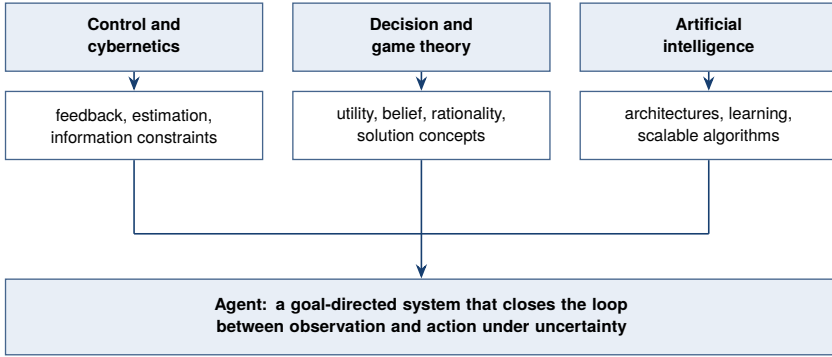


Figure 1.1. Three traditions converge on a common notion of an agent. Each contributes a distinct set of concepts on which the book draws.

producing an observation and a scalar feedback signal, and $\rho_0 \in \Delta(\mathcal{S})$ an initial distribution. Given s_t and a_t , the environment draws $s_{t+1} \sim P(\cdot | s_t, a_t)$ and $(o_{t+1}, r_{t+1}) \sim O(\cdot | s_{t+1}, a_t)$.

The state s_t is in general *not* observed. The feedback r_{t+1} may be a reward, a loss, a verdict from a test, or a message from a user; it need not be observed either. What an agent has at its disposal at time t is its *history*

$$h_t := (o_0, a_0, r_1, o_1, \dots, a_{t-1}, r_t, o_t) \in \mathcal{H}_t.$$

Definition 1.2 (Agent). An *agent* is a tuple $\text{Ag} = (\mathcal{Z}, z_0, F, \pi)$, where $\mathcal{Z}$ is a set of *internal states*, $z_0 \in \mathcal{Z}$ an initial internal state, $F : \mathcal{Z} \times \mathcal{A} \times \mathcal{O} \times \mathbb{R} \rightarrow \mathcal{Z}$ an *update map*, and $\pi : \mathcal{Z} \rightarrow \Delta(\mathcal{A})$ a *decision rule*. The agent selects $a_t \sim \pi(\cdot | z_t)$ and, after receiving (o_{t+1}, r_{t+1}) , updates

$$z_{t+1} = F(z_t, a_t, o_{t+1}, r_{t+1}). \quad (1.1)$$

An agent and an environment together generate a stochastic process $(s_t, z_t, a_t, o_t, r_t)_{t \geq 0}$ with law $\mathbb{P}^{\text{Ag}, \text{E}}$. Goals are expressed by a performance criterion, most often the expected discounted return

$$J(\text{Ag}; \text{E}) := \mathbb{E}^{\text{Ag}, \text{E}} \left[\sum_{t=0}^{\infty} \gamma^t r_{t+1} \right], \quad \gamma \in [0, 1). \quad (1.2)$$

Definition 1.2 is intentionally minimal, and it says three things. First, an agent is *causal*: its action depends on the past only through its internal state, which is computed from the history. Second, the internal state is the *only* place where the agent can store anything, so it is where perception, memory, belief, and knowledge must live. Third, the definition says nothing about how F and π are obtained; they may be designed, optimized, or learned. Chapter 5 refines the

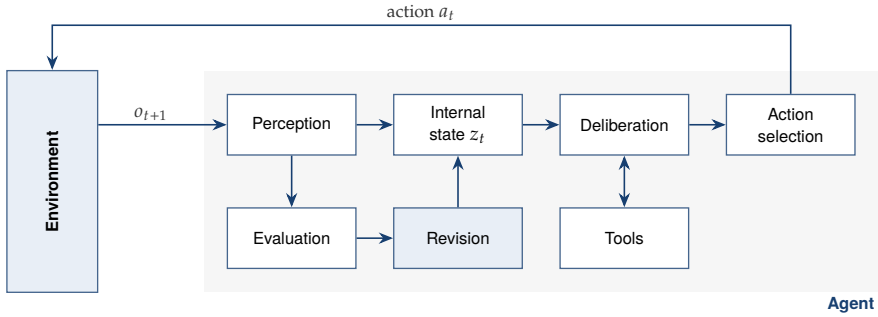


Figure 1.2. Functional anatomy of an agent. Perception turns observations into evidence, the internal state stores what is retained, deliberation (possibly calling tools) selects an action, evaluation scores experience, and revision changes the internal state. The lower loop, evaluation and revision, is what distinguishes a learning agent from a fixed one.

definition by giving the internal state a structure and allowing the update to be stochastic (Definition 5.1).

The internal state costs no generality, in the following sense.

Proposition 1.3 (Internal-state realization). *For every history-dependent policy $\sigma = (\sigma_t)_{t \geq 0}$, $\sigma_t : \mathcal{H}_t \rightarrow \Delta(\mathcal{A})$, there is an agent whose interaction with every environment produces the same law of $(s_t, a_t, o_t, r_t)_{t \geq 0}$ as σ . Conversely, every agent induces a history-dependent policy with the same law.*

Proof. Take $\mathcal{Z} = \bigcup_t \mathcal{H}_t$ and $z_0 = (o_0)$, let F append the new action, feedback, and observation to the history, and set $\pi(\cdot | h_t) = \sigma_t(\cdot | h_t)$. The conditional law of each action given the past is the same under σ and under the agent, and the environment kernels are shared, so the laws agree by induction on t . Conversely, iterating (1.1) expresses z_t as a randomized function of h_t , and $\sigma_t(\cdot | h_t) := \mathbb{E}[\pi(\cdot | z_t) | h_t]$ has the same law. ■

The interest of Proposition 1.3 lies in its converse reading. An agent that remembers everything can do anything, but real agents do not remember everything. Their internal states are compressions: a belief vector, a set of value estimates, a bounded context window, a few retrieved documents. The choice of compression determines what the agent *can* learn, a theme that returns when we study memory as an information state (Proposition 5.4) and bounded rationality (Chapter 10).

1.1.3 The anatomy of an agent

It is useful to open the box of Definition 1.2 and name the functions that an agent performs in each cycle of interaction. Figure 1.2 shows the functional anatomy used throughout the book.

Perception maps raw observations, such as packets, images, or text, to a representation the agent can use. For a Bayesian agent perception is a likelihood; for a neural agent it is an encoder.

Internal state retains what the agent carries from the past: a belief over hidden states, value estimates, a memory of episodes, a model of the world.

Deliberation computes an action from the internal state. It may be a table lookup, a search over future trajectories, or a chain of reasoning expressed in language.

Tools are external procedures that deliberation may call, such as a calculator, a database, a simulator, or another agent, whose results return to deliberation.

Action selection commits to an action in the environment.

Evaluation scores experience through a reward, a prediction error, or a verdict.

Revision changes the internal state in light of the evaluation.

The upper loop of Figure 1.2 is present in every agent. The lower loop makes an agent a *learning* agent, and much of this book concerns what the revision box is allowed to change.

Example 1.4 (The defender as an agent). For the network defender, observations o_t are flow records, logs, and alerts; actions a_t include *allow*, *block*, *isolate host*, *query threat intelligence*, and *escalate*; and the feedback r_{t+1} combines the cost of disruption with the damage of undetected compromise, observed only partially and with delay. The hidden state s_t includes which hosts are compromised and what the attacker intends. The internal state holds a belief over compromise, a record of past incidents, a list of available tools, and a model of how attacks unfold. The threat-intelligence service is a tool: calling it is an action whose only effect is to produce information. ◀

1.1.4 What makes a system an agent

Every feedback law satisfies Definition 1.2, a thermostat included, so the definition alone does not capture what is usually meant by an agent. Wooldridge and Jennings (1995) proposed four properties that distinguish agents from ordinary programs. Each has a precise meaning in our framework, and each admits degrees.

Autonomy The agent's actions are generated by its own decision rule $\pi(\cdot | z_t)$, without external override. Autonomy is measured by the range of situations in which no human or supervisory system intervenes and by the length of the horizon over which the agent acts on its own.

Reactivity The internal state responds to observations: z_{t+1} depends nontrivially on o_{t+1} , and quickly enough that actions remain appropriate as the environment changes (Franklin and Graesser, 1997).

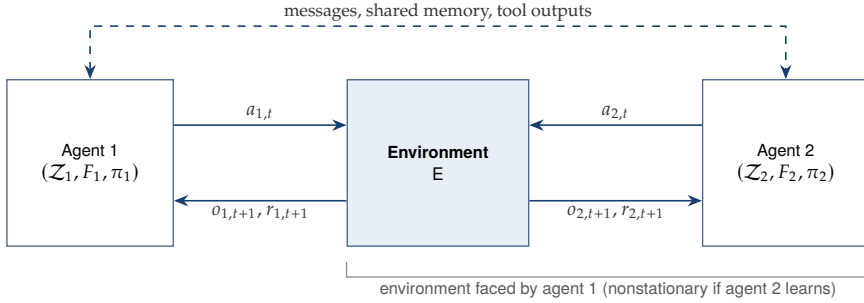


Figure 1.3. Several agents sharing an environment. From the standpoint of agent 1, agent 2 is part of the environment; if agent 2 learns, the environment faced by agent 1 changes over time. Messages, shared memory, and tool outputs form an additional information channel.

Proactiveness The agent is evaluated by a criterion, such as (1.2), that looks ahead, so its actions are chosen for their effect on future states and observations. A proactive agent may take actions whose only purpose is to set up later ones, including actions that gather information.

Social ability Some components of the environment are themselves agents, and the decision rule takes their behavior into account, possibly through communication.

A thermostat is reactive but neither proactive nor social, and its autonomy is confined to a set point. A chess program is autonomous, reactive, and proactive within the game. A defender that plans investigations, calls tools to gather evidence, coordinates with other defenders, and anticipates attackers has all four properties. Agentic AI systems, as we shall see, are distinguished less by any one property than by the breadth of the action spaces over which all four are exercised.

1.1.5 Many agents

Let $\mathcal{N} = \{1, \dots, N\}$ index agents $\text{Ag}_i = (Z_i, z_{i,0}, F_i, \pi_i)$ acting in a common environment with joint action $a_t = (a_{1,t}, \dots, a_{N,t})$, private observations $o_{i,t}$, and individual feedback $r_{i,t}$. From the standpoint of agent i the others are part of its environment: the environment *faced by* agent i is E composed with Ag_{-i} , with augmented hidden state $(s_t, z_{-i,t})$, as depicted in Figure 1.3. Two consequences shape the book.

Remark 1.5 (Endogenous nonstationarity). If the other agents learn, their update maps F_{-i} change the decision rules they apply, and the environment faced by agent i is nonstationary *even if* E is stationary. Guarantees derived for stationary environments, including most of single-agent reinforcement learning, then

no longer apply. Chapter 6 studies when learning nevertheless converges to equilibrium, and Chapter 7 how to describe it when it does not.

Remark 1.6 (Information structure). What agent i can condition on is determined by how its observations depend on the state and on other agents' actions. This *information structure* determines which solution concepts are meaningful and which problems are tractable (Chapter 2). In agentic systems it is often a design choice: which agent reads which message, which tool output is shared, which memory is common.

Multi-agent systems. A *multi-agent system* is a collection of agents whose decisions jointly determine what each of them observes and achieves. Such systems differ along four dimensions that recur throughout the book.

Interaction Agents may share a common goal (a team), have opposed goals (a zero-sum game), or have partly aligned goals (a general-sum game). The first is the setting of team decision theory, the second of adversarial settings such as cybersecurity, and the third of most economic and social systems.

Organization Control may be centralized in an orchestrator that delegates tasks to workers, arranged hierarchically, or fully decentralized among peers.

Communication Agents may not communicate at all, may exchange messages over a communication graph, or may read and write a shared memory.

Information Each agent observes a different part of the world and of the others' behavior; the resulting information structure (Remark 1.6) determines what the agents can jointly achieve.

Figure 1.4 shows four common organizations, and Figure 1.5 shows how they combine in a single system: a security operations team in which specialized agents cooperate with each other, share a case memory, consult tools and a human analyst, and together face an attacker who adapts to their behavior. The team is a cooperative multi-agent system internally and one side of a strategic game externally, a dual character that Chapter 13 analyzes in detail and that Chapter 15 generalizes to networked agents.

1.2 Agentic AI

The term *agentic AI* refers to systems in which large generative models are given the ability to pursue goals over many steps. They decide what to do next, call tools, read and write memory, observe the results, and continue until they judge the task complete (Park et al., 2023; Schick et al., 2023; Wang et al., 2024; Yao et al., 2023). This section explains how such systems instantiate the notion of an agent and what is genuinely new about them.

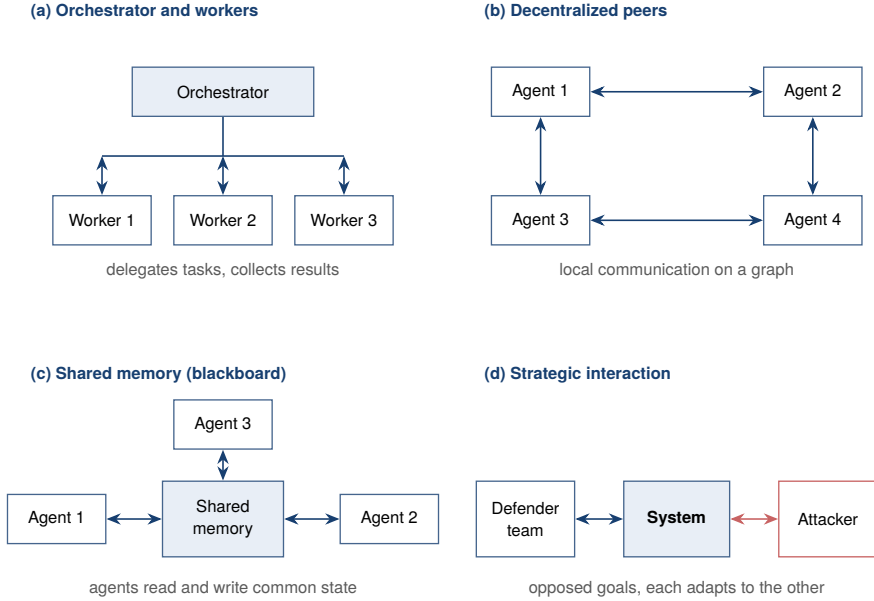


Figure 1.4. Four organizations of multi-agent systems. (a) An orchestrator delegates tasks to workers. (b) Peers communicate locally over a graph. (c) Agents coordinate through a shared memory. (d) Agents with opposed goals interact through a common system, each adapting to the other.

1.2.1 From generative models to agents

A large language model with parameters ϕ is a conditional distribution $p_\phi(y|x)$ over token sequences y given a context x , trained on large corpora to predict text and then adapted to follow instructions and human preferences (Chapter 4). By itself it is not an agent in the sense of [Definition 1.2](#): it has no environment to act on, no internal state that persists between calls, and no goal beyond producing a likely continuation. Yet it contains three things that classical agents had to be given by hand:

- (i) a *broad prior over plausible worlds*, since predicting text written about the world requires representing regularities of the world;
- (ii) a *policy expressible in language*, since instructions, plans, and tool invocations are text, and the model can produce them conditionally on a goal;
- (iii) an *approximate world model*, since the model can be queried about what would happen if an action were taken.

An agent is obtained by placing the model in a loop. Let T be a set of tools, each a map from arguments to results, and let m_t be an external memory. The internal

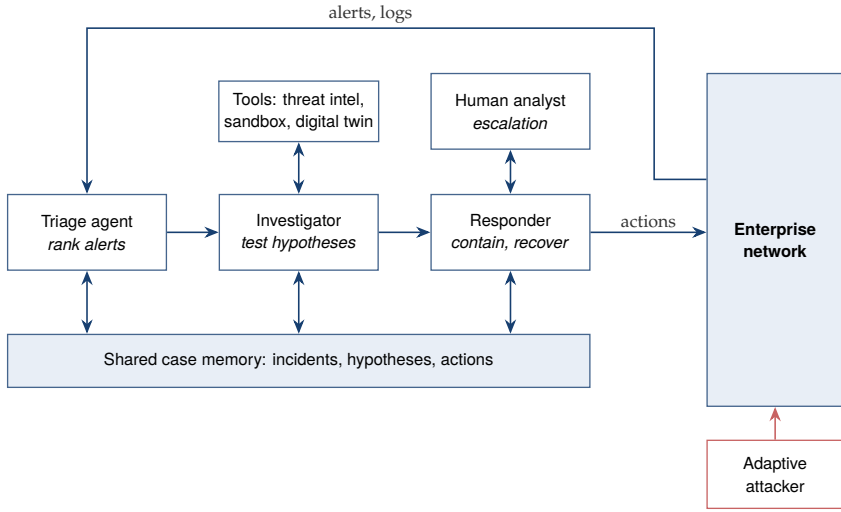


Figure 1.5. A multi-agent security operations team. Triage, investigation, and response agents pass work along a pipeline, share a case memory, call tools, and escalate to a human analyst, while an adaptive attacker acts on the same network. Internally the team is cooperative; externally it plays a game against the attacker.

state is $z_t = (c_t, m_t)$, where c_t is the current context, and one cycle of interaction is

$$\begin{aligned}
 a_t &\sim p_\phi(\cdot | \text{prompt}(c_t, m_t)), \\
 (o_{t+1}, r_{t+1}) &= \text{result of executing } a_t \text{ in the environment or a tool,} \\
 (c_{t+1}, m_{t+1}) &= F(c_t, m_t, a_t, o_{t+1}, r_{t+1}).
 \end{aligned} \tag{1.3}$$

The model supplies the decision rule π ; the prompt construction, the memory, the tools, and the orchestration code supply $\mathcal{Z}$ and F . Figure 1.6 shows the resulting architecture, and Algorithm 1.1 makes the loop explicit.

Three features of Algorithm 1.1 deserve emphasis. First, the action space is *open-ended*: an action is any string, including a plan, a query, or a program. Second, *reasoning is part of acting*: the text the model generates before committing to a tool call changes the context on which later actions are conditioned, so it is itself a sequence of internal actions (Wei et al., 2022). Third, *tools reshape the information structure*: a tool call is an action whose consequence is information, and choosing which tools to call is choosing what to observe (Chapter 5).

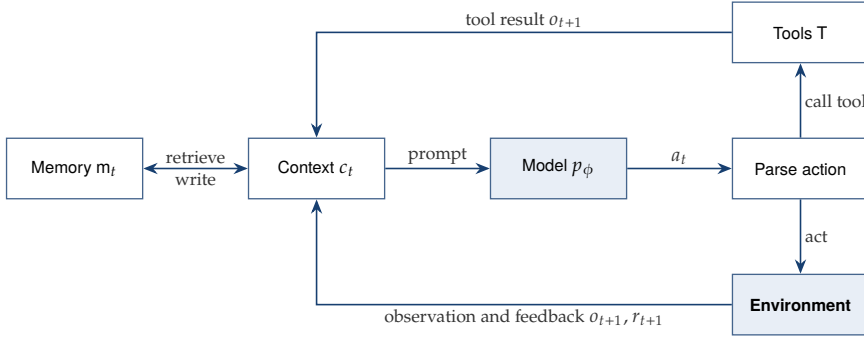


Figure 1.6. Architecture of a language model agent. The model is the decision rule; the context, memory, parser, and tools are supplied by orchestration code. Tool results and environment responses both re-enter the context, which is how the agent observes the consequences of its actions.

Algorithm 1.1 A language model agent

Require: model p_ϕ , tools T , memory m_0 , goal g , step budget T

```

1:  $c_0 \leftarrow \text{init}(g)$ 
2: for  $t = 0, 1, \dots, T - 1$  do
3:    $x_t \leftarrow \text{prompt}(c_t, \text{retrieve}(m_t, c_t))$  ▷ assemble context
4:    $a_t \sim p_\phi(\cdot | x_t)$  ▷ reason, then choose an action
5:   if  $a_t$  is FINISH then
6:     return the answer contained in  $a_t$ 
7:   else if  $a_t$  calls a tool  $\tau \in T$  then
8:      $o_{t+1} \leftarrow \tau(\text{args}(a_t))$ 
9:   else
10:     $o_{t+1} \leftarrow \text{response of the environment to } a_t$ 
11:   end if
12:    $c_{t+1} \leftarrow \text{append}(c_t, a_t, o_{t+1}), \text{ summarized if too long}$ 
13:    $m_{t+1} \leftarrow \text{write}(m_t, c_{t+1})$  ▷ optional
14: end for

```

1.2.2 Degrees of agency

Systems described as agentic differ in how much of the loop is delegated to the model. Figure 1.7 orders them pictorially, and Table 1.2 in detail, by who decides the next step and by what persists and changes across tasks.

A *workflow* executes a control flow fixed by its designer, with the model filling in individual steps, and is autonomous only within each step. An *autonomous agent* decides its own sequence of steps and when to stop, but approaches each new task afresh. A *learning agent* carries the consequences of past tasks into future

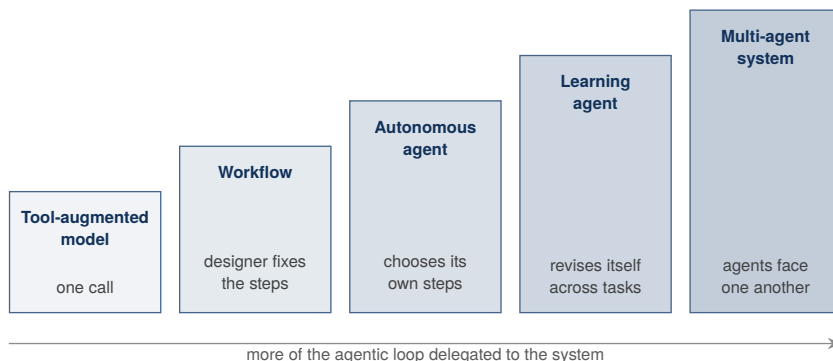


Figure 1.7. Degrees of agency. Moving to the right, more of the loop of Figure 1.2 is delegated to the system: first the choice of steps, then revision of the agent itself, then interaction among several agents.

ones. A *multi-agent system* composes agents with distinct roles, memories, and tools, so that the information structure of Remark 1.6 becomes an object of design.

1.2.3 What is new, and what is not

It is tempting to regard agentic AI as a new field requiring new foundations. We take a more measured view. The loop of Algorithm 1.1 is an agent in the sense of Definition 1.2, and decision making under uncertainty, partially observed control, and game theory apply to it without modification. What is new is the *scale and form* of the objects that enter those theories.

- *Priors and policies come pretrained.* The agent begins with a broad prior and a competent policy that it did not learn from its own experience and that may encode assumptions its designers cannot fully inspect.
- *Action and observation spaces are linguistic.* Actions include plans and queries; observations include documents and messages written by other agents, which may be mistaken or adversarial (Chapter 15; Greshake et al., 2023).
- *Information acquisition is a first-class decision.* Tools make the value of information, a classical but often peripheral concept, central (Chapter 2).
- *Errors compound through the context.* Because the agent’s own outputs re-enter its context, a mistaken belief can be reinforced by the agent’s own reasoning, an agentic form of the exposure bias of sequence models (Chapter 4; Bengio et al., 2015).
- *Multi-agent composition is cheap.* Teams of agents are assembled by configuration rather than training, which makes coordination, incentive design, and security questions immediate (Chapter 15).

Table 1.2. Degrees of agency in systems built on generative models. Each row delegates more of the loop of Figure 1.2 to the system, and each acquires the reach of the row below it only by giving something up.

System	Who decides the next step	What survives the task (defender example)	Characteristic limit
Tool-augmented model	the model, once per query	nothing; each query starts clean (look up an address)	cannot pursue a goal that needs several steps
Workflow	the designer fixes the control flow; the model fills in each step	logs only (a fixed triage pipeline)	handles only the situations the designer anticipated
Autonomous agent	the model, until it decides to stop	context and memory within the task (investigating one incident)	meets every new incident with the same competence as the last
Learning agent	the model; outcomes then revise the agent itself	policy, memory, tools, world model, and vocabulary (a defender that improves across incidents)	improves only along the dimensions its revision operator can reach
Multi-agent system	several agents, each facing the others	all of the above per agent, plus the protocol between them (a security operations team)	each agent's environment is nonstationary because the others learn (Remark 1.5)

In consequence, questions that were once of mainly theoretical interest, such as what happens when an agent's model of the world is wrong in a way it cannot detect, have become questions of engineering practice.

Agentic AI and agentic learning

Agentic AI concerns *autonomous goal-directed action*: what the agent does, with which tools, and for how long. Agentic learning concerns the *improvement of that action through experience the agent itself generates*. An agent can be highly autonomous and learn nothing, as a frozen model executing a workflow does; and a system can learn a great deal without being agentic, as a model trained on a fixed corpus does. This book is about the intersection.

1.3 Agentic Learning

1.3.1 What agentic learning is

Definition 1.7 (Agentic learning, informal). *Agentic learning* is a closed-loop learning process in which a goal-directed agent improves its future decisions by acting in

an environment, observing the consequences, evaluating the results, and revising its internal knowledge or behavior.

Each part of the definition matters. *Closed-loop* means that the data from which the agent learns are produced by its own decisions. *Goal-directed* means that improvement is measured against a criterion, not merely by a better fit to data. *Acting, observing, evaluating, and revising* are the four phases of the lower and upper loops of Figure 1.2. And *internal knowledge or behavior* signals that what is revised need not be the policy.

To see why the last point matters, return to the defender, and suppose an intrusion it classified as benign turns out to have exfiltrated data. It might respond in quite different ways:

- (i) update its policy, for example by reinforcement learning, so that similar traffic is blocked in future;
- (ii) change which tools it queries, for example by consulting a sandbox before allowing unfamiliar executables;
- (iii) store the incident in episodic memory so that the pattern is recognized when it recurs, or revise its investigation plan;
- (iv) change the rule by which it adapts, for example by adapting faster to newly observed attacker behavior;
- (v) revise its model of how attacks unfold, within the family of attack models it already entertains;
- (vi) inspect why the failure occurred and request measurements it does not currently collect;
- (vii) conclude that its model lacks a variable altogether, for example a covert channel it had never conceived of, and enlarge its vocabulary.

These responses differ in *what* they revise. A system confined to the first is doing reinforcement learning. A system that can choose among all seven, and decide which the evidence warrants, is doing agentic learning. The rest of this section makes the idea precise.

1.3.2 The internal state of a learning agent

We give the internal state of a learning agent the structure

$$z_t = (\pi_t, m_t, T_t, M_t, H_t), \quad (1.4)$$

consisting of the current decision rule π_t ; the memory m_t , including episodic records, retrieved facts, and summaries; the tools T_t currently available or preferred; the world model M_t ; and the *hypothesis space* H_t . Not every agent has every component: a model-free reinforcement learner has no world model, and a frozen language model agent has no revisable policy. The decomposition is a lens, and Chapter 5 develops each component.

The hypothesis space is the component that classical accounts of learning leave implicit, and it is the one on which the most important distinctions in this book depend.

Definition 1.8 (Vocabulary and model class). A *vocabulary* H is a collection of random variables that the agent is able to represent, and $\sigma(H)$ is the σ -algebra they generate. The *model class* $\mathfrak{M}(H)$ is the set of probability laws for the agent's observations whose dependence on the unobserved world is expressed through $\sigma(H)$ -measurable quantities. When a parametrization is available we write $\mathfrak{M}(H) = \{Q_\theta : \theta \in \Theta(H)\}$.

Larger vocabularies express more: $H \subseteq H'$ implies $\mathfrak{M}(H) \subseteq \mathfrak{M}(H')$. The world model M_t is an element of $\mathfrak{M}(H_t)$, or a belief over it. Informally, the vocabulary is the set of concepts the agent has; Chapter 10 develops the idea using σ -algebras, Aumann's model of knowledge, and knowledge graphs.

1.3.3 The learning loop

Definition 1.9 (Agentic learner). An *agentic learner* is an agent whose internal state has the form (1.4) and whose update map factors into evaluation and revision,

$$e_{t+1} = \text{Eval}(z_t, h_{t+1}), \quad z_{t+1} = \text{Rev}(z_t, h_{t+1}, e_{t+1}; L_t), \quad (1.5)$$

where e_{t+1} is an *evaluation signal*, Rev is the *revision operator*, and L_t is the *learning rule* in force at time t , which may itself be revised. Actions are drawn as $a_t \sim \pi_t(\cdot | h_t)$.

Figure 1.8 shows the loop; its formal version, with stochastic revision, is Definition 5.20.

Evaluation signals. Evaluation is where agentic learning departs most visibly from reward maximization. A reward is one possible signal. Others include a temporal-difference error, the verdict of a verifier or test suite, a critique produced by another agent or by the agent itself (Shinn et al., 2023), a constraint violation, and the *surprise*

$$\varsigma_{t+1} := -\log p_{M_t}(o_{t+1} | h_t, a_t), \quad (1.6)$$

the negative log-likelihood of what happened under the agent's own model, a quantity that is also central to theories of biological agency (Friston, 2010). Different signals naturally trigger different revisions. A reward prediction error suggests adjusting behavior; a failed tool call suggests revising tool selection; persistent surprise suggests that the world model, or the vocabulary in which it is written, is wrong.

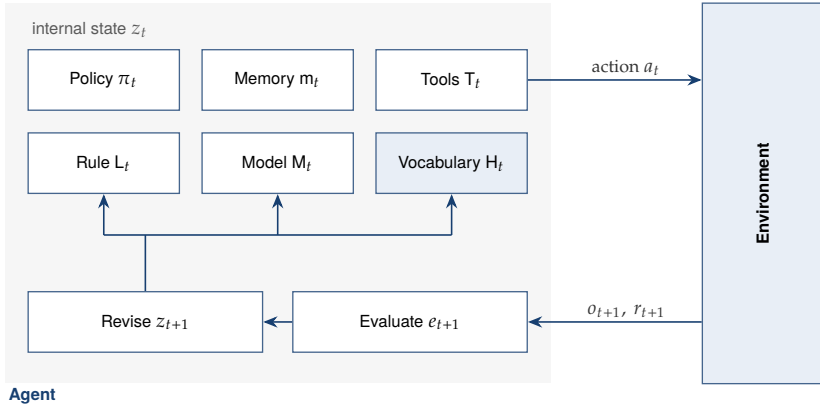


Figure 1.8. The agentic learning loop. By acting, the agent generates the observations from which it learns. Evaluation may lead to revision of any component of the internal state: policy, memory, tools, learning rule, world model, or vocabulary.

Endogenous data. The feature that most sharply separates agentic learning from learning on a fixed dataset appears in the law of the history. For a fixed sequence of policies,

$$\mathbb{P}(h_T) = \rho(o_0) \prod_{t=0}^{T-1} \underbrace{\pi_t(a_t | h_t)}_{\text{agent}} \underbrace{\mathbb{P}^E(r_{t+1}, o_{t+1} | h_t, a_t)}_{\text{environment}}, \quad (1.7)$$

where $\mathbb{P}^E$ is the environment's predictive law. The distribution of the agent's data depends on its own past revisions. This is the source of the power of agentic learning, since the agent can seek the information it needs, and of its characteristic failure, since the agent can avoid the information that would correct it (Figure 1.9).

1.3.4 Agentic learning and other forms of learning

Table 1.3 compares agentic learning with the forms of learning from which it draws.

Proposition 1.10 (Reinforcement learning as a special case). *Reinforcement learning is the special case of Definition 1.9 in which Eval returns the reward or a function of it, Rev modifies only the policy and the statistics used to compute it, and T_t , L_t , and H_t are constant.*

Proof. A reinforcement learning algorithm maintains policy parameters, value estimates, possibly a replay buffer, and in the model-based case an estimated model in a fixed parametric class, and it updates them after each transition by a fixed rule. Placing these in (π_t, m_t, M_t) gives an update of the form (1.5) with the stated restrictions. ■

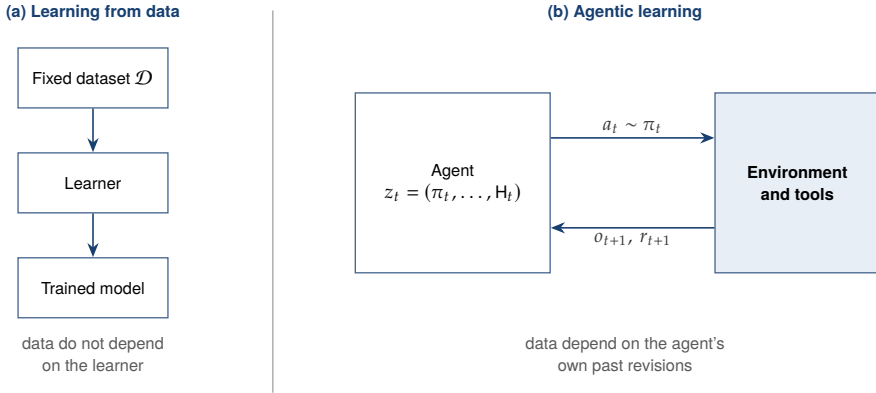


Figure 1.9. Where the data come from. (a) In learning from data, the dataset is fixed before learning begins. (b) In agentic learning, the data are generated by the agent's own actions, so what the agent learns depends on what it chose to do.

Proposition 1.10 is deliberately modest. Reinforcement learning is a *mechanism* inside agentic learning, not a competitor to it. The restrictions it lists are exactly the freedoms an agentic learner may exercise, and exercising them raises four questions that the rest of the book addresses.

What to revise Which component of z_t should change in response to a given piece of evidence?

When to revise How much evidence should accumulate before a deep revision, such as a change of vocabulary, is warranted?

How to revise Which operators implement revision at each level, and at what computational and statistical cost?

Where it leads What is true of the agent's beliefs and behavior when revision stops, or when it continues indefinitely?

The last question is the subject of solution concepts.

1.4 Solution Concepts

1.4.1 What a solution concept is

In game theory a *solution concept* assigns to each game a set of outcomes, regarded as predictions of how rational players behave or recommendations of how they should. Nash equilibrium selects profiles at which no player gains by a unilateral deviation (Nash, 1950); Bayesian Nash equilibrium extends the idea to incomplete information (Harsanyi, 1967). A solution concept is useful because it describes the result of a process without describing the process itself.

Table 1.3. Agentic learning compared with the paradigms it draws on. The decisive column is the second: as one reads down, the learner has more influence over the data it will see, until in the last row it also chooses the terms in which those data are described.

Paradigm	The data come from	What is revised	What is guaranteed, and where
Supervised learning	a fixed sample, drawn before learning begins	model parameters	generalization to the same distribution
Online learning	a sequence chosen by nature or an adversary, not by the learner	the predictor	sublinear regret against the best fixed comparator (Chapter 6)
Reinforcement learning	the learner’s own policy, through exploration	policy and value estimates	convergence to an optimal policy in a stationary environment (Chapter 3)
Learning in games	all agents’ policies at once, so the target moves	strategies and beliefs about others	equilibrium, or no-regret guarantees when it fails (Chapters 6 and 7)
Meta-learning	a distribution of tasks, sampled repeatedly	the learning rule and the prior it starts from	fast adaptation within the task distribution (Chapters 8 and 9)
Agentic learning	the agent’s own actions, its tools, and other agents	any component of z_t , including the vocabulary H_t	consistency at the level revised (Chapters 6 and 12)

For learning systems, solution concepts play a second role. A learning process that converges has a limit, and the limit typically satisfies a consistency condition: behavior is optimal given beliefs, and beliefs are not contradicted by the data that behavior produces. Solution concepts are the language in which such limits are described. The theory of learning in games (Fudenberg and Levine, 1998) showed that simple learning rules lead to Nash, correlated, or self-confirming equilibria depending on what players observe and how they update. Agentic learning requires extending this correspondence to revisions that reach beyond behavior.

1.4.2 The triad of information, belief, and policy

Following Li et al. (2022), every solution concept in this book can be read as a set of consistency conditions among three objects attached to each agent: the *information* it receives, the *belief* it holds, and the *policy* it follows (Figure 1.10). Policies generate information, information shapes beliefs, and beliefs determine policies; a fourth object, the model class, limits which beliefs can be held at all.

Optimality The policy is a best response to the belief.

Consistency The belief agrees with the information generated by the policies actually played: exactly, only along the path of play, or only as closely as the model class allows.

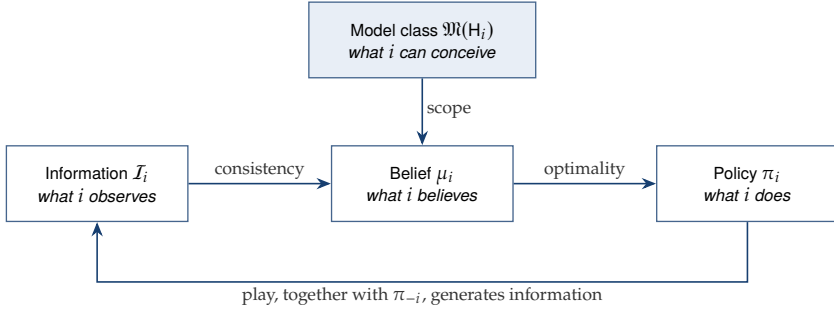


Figure 1.10. The triad of information, belief, and policy. Solution concepts differ in how strictly beliefs must agree with information (consistency), how policies relate to beliefs (optimality), and which beliefs are admissible (scope).

Scope The belief ranges over the model class determined by the agent’s vocabulary.

Chapter 6 formalizes the triad for games. Here we state single-agent versions of three consistency conditions, which already display the distinctions that matter.

1.4.3 Three consistency conditions

An agent chooses a mixed action $\pi \in \Delta(\mathcal{A})$, $\mathcal{A}$ finite, and after action a observes $y \in \mathcal{Y}$, $\mathcal{Y}$ finite, with true law $P^*(\cdot | a)$, receiving utility $u(a, y)$. It entertains a model class $\{Q_\theta(\cdot | a) : \theta \in \Theta\}$, Θ finite, with $Q_\theta(y | a) > 0$, and holds a belief $\mu \in \Delta(\Theta)$. Its subjective expected utility is

$$U(\pi, \mu) := \sum_a \pi(a) \sum_\theta \mu(\theta) \sum_y Q_\theta(y | a) u(a, y).$$

Definition 1.11 (Optimal, self-confirming, and Berk–Nash policies). A pair (π, μ) satisfies *optimality* if $\pi \in \arg \max_{\pi' \in \Delta(\mathcal{A})} U(\pi', \mu)$. It is

- (i) an *objective optimum* if it satisfies optimality and $Q_\theta(\cdot | a) = P^*(\cdot | a)$ for all $a \in \mathcal{A}$ and $\theta \in \text{supp } \mu$;
- (ii) *self-confirming* if it satisfies optimality and $Q_\theta(\cdot | a) = P^*(\cdot | a)$ for all $a \in \text{supp } \pi$ and $\theta \in \text{supp } \mu$;
- (iii) *Berk–Nash* if it satisfies optimality and

$$\text{supp } \mu \subseteq \arg \min_{\theta \in \Theta} \sum_a \pi(a) D_{\text{KL}}(P^*(\cdot | a) \| Q_\theta(\cdot | a)).$$

An objective optimum requires beliefs to be correct everywhere. A self-confirming policy (Fudenberg and Levine, 1993) requires correctness only along the path of play, because actions never taken generate no information that could

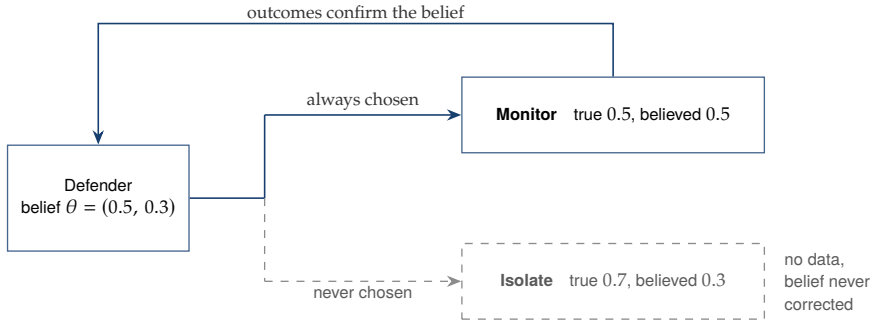


Figure 1.11. A self-confirming trap. Only monitoring is ever chosen, so only its outcomes are observed, and they confirm the belief. The mistaken belief about isolation generates no data that could correct it.

contradict the belief. A Berk–Nash policy (Esponda and Pouzo, 2016) allows the model class to contain no correct model at all, and asks only that beliefs concentrate on the best fits to the data the policy generates.

Proposition 1.12 (Nesting of consistency conditions). *Every objective optimum is self-confirming, and every self-confirming pair is Berk–Nash. If every $\theta \in \Theta$ disagrees with $P^\star$ at some action in $\text{supp } \pi$, then no self-confirming pair with policy π exists.*

Proof. The first claim holds because $\text{supp } \pi \subseteq \mathcal{A}$. For the second, if (π, μ) is self-confirming, then for $\theta \in \text{supp } \mu$ every term of the weighted divergence

$$\sum_a \pi(a) D_{\text{KL}}(P^\star(\cdot | a) \| Q_\theta(\cdot | a))$$

vanishes, so θ attains the minimum value zero. The third claim is immediate from the definition. ■

Example 1.13 (A self-confirming trap). When the defender suspects a compromised host it may *monitor* or *isolate* it. Containment succeeds with probability 0.5 under monitoring and 0.7 under isolation, but the defender, from a poorly chosen prior, believes that isolation succeeds with probability 0.3; its belief is a point mass at $\theta = (0.5, 0.3)$. Optimality selects monitoring. Only monitoring is ever tried, and its outcomes are exactly as believed, so the pair is self-confirming. It is not an objective optimum, and the defender forgoes 0.2 in success probability on every incident. Its belief about isolation is wrong in a way it has arranged never to observe (Figure 1.11). ◀

The formal treatment of self-confirming and Berk–Nash equilibria in games, and their generalization to *knowledge–behavior consistency*, is the subject of Chapter 11.

1.4.4 Beyond fixed points

Two further notions complete the picture. First, many learning processes have no fixed point to converge to: independent learners in a nonstationary environment may track a moving target or cycle indefinitely. For such processes we replace convergence by a statistical description of where the process lies over finite windows, the notion of *non-equilibrium* developed in Chapter 7 (Pan et al., 2024). Second, when agents revise their learning rules (Pan et al., 2026), the natural consistency condition is on the rules themselves: a profile of rules is a *meta-equilibrium* if no agent can improve its expected post-adaptation performance, across the distribution of tasks or opponents, by changing its own rule (Chapter 8).

1.5 Behavioral, Cognitive, and Epistemic Learning

1.5.1 Three levels of revision

We can now state the organizing principle of the book. Order the components of the internal state (1.4), together with the learning rule, by how deeply they reach into the agent’s representation of the world. Revising a policy changes what the agent does. Revising a learning rule changes how it comes to do it. Revising a world model changes what it believes. Revising a vocabulary changes what it is able to believe.

Definition 1.14 (Levels of agentic learning). An agentic learner is

- (i) *behavioral* if revision acts on (π_t, m_t, T_t) only, with the learning rule L_t , the model class, and the vocabulary H_t fixed;
- (ii) *cognitive* if, in addition, revision may change the learning rule L_t or the world model M_t within the model class $\mathfrak{M}(H_0)$;
- (iii) *epistemic* if, in addition, revision may change the vocabulary, $H_{t+1} \neq H_t$, and with it the model class.

The formal counterpart is Definition 5.22. The levels are nested, and they typically operate on separated timescales: behavior is revised quickly, learning rules and models more slowly, and vocabularies most slowly of all (see Borkar, 2008). Figure 1.12 shows the resulting structure of nested loops, and Figure 1.13 places the seven responses of Section 1.3.1 on the levels.

1.5.2 Known unknowns and unknown unknowns

The levels correspond to a classification of what the agent does not know. Let P^* denote the true law of the agent’s observations under its actions.

Definition 1.15 (Well-specified and misspecified learners). A learner with vocabulary H is *well specified* if $P^* \in \mathfrak{M}(H)$, that is, $P^* = Q_{\theta^*}$ for some $\theta^* \in \Theta(H)$, and

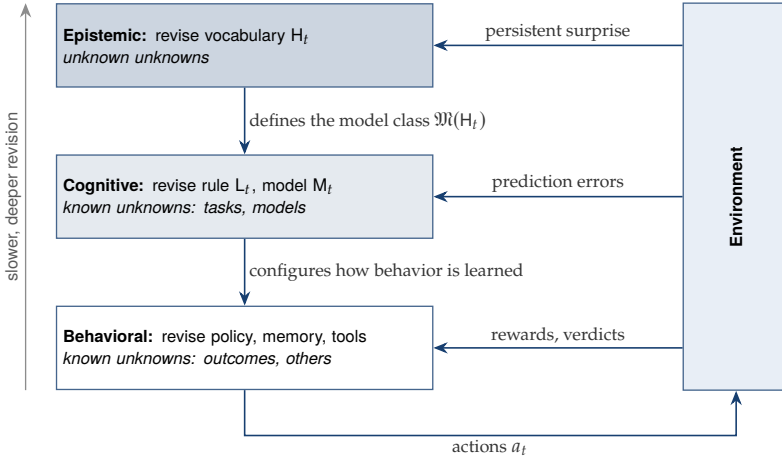


Figure 1.12. The three levels of agentic learning as nested loops. Each level receives evidence from the environment and revises a deeper component of the internal state, on a slower timescale, and each level constrains the level below it.

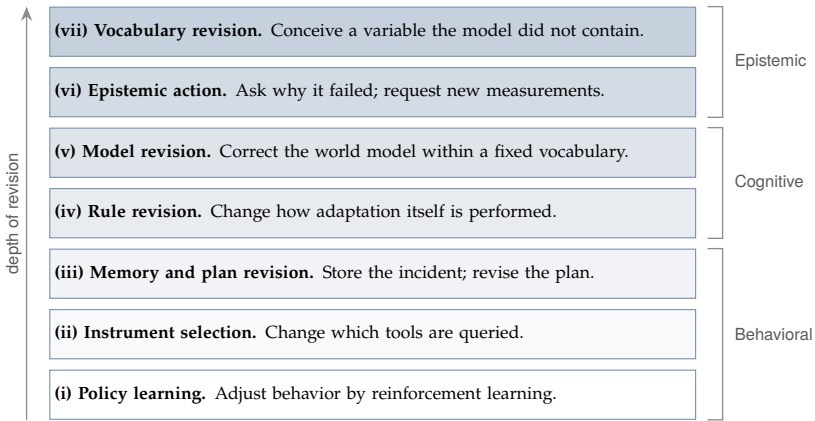


Figure 1.13. The repertoire of responses available to the defender after a missed intrusion, ordered by depth of revision and grouped by level.

misspecified otherwise. Uncertainty about θ^* within a well-specified class is a *known unknown*; the gap $P^* \notin \mathcal{M}(H)$ is an *unknown unknown*.

The distinction is depicted in Figure 1.14. Known unknowns come in two kinds, distinguished in Chapter 2: *aleatoric* uncertainty, the randomness that remains when θ^* is known, and *epistemic* uncertainty, ignorance of θ^* , which data can reduce. Both are handled in principle by a prior over $\Theta(H)$ and Bayes'

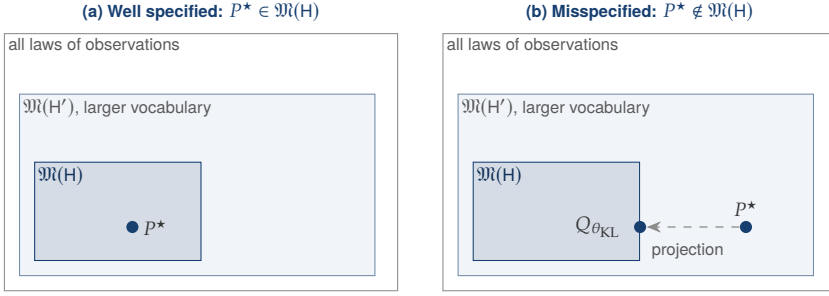


Figure 1.14. Known unknowns and unknown unknowns. (a) When the truth P^* lies in the model class, learning within the vocabulary can find it. (b) When it does not, learning converges to the best approximation within the class, and only a larger vocabulary H' contains the truth.

rule, and the behavioral and cognitive levels are built to handle them. Unknown unknowns are of a different character, as the following classical result shows.

Proposition 1.16 (Posterior concentration under misspecification). *Let $y_1, y_2, \dots$ be i.i.d. with law P^* on a finite set $\mathcal{Y}$, let $\{Q_\theta\}_{\theta \in \Theta}$ with Θ finite satisfy $Q_\theta(y) > 0$, and let the prior have full support. Then the posterior concentrates almost surely on $\Theta_{KL} := \arg \min_\theta D_{KL}(P^* \| Q_\theta)$.*

Proof. For $\theta \in \Theta_{KL}$ and $\theta' \notin \Theta_{KL}$, Bayes' rule and the strong law of large numbers give

$$\frac{1}{t} \log \frac{\mu_t(\theta')}{\mu_t(\theta)} \xrightarrow{\text{a.s.}} D_{KL}(P^* \| Q_\theta) - D_{KL}(P^* \| Q_{\theta'}) < 0,$$

so $\mu_t(\theta')/\mu_t(\theta) \rightarrow 0$, and finiteness of Θ completes the argument. $\blacksquare$

When the learner is misspecified, the posterior still concentrates, with increasing confidence, on the best approximation *available in the vocabulary*. More data make the agent more certain, not more correct. The general statement is given in Chapter 10. The next example shows that the difficulty is not merely one of approximation: an agent may have no statistical means of discovering that its vocabulary is inadequate.

Example 1.17 (A missing variable that no test can reveal). The defender provisions inspection capacity $a_t \in \mathbb{R}$ for the traffic load y_t it will face, incurring loss $(a_t - y_t)^2$. The load is

$$y_t = w_t + \varepsilon_t, \quad w_t \stackrel{\text{i.i.d.}}{\sim} N(m, \tau^2), \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} N(0, \eta^2),$$

where w_t is the volume of a scheduled backup job, recorded in a configuration log before a_t is chosen but absent from the defender's vocabulary $H = \{y\}$. Its model class for the load is $\mathfrak{M}(H) = \{N(c, s^2) : c \in \mathbb{R}, s > 0\}$.

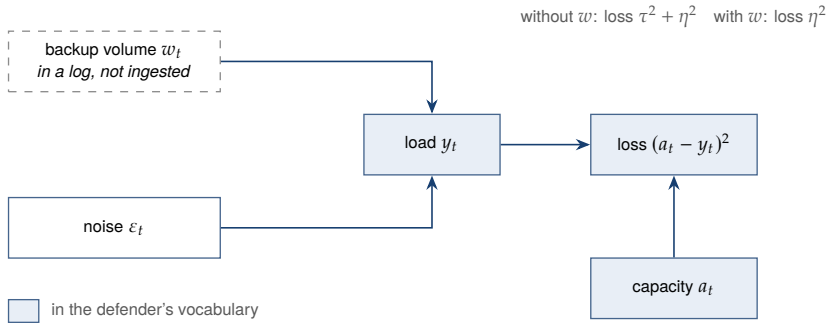


Figure 1.15. A missing variable. The backup volume w_t drives the load but lies outside the defender's vocabulary. The defender's model of y_t is exactly calibrated, yet its expected loss exceeds that of a defender who can represent w_t by τ^2 .

The marginal law of y_t is $N(m, \tau^2 + \eta^2)$, which belongs to $\mathfrak{M}(\mathcal{H})$. The defender's estimates converge to $c = m$ and $s^2 = \tau^2 + \eta^2$, its predictive distribution becomes exactly correct, every calibration test on (a_t, y_t) passes, and it settles on $a_t = m$ with expected loss $\tau^2 + \eta^2$. A defender whose vocabulary contains w would set $a_t = w_t$ and incur expected loss η^2 . The value of the missing variable is τ^2 , which may be arbitrarily large, yet it is invisible: the likelihood of every observed history under the fitted model equals its likelihood under the truth, so no procedure using only (a_t, y_t) can detect the gap. Only an epistemic action, examining the configuration log and asking whether it predicts y_t , can reveal it (Figure 1.15). ◀

Examples 1.13 and 1.17 are instructive in combination. The first is a failure of exploration and has a behavioral remedy: try isolation occasionally. The second is not a failure of exploration, estimation, optimization, or the learning rule, all of which are flawless. It is a failure of *conception*, and no revision within the vocabulary can repair it. This is the phenomenon the epistemic level is built to address. Chapter 12 shows that when a missing variable does leave a trace in the data, in the form of a persistent likelihood gap, the gap can be detected and used to trigger an expansion of the vocabulary.

1.5.3 Levels, uncertainty, and solution concepts

Table 1.4 summarizes the correspondence between the levels, the uncertainty each addresses, the mathematical tools each requires, and the solution concepts that describe its outcomes.

Classical agent-based learning is behavioral almost without exception: the learning-in-games literature (Fudenberg and Levine, 1998), multi-agent reinforcement learning, and most adaptive control revise behavior, or parameter estimates in a fixed class, under a fixed rule. Meta-learning (Finn et al., 2017; Hospedales et al., 2022) and learned world models (Ha and Schmidhuber, 2018) are cognitive.

Table 1.4. The three levels of agentic learning. Reading down a column gives a complete account of one level: what it changes, what it must hold still to work, the ignorance it can repair, and the kind of statement that describes where it ends up.

	Behavioral	Cognitive	Epistemic
Revises	policy, memory, choice of tools	the learning rule, and the world model inside $\mathcal{M}(H)$	the vocabulary H , and with it the model class
Holds fixed	the rule, the model class, the vocabulary	the vocabulary	nothing in principle
Repairs	wrong behavior, and behavior made stale by others	a rule that adapts too slowly, or a model that fits the wrong member of its class	a model class that excludes the truth
Cannot repair	a model that is wrong	a variable the vocabulary lacks	nothing within reach of evidence; the cost is search
Defender	learns which responses contain an intrusion	retunes how fast it adapts, and recalibrates its model of attack spread	adds the covert channel it had never represented
Main tools	dynamic programming, stochastic approximation, learning in games	compositional optimization, online model adaptation, digital twins	knowledge representation, KL projections, conjectures, causal inference
Ends at	Nash and correlated equilibrium; non-equilibrium under drift	meta-equilibrium	self-confirming and Berk–Nash equilibrium; knowledge–behavior consistency
Chapters	6 and 7	8 and 9	10 to 12

The epistemic level is the least developed, and it is where the book makes its principal contributions.

1.5.4 Choosing among levels

Agentic learning subsumes the classical case: behavioral algorithms remain, but they become one option among several that the architecture can invoke. An architecture is agentic to the extent that it can *choose* among the levels, and the choice is itself a decision under uncertainty. Deeper revisions are more expensive, since they require more evidence and computation and discard accumulated estimates, and more disruptive, since a change of vocabulary invalidates beliefs formulated in the old one. They are also the only remedy for the failures of the shallower levels. Deciding which level to revise, and when, is a metareasoning problem in the sense of Russell and Wefald (1991), studied in Chapter 5.

The central claim of the book

Game theory supplies the right foundation for agentic learning, but behavioral learning alone is not sufficient. The outcomes of agentic learning systems should be analyzed through solution concepts matched to the level at which agents revise themselves: equilibrium and non-equilibrium at the behavioral level, meta-equilibrium at the cognitive level, and knowledge-behavior consistency at the epistemic level, where the central mechanism is conjecture, acting on a surrogate for what cannot be modeled and revising it when observation contradicts it.

1.6 Organization of the Book

The book has five parts following this chapter (Figure 1.16). **Part I** supplies the mathematical and computational foundations. Parts II to IV take one level each, in order of the depth of revision. Part V turns the theory into systems and applications.

Part I: Foundations. Chapter 2 develops decision making under aleatoric and epistemic uncertainty, Markov decision processes and partially observed models, games and information structures, and the foundations of utility and rationality. Chapter 3 presents dynamic programming, Monte Carlo methods and rollout, reinforcement learning, and Bayesian decision making as the computational methods that solve these models. Chapter 4 reads generative models and large language models probabilistically: as divergence minimizers, as policies obtained by regularized alignment, as implicit Bayesian inference in context, and as candidate world models. Chapter 5 assembles memory, planning, and tools into the agentic learning loop.

Part II: Behavioral agentic learning. Chapter 6 develops the triad of Figure 1.10 for games and the conditions under which fictitious play, no-regret learning, and gradient play converge to equilibrium. Chapter 7 explains why equilibrium is insufficient for independent learners under nonstationarity and develops a stochastic-process notion of non-equilibrium, with multi-agent mirror descent as the principal example.

Part III: Cognitive agentic learning. Chapter 8 formulates meta-learning on policies as stochastic compositional programming, in model-free and model-based forms, and introduces meta-equilibrium. Chapter 9 treats online adaptation of world models and the calibration of digital twins.

Part IV: Epistemic agentic learning. Chapter 10 revisits knowledge and rationality through σ -algebras, Aumann's model, and knowledge graphs, studies misspecification and subjective rationality, and treats bounded rationality. Chapter 11 develops conjectural learning, knowledge-behavior consistency, and the

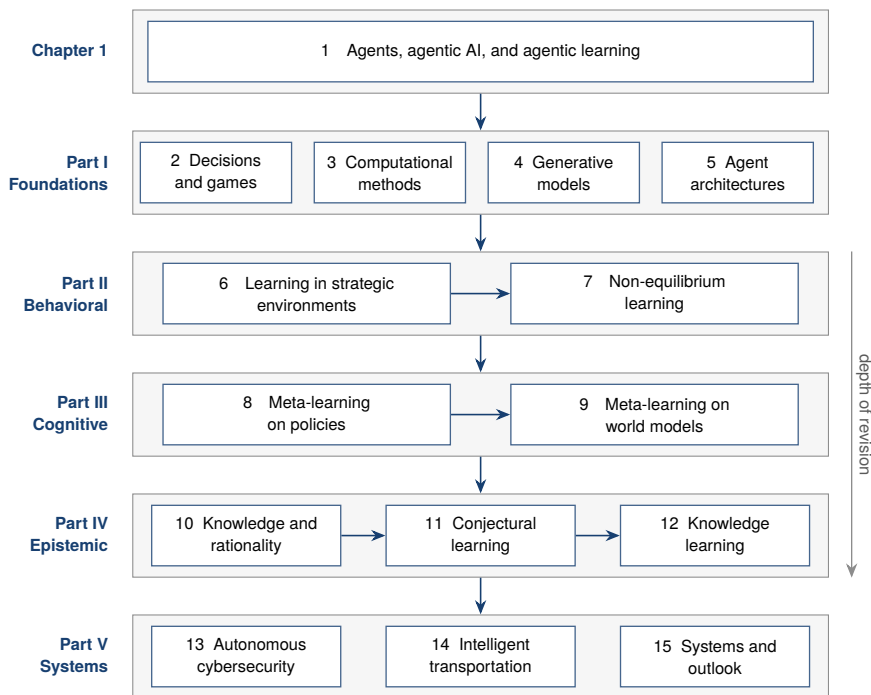


Figure 1.16. Organization of the book. Chapters within a part are read in order, and each part builds on the parts above it. Parts II, III, and IV take the behavioral, cognitive, and epistemic levels in turn.

calibration of language model conjectures in the logit space. Chapter 12 addresses epistemic revision through knowledge graph learning under online causal inference.

Part V: Systems and applications. Chapters 13 and 14 apply the theory to autonomous cybersecurity and intelligent transportation, and Chapter 15 treats coordination, decentralization, and the security and privacy of agentic learning. [Table 1.5](#) shows which levels each application exercises.

Reading paths. The book can be read in several ways, summarized in [Table 1.6](#). Throughout, definitions, theorems, and examples share a single counter within each chapter, notation is summarized at the front of the book, and each chapter ends with notes on the literature and exercises.

Table 1.5. The applications of Part V and the levels they exercise (● primary, ○ secondary). The mechanism column names what actually does the revising, which is what makes the marks more than a classification.

Application	Mechanism of revision	B	C	E
Meta-RL defense in zero-trust networks (Chapter 13)	a learning rule adapted to the attacker type met within the episode	○	●	
Digital twin intrusion response (Chapter 13)	the twin is recalibrated from telemetry, then used for rollout	○	●	○
Multi-agent language model security operations (Chapter 13)	roles, debate, and aggregation across agents; conjectures about the attacker	●		○
Cooperative camera sensing (Chapter 14)	coordinated selection of what to observe, under a submodular objective	●		
Knowledge-infused forecasting (Chapter 14)	a language model proposes causal structure that data then test		○	●
Physics-grounded traffic management (Chapter 14)	a conservation-law model is fitted online and used for predictive control	○	●	
Coordination and alignment (Chapter 15)	payoffs and information are designed so that learning selects the intended convention	●		○
Decentralized and networked agents (Chapter 15)	consensus over a communication graph; local revision, shared statistics	●	○	
Security and privacy (Chapter 15)	attacks and defenses aimed at each component of the loop in turn	●	●	●

Notes and References

The definition of a rational agent as a map from percept sequences to actions is standard in artificial intelligence (Russell and Norvig, 2021). The four properties of Section 1.1.4 are due to Wooldridge and Jennings (1995), a taxonomy of autonomous agents is given by Franklin and Graesser (1997), and Wooldridge (2009) and Shoham and Leyton-Brodie (2009) give comprehensive treatments of multi-agent systems from the logical and game-theoretic sides. The feedback view of purposive behavior goes back to Wiener (1948), and the decision-theoretic view to von Neumann and Morgenstern (1944) and Savage (1954). The internal-state realization of Proposition 1.3 is the standard sufficient-statistic construction of stochastic control (Bertsekas, 2017; Hernández-Lerma and Lasserre, 1996).

For language model agents, Yao et al. (2023) interleave reasoning and acting, Schick et al. (2023) train models to decide when to call tools, Park et al. (2023) equip agents with memory and reflection, Shinn et al. (2023) use verbal self-critique as an evaluation signal, and Wang et al. (2024) survey the area. Treating intermediate reasoning as internal actions goes back to chain-of-thought prompting (Wei et al., 2022). The taxonomy of Table 1.2 is ours; the distinction between workflows, whose control flow is fixed in code, and agents, which direct their own steps,

Table 1.6. Suggested reading paths. Every path begins with this chapter, which fixes the vocabulary the others use.

Reader or course	Chapters	What the path delivers
Learning in games and multi-agent reinforcement learning	1, 2, 3, 6, 7, 8	when independent learners converge, what to say when they do not, and how a learning rule itself becomes an object of choice
Language model agents	1, 4, 5, 11, 13, 15 (2 as needed)	a probabilistic reading of generative models, the architecture around them, and why such agents can confirm their own mistakes
Epistemic theory of learning	1, 2, 10, 11, 12 (6 and 8 for background)	what an agent knows, what misspecification costs, and what it takes to change the vocabulary in which a model is written
Applications in security and transportation	1, 5, 9, 13, 14	the architecture, the world model and its calibration, and two deployments that test them
Two-semester graduate course	1–8, then 9–15	foundations and behavioral learning in the first term; cognitive and epistemic learning, with applications, in the second

follows [Schluntz and Zhang \(2024\)](#). [Sumers et al. \(2024\)](#) organize language agents by memory, action space, and decision procedure, a decomposition close to (1.4). The view of reinforcement learning as a mechanism rather than an architecture follows [Sutton and Barto \(2018\)](#), who define reinforcement learning by its problem.

The distinction between risk and uncertainty is due to [Knight \(1921\)](#); aleatoric and epistemic uncertainty are discussed by [Der Kiureghian and Ditlevsen \(2009\)](#) and [Hüllermeier and Waegeman \(2021\)](#). [Definition 1.15](#) gives the informal terms “known unknowns” and “unknown unknowns” a precise meaning through the well-specification of the model class. [Proposition 1.16](#) is a finite version of [Berk \(1966\)](#).

Solution concepts as limits of learning processes are the subject of [Fudenberg and Levine \(1998\)](#). Self-confirming equilibrium is due to [Fudenberg and Levine \(1993\)](#); related notions of conjectural equilibrium originate in [Battigalli and Guaitoli \(1997\)](#) and, earlier, in the conjectural variations tradition of [Bowley \(1924\)](#) and [Hahn \(1977\)](#). Berk–Nash equilibrium is due to [Esponda and Pouzo \(2016\)](#). The triad of information, belief, and policy ([Section 1.4.2](#)) is developed for game-theoretic multi-agent learning in [Li et al. \(2022\)](#). The conjectural learning framework of Chapter 11 builds on the forecaster–actor–critic architecture of [Li et al. \(2024\)](#) and on belief-conditioned adaptation in [Li et al. \(2023\)](#); the finite-time view of mirror play in Chapter 7 builds on [Pan et al. \(2024\)](#); and the zeroth-order meta-policy optimization of Chapter 8 on [Pan et al. \(2026\)](#). Conditions under which Bayesian learners converge to Nash equilibrium are given by [Kalai and Lehrer \(1993\)](#). That bounded agents should be analyzed through the procedures they use goes back to [Simon \(1955\)](#); see also [Rubinstein \(1998\)](#).

Exercises

1.1 (Agents or not) For each of the following, identify $(\mathcal{Z}, F, \pi)$ in [Definition 1.2](#), decide which of the four properties of [Section 1.1.4](#) it has, and justify your answer: (a) a proportional–integral controller; (b) a spam filter retrained nightly on user reports; (c) a language model answering a single question with web search; (d) the defender of [Example 1.4](#).

1.2 (Finite memory) Let $\mathcal{A} = \mathcal{O} = \{0, 1\}$ and consider agents with $|\mathcal{Z}| = k$. Construct an environment and a history-dependent policy σ such that no agent with $k = 2$ realizes σ in the sense of [Proposition 1.3](#), but one with $k = 4$ does.

1.3 (Levels in practice) Classify each of the following as behavioral, cognitive, or epistemic in the sense of [Definition 1.14](#), identifying which components of (1.4) are revised: (a) ϵ -greedy Q-learning; (b) Q-learning whose step size is adapted by a second, slower learner; (c) a language model agent that writes a note to memory after each failed tool call; (d) a system identification procedure that adds a state variable when residuals exhibit structure; (e) a language model agent that proposes a new causal variable and tests it against logged data.

1.4 (Escaping the trap) In [Example 1.13](#), suppose the defender holds independent Beta priors on the two success probabilities and acts greedily with respect to posterior means. Show that with positive probability isolation is tried only finitely often. Show that any policy that tries both responses infinitely often escapes the trap, and explain why no analogous behavioral modification removes the loss in [Example 1.17](#).

1.5 (Pricing an epistemic action) In [Example 1.17](#), suppose that examining the configuration log costs $\kappa > 0$ per period, and that the defender assigns prior probability q to the existence of a predictive variable with variance at least τ_0^2 in that log. Formulate the decision to examine the log as a Bayesian decision problem ([Chapter 2](#)), and find the values of $(q, \tau_0^2, \kappa, \gamma)$ for which it is worthwhile.

1.6 (Rate of misspecified confidence) In the setting of [Proposition 1.16](#), show that $\mu_t(\theta')/\mu_t(\theta) = \exp(-t\Delta + o(t))$ almost surely for $\theta \in \Theta_{\text{KL}}$ and $\theta' \notin \Theta_{\text{KL}}$, where $\Delta = D_{\text{KL}}(P^\star \| Q_{\theta'}) - D_{\text{KL}}(P^\star \| Q_\theta)$, and interpret Δ as the rate at which a misspecified agent becomes confident.

PART I

**Mathematical and
Computational Foundations of
Agentic Learning**

Decision Making and Games under Uncertainty

This chapter develops the formal language of the book. We begin with a single decision maker facing an unknown parameter and random outcomes, separate aleatoric from epistemic uncertainty, and show through the value of information why the distinction changes what an agent should do. We then extend the analysis in time, to Markov decision processes and their partially observed versions, and show that epistemic uncertainty about a model is itself a form of partial observability. Next we introduce several decision makers: normal-form, Bayesian, extensive-form, and Markov games, together with the information structures that determine what each player may condition on. We end with the foundations on which all of this rests: preferences, expected utility, rationality, and the comparison of information, and with the points at which these foundations begin to strain.

In this chapter

- Formulate Bayesian decision problems and derive Bayes decision rules.
- Decompose predictive uncertainty into aleatoric and epistemic parts, and prove that only the latter vanishes with data.
- Define the value of information and prove that it is nonnegative for a single decision maker.
- State and prove the Bellman optimality theorem, and reduce partially observed problems to belief MDPs.
- Define Nash and Bayesian Nash equilibrium, Markov games, and information structures, and recognize classical and nonclassical structures.
- Relate expected utility, rationality, and Blackwell's comparison of experiments, and identify where a single prior ceases to be adequate.

Every chapter that follows studies an agent that is uncertain and must act anyway. We keep the network defender of [Chapter 1](#) as a running example: its uncertainty about whether a flow is malicious, its partial view of compromised hosts, and its interaction with an attacker of unknown type illustrate, in turn, each of the models below. The purpose of this chapter is to say precisely what

“uncertain” means, what “act optimally” means relative to that uncertainty, and how both notions change when time passes, when observations are partial, and when other decision makers are present. Much of the material is classical, and we present it with an eye to the uses we make of it later: the belief state reappears as the internal state z_t of an agent (Chapter 5); information structures determine which learning dynamics converge (Chapter 6); and the requirement that the true model belong to the agent’s model class, implicit throughout this chapter, is precisely what Part IV abandons.

2.1 Decision Making under Aleatoric and Epistemic Uncertainty

2.1.1 Bayesian decision problems

We work with finite sets whenever doing so costs no insight, and indicate the measure-theoretic extensions in the notes.

Definition 2.1 (Bayesian decision problem). A Bayesian decision problem is a tuple $\mathcal{D} = (\mathcal{A}, \Theta, \mathcal{Y}, Q, u, \mu)$, where $\mathcal{A}$ is a set of actions, Θ a set of parameters, $\mathcal{Y}$ a set of outcomes, $Q : \Theta \times \mathcal{A} \rightarrow \Delta(\mathcal{Y})$ an outcome model, $u : \mathcal{A} \times \mathcal{Y} \rightarrow \mathbb{R}$ a utility function, and $\mu \in \Delta(\Theta)$ a prior. The *expected utility* of action a under belief μ is

$$U(a, \mu) := \sum_{\theta \in \Theta} \mu(\theta) \sum_{y \in \mathcal{Y}} Q_{\theta}(y | a) u(a, y), \quad (2.1)$$

and an action is *Bayes optimal* if it maximizes $U(\cdot, \mu)$.

The structure of (2.1) contains the whole distinction of this section. The inner sum averages over outcomes for a fixed parameter; its randomness would remain even for an agent who knew θ . The outer sum averages over parameters; its randomness reflects only what the agent does not know. It is often convenient to write $\bar{u}(a, \theta) := \sum_y Q_{\theta}(y | a) u(a, y)$ and $U(a, \mu) = \sum_{\theta} \mu(\theta) \bar{u}(a, \theta)$. Figure 2.1 shows how the pieces fit together: the prior, the parameter, and the experiment carry what the agent does not know, while the outcome model carries the randomness that would remain even if it knew θ .

Before acting, the agent may observe the outcome of an *experiment*, a kernel $E : \Theta \rightarrow \Delta(\mathcal{X})$ on a finite set $\mathcal{X}$. A *decision rule* is a map $\delta : \mathcal{X} \rightarrow \Delta(\mathcal{A})$, and its Bayes risk, stated here as a reward, is

$$\mathcal{U}(\delta) := \sum_{\theta} \mu(\theta) \sum_x E(x | \theta) \sum_a \delta(a | x) \bar{u}(a, \theta).$$

For x with positive marginal probability $p_{\mu}(x) := \sum_{\theta} \mu(\theta) E(x | \theta)$, the posterior is $\mu(\theta | x) = \mu(\theta) E(x | \theta) / p_{\mu}(x)$.

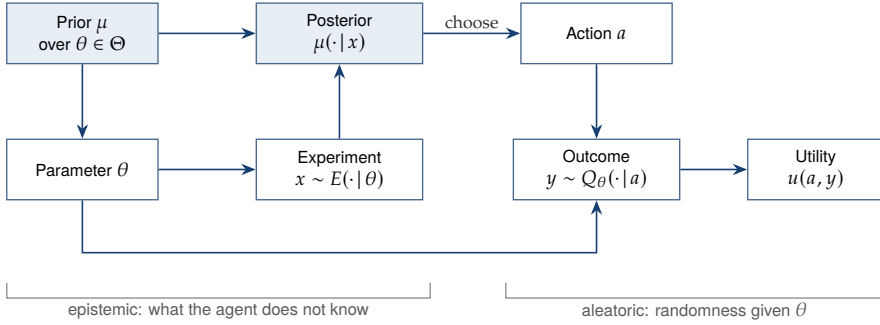


Figure 2.1. Anatomy of a Bayesian decision problem. The agent updates its prior with the outcome of an experiment and chooses an action; the outcome depends on the unknown parameter and on chance. Uncertainty in the left half of the figure is epistemic; uncertainty in the right half is aleatoric.

Theorem 2.2 (Bayes decision rules are posterior-optimal). *A decision rule δ maximizes $\mathcal{U}$ if and only if, for every x with $p_\mu(x) > 0$, $\delta(\cdot|x)$ is supported on $\arg \max_{a \in \mathcal{A}} U(a, \mu(\cdot|x))$.*

Proof. Exchanging the order of summation,

$$\mathcal{U}(\delta) = \sum_x p_\mu(x) \sum_a \delta(a|x) U(a, \mu(\cdot|x)).$$

Each inner sum depends on δ only through $\delta(\cdot|x)$, and it is at most $\max_a U(a, \mu(\cdot|x))$, with equality exactly when $\delta(\cdot|x)$ is supported on the maximizers. Summing with the nonnegative weights $p_\mu(x)$ gives both directions. ■

Theorem 2.2 is the equivalence of the “normal form” of Bayesian analysis, choosing a rule before seeing data, and its “extensive form,” updating and then optimizing. It is the reason an agent may be represented by a belief and a best response, and it fails in exactly the settings, such as ambiguity (Section 2.4.4) and games (Remark 2.40), that break the link between the two forms.

2.1.2 Aleatoric and epistemic uncertainty

Fix an action a and drop it from the notation. The agent’s *predictive distribution* over outcomes is the mixture

$$p_\mu(y) := \sum_{\theta} \mu(\theta) Q_{\theta}(y). \quad (2.2)$$

Regard $\theta \sim \mu$ and $y|\theta \sim Q_{\theta}$ as jointly distributed random variables.

Definition 2.3 (Aleatoric and epistemic uncertainty). Uncertainty about y that remains when θ is known, measured by the conditional spread of Q_{θ} , is *aleatoric*.

Uncertainty about y that is due to ignorance of θ , measured by the spread of Q_θ across $\theta \sim \mu$, is *epistemic*.

Two standard measures make the definition quantitative, and both decompose additively.

Proposition 2.4 (Uncertainty decompositions). *For real-valued y with finite second moment,*

$$\text{Var}(y) = \underbrace{\mathbb{E}_\mu [\text{Var}_{Q_\theta}(y)]}_{\text{aleatoric}} + \underbrace{\text{Var}_\mu (\mathbb{E}_{Q_\theta}[y])}_{\text{epistemic}}, \quad (2.3)$$

and for y on a finite set,

$$H(p_\mu) = \underbrace{\mathbb{E}_\mu [H(Q_\theta)]}_{\text{aleatoric}} + \underbrace{I(\theta; y)}_{\text{epistemic}}. \quad (2.4)$$

Proof. (2.3) is the law of total variance. For (2.4), $H(p_\mu) = H(y)$ and $\mathbb{E}_\mu [H(Q_\theta)] = H(y | \theta)$, and $H(y) = H(y | \theta) + I(\theta; y)$ by definition of mutual information (Cover and Thomas, 2006). ■

The epistemic term in (2.4) has an operational meaning: it is the expected reduction in the agent's uncertainty about θ from observing one outcome, $I(\theta; y) = H(\theta) - H(\theta | y)$. Outcomes are informative about the parameter precisely to the extent that the agent's predictive uncertainty is epistemic. This is the formal reason that curiosity bonuses based on disagreement among models, or on information gain, direct exploration toward reducible uncertainty. The following result makes precise the sense in which only epistemic uncertainty is reducible.

Proposition 2.5 (Epistemic uncertainty vanishes with data). *Let Θ be finite, $\theta \sim \mu$, and let $y_1, y_2, \dots$ be conditionally i.i.d. with law Q_θ given θ . Then*

$$\sum_{n=0}^{\infty} I(\theta; y_{n+1} | y_{1:n}) \leq H(\mu) < \infty,$$

so the epistemic component of the predictive uncertainty about the next outcome tends to zero, while the aleatoric component does not depend on n :

$$I(\theta; y_{n+1} | y_{1:n}) \rightarrow 0, \quad H(y_{n+1} | \theta, y_{1:n}) = \mathbb{E}_\mu [H(Q_\theta)].$$

Proof. By the chain rule for mutual information and the bound $I(\theta; y_{1:N}) \leq H(\theta)$,

$$\sum_{n=0}^{N-1} I(\theta; y_{n+1} | y_{1:n}) = I(\theta; y_{1:N}) \leq H(\mu).$$

The partial sums of a series of nonnegative terms are bounded, so the terms tend to zero. The second claim follows from conditional independence of y_{n+1} and $y_{1:n}$ given θ . ■

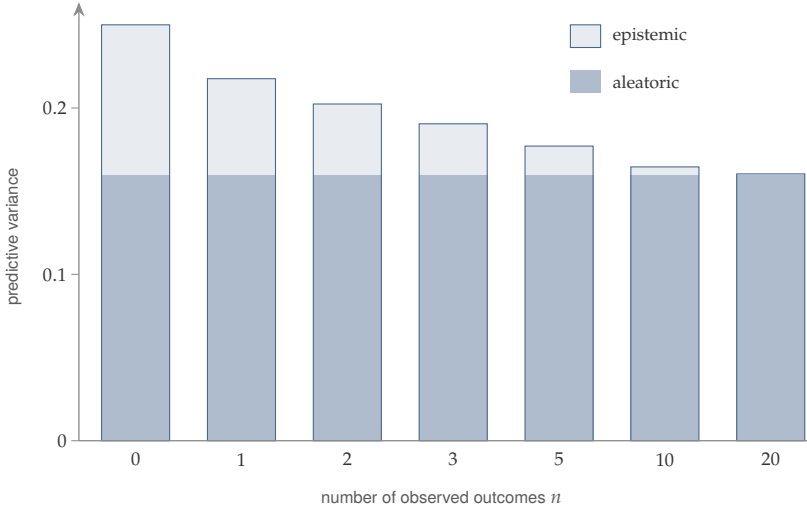


Figure 2.2. Predictive variance in the two-coins example (Example 2.6), averaged over the data, as a function of the number n of observed outcomes. The epistemic part $\mathbb{E}[\text{Var}_{\mu_n}(\theta)]$ decays to zero; the aleatoric part $\mathbb{E}_{\mu}[\theta(1 - \theta)] = 0.16$ does not.

Example 2.6 (Two coins). An outcome $y \in \{0, 1\}$ is Bernoulli with mean $\theta \in \{0.2, 0.8\}$, and μ is uniform. The predictive mean is 0.5 and $\text{Var}(y) = 0.25$. The aleatoric term is $\mathbb{E}_{\mu}[\theta(1 - \theta)] = 0.16$ and the epistemic term is $\text{Var}_{\mu}(\theta) = 0.09$. An agent that learns θ reduces its predictive variance to 0.16, never to zero. An agent that confuses the two parts, treating all 0.25 as irreducible, will not pay to learn θ ; one that treats all of it as reducible will keep paying to learn after θ is known. ◀

Figure 2.2 shows how, in this example, the epistemic part of the predictive variance shrinks as outcomes are observed while the aleatoric part stays fixed.

Remark 2.7 (The distinction is relative to the model). Whether a source of randomness is aleatoric or epistemic is not a property of the world but of the model class. In Example 1.17 of Chapter 1, the variable w_t contributes to aleatoric uncertainty for an agent whose vocabulary omits it and to epistemic uncertainty (or none at all) for an agent that can measure it. The decompositions of Proposition 2.4 are exact given $\{Q_{\theta}\}$; they say nothing about uncertainty the model class cannot express.

2.1.3 The value of information

The distinction between aleatoric and epistemic uncertainty matters for decisions because information about θ can be acquired and has value, while the aleatoric part of y is revealed only by acting.

Definition 2.8 (Value of information). The *value* of an experiment $E : \Theta \rightarrow \Delta(\mathcal{X})$ in decision problem $\mathcal{D}$ is

$$\text{VoI}(E) := \sum_x p_\mu(x) \max_{a \in \mathcal{A}} U(a, \mu(\cdot | x)) - \max_{a \in \mathcal{A}} U(a, \mu).$$

The *expected value of perfect information* is $\text{EVPI} := \sum_\theta \mu(\theta) \max_a \bar{u}(a, \theta) - \max_a U(a, \mu)$, the value of the experiment that reveals θ .

Theorem 2.9 (Information has nonnegative value). For every single-agent Bayesian decision problem and every experiment E , $0 \leq \text{VoI}(E) \leq \text{EVPI}$.

Proof. The map $v \mapsto V(v) := \max_a U(a, v)$ is convex on $\Delta(\Theta)$, being a pointwise maximum of functions linear in v . The prior is the average of the posteriors, $\mu = \sum_x p_\mu(x) \mu(\cdot | x)$, so Jensen's inequality gives

$$\sum_x p_\mu(x) V(\mu(\cdot | x)) \geq V(\mu),$$

which is the lower bound. For the upper bound, $V(v) \leq \sum_\theta v(\theta) \max_a \bar{u}(a, \theta)$ for every v ; apply this to each posterior and average. ■

Example 2.10 (Value of an intrusion detector). A network flow is benign or malicious, $\theta \in \{\text{b}, \text{m}\}$, with prior $\mu(\text{m}) = 0.02$. The defender may *allow* or *block* it. Blocking a benign flow costs 5; allowing a malicious one costs 100; other outcomes cost nothing. Without information, allowing has expected cost 2 and blocking 4.9, so the defender allows.

A detector raises an alarm with probability 0.9 on malicious flows and 0.1 on benign ones. Then $p_\mu(\text{alarm}) = 0.116$, and the posterior probability of malice is 0.155 after an alarm and about 0.0023 after silence. The Bayes rule blocks on alarm and allows on silence, with expected cost $5 \cdot 0.098 + 100 \cdot 0.002 = 0.69$. The detector is therefore worth $2 - 0.69 = 1.31$ per flow, while perfect information would be worth $\text{EVPI} = 2$. A defender that cannot tell the aleatoric variability of benign traffic from its own ignorance of the attack rate will misprice both. ◀

Figure 2.3 gives the geometry behind the example and behind Theorem 2.9.

Theorem 2.9 is a theorem about a *single* decision maker who processes information by Bayes' rule and faces a fixed environment. Each hypothesis matters. In games, information can hurt (Remark 2.40); under ambiguity, dynamic consistency can fail (Section 2.4.4); and under misspecification, the “posterior” of Definition 2.8 is computed with the wrong likelihood, so an experiment's subjective value may bear no relation to its actual value.

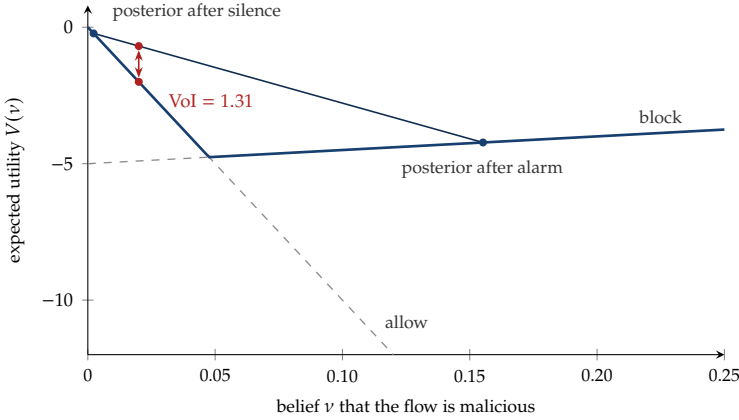


Figure 2.3. Value of information in Example 2.10. The value $V(v) = \max_a U(a, v)$ of a belief is the upper envelope of the utilities of allowing and blocking, hence convex. The prior $v = 0.02$ (red points) is the average of the two posteriors, so the chord joining the posteriors passes above $V(0.02)$; the gap is the value of information.

2.2 Markov Decision Processes and Partially Observed Models

2.2.1 Markov decision processes

Definition 2.11 (Markov decision process). A (discounted) *Markov decision process* is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma, \rho_0)$ with finite state and action sets $\mathcal{S}, \mathcal{A}$, transition kernel $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, bounded reward $r : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, discount factor $\gamma \in [0, 1)$, and initial distribution ρ_0 . The state is observed.

A *history-dependent randomized* policy $\sigma = (\sigma_t)_{t \geq 0}$ maps histories $h_t = (s_0, a_0, \dots, s_t)$ to $\Delta(\mathcal{A})$. A *stationary* policy $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$ depends only on the current state, and it is *deterministic* if each $\pi(\cdot | s)$ is a point mass. Every policy and initial state s determine a law $\mathbb{P}_s^\sigma$ on trajectories, and

$$V^\sigma(s) := \mathbb{E}_s^\sigma \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right], \quad V^*(s) := \sup_{\sigma} V^\sigma(s),$$

the supremum being over all history-dependent randomized policies. For a stationary π write $r_\pi(s) = \sum_a \pi(a | s) r(s, a)$ and $P_\pi(s' | s) = \sum_a \pi(a | s) P(s' | s, a)$. Define on $\mathbb{R}^{\mathcal{S}}$ the *policy evaluation* and *Bellman optimality* operators

$$\mathcal{T}^\pi V := r_\pi + \gamma P_\pi V, \quad (\mathcal{T}V)(s) := \max_{a \in \mathcal{A}} \left[r(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right].$$

Lemma 2.12 (Monotone contractions). $\mathcal{T}^\pi$ and $\mathcal{T}$ are monotone ($V \leq W$ implies $\mathcal{T}V \leq \mathcal{T}W$) and are γ -contractions in the supremum norm. Consequently each has a unique fixed point, and V^π is the fixed point of $\mathcal{T}^\pi$.

Proof. Monotonicity is clear because P is a nonnegative kernel. For the contraction, $|\max_a f(a) - \max_a g(a)| \leq \max_a |f(a) - g(a)|$ gives

$$\begin{aligned} |(\mathcal{T}V)(s) - (\mathcal{T}W)(s)| &\leq \gamma \max_a \sum_{s'} P(s' | s, a) |V(s') - W(s')| \\ &\leq \gamma \|V - W\|_\infty, \end{aligned}$$

and similarly for $\mathcal{T}^\pi$. Banach's fixed-point theorem gives existence and uniqueness. Conditioning on the first transition shows $V^\pi = r_\pi + \gamma P_\pi V^\pi$. ■

Theorem 2.13 (Bellman optimality). Let $\bar{V}$ be the unique fixed point of $\mathcal{T}$. Then $V^* = \bar{V}$, and every deterministic stationary policy π satisfying

$$\pi(s) \in \arg \max_{a \in \mathcal{A}} \left[r(s, a) + \gamma \sum_{s'} P(s' | s, a) \bar{V}(s') \right] \quad \text{for all } s \quad (2.5)$$

is optimal: $V^\pi = V^*$. In particular the supremum over history-dependent randomized policies is attained by a deterministic stationary policy.

Proof. Upper bound. Let σ be any policy. For each t ,

$$\begin{aligned} \mathbb{E}_s^\sigma \left[r(s_t, a_t) + \gamma \bar{V}(s_{t+1}) \mid h_t \right] \\ = \sum_a \sigma_t(a \mid h_t) \left[r(s_t, a) + \gamma \sum_{s'} P(s' | s_t, a) \bar{V}(s') \right] \leq (\mathcal{T}\bar{V})(s_t) = \bar{V}(s_t). \end{aligned}$$

Multiplying by γ^t , taking expectations, and summing over $t < n$ gives, after telescoping, $\mathbb{E}_s^\sigma \left[\sum_{t < n} \gamma^t r(s_t, a_t) \right] + \gamma^n \mathbb{E}_s^\sigma [\bar{V}(s_n)] \leq \bar{V}(s)$. Letting $n \rightarrow \infty$ and using boundedness, $V^\sigma(s) \leq \bar{V}(s)$.

Attainment. If π satisfies (2.5) then $\mathcal{T}^\pi \bar{V} = \mathcal{T}\bar{V} = \bar{V}$, so $\bar{V}$ is a fixed point of $\mathcal{T}^\pi$, and by Lemma 2.12 $\bar{V} = V^\pi$. Combining, $V^\pi = \bar{V} \geq V^\sigma$ for all σ , so $V^* = \bar{V} = V^\pi$. ■

The action-value function $Q^*(s, a) := r(s, a) + \gamma \sum_{s'} P(s' | s, a) V^*(s')$ turns (2.5) into $\pi(s) \in \arg \max_a Q^*(s, a)$, and satisfies its own fixed point equation $Q^*(s, a) = r(s, a) + \gamma \sum_{s'} P(s' | s, a) \max_{a'} Q^*(s', a')$. The algorithms that compute or learn these fixed points, value iteration, policy iteration, rollout, and temporal difference methods, are the subject of Chapter 3. For finite horizons T , the same argument by backward induction gives $V_t^* = \mathcal{T}_t V_{t+1}^*$ with $V_T^* = 0$, and optimal policies are deterministic and Markov but in general nonstationary.

2.2.2 Partially observed Markov decision processes

Definition 2.14 (POMDP). A *partially observed Markov decision process* is a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{O}, P, O, r, \gamma, b_0)$ in which $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ is as in [Definition 2.11](#), $\mathcal{O}$ is a finite set of observations, $O : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{O})$ is an observation kernel with $o_{t+1} \sim O(\cdot | s_{t+1}, a_t)$, and $b_0 \in \Delta(\mathcal{S})$ is the law of s_0 . The agent observes $h_t = (a_0, o_1, \dots, a_{t-1}, o_t)$ but not the states.

The natural statistic of the history is the *belief* $b_t(s) := \mathbb{P}(s_t = s | h_t)$, the posterior over the hidden state.

Lemma 2.15 (Belief update). The belief evolves by the deterministic map $b_{t+1} = \tau(b_t, a_t, o_{t+1})$,

$$\begin{aligned} \tau(b, a, o)(s') &:= \frac{O(o | s', a) \sum_s P(s' | s, a) b(s)}{\sigma(o | b, a)}, \\ \sigma(o | b, a) &:= \sum_{s'} O(o | s', a) \sum_s P(s' | s, a) b(s), \end{aligned} \quad (2.6)$$

whenever $\sigma(o | b, a) > 0$, and $\mathbb{P}(o_{t+1} = o | h_t, a_t) = \sigma(o | b_t, a_t)$.

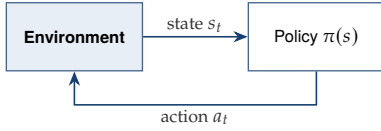
Proof. Bayes' rule applied to the joint law of (s_t, s_{t+1}, o_{t+1}) given (h_t, a_t) , noting that the agent's action is a function of h_t and its own randomization, which is independent of s_t given h_t . ■

Theorem 2.16 (Belief MDP reduction). Define the MDP on the state space $\Delta(\mathcal{S})$ with actions $\mathcal{A}$, reward $\bar{r}(b, a) := \sum_s b(s) r(s, a)$, and transition kernel that moves from b to $\tau(b, a, o)$ with probability $\sigma(o | b, a)$. For every policy of the POMDP there is a policy of this belief MDP with the same expected discounted reward, and conversely. Hence the optimal value of the POMDP equals that of the belief MDP, and optimal policies may be taken to depend on the history only through b_t .

Proof sketch. By [Lemma 2.15](#), b_t is a function of h_t , and the conditional law of (b_{t+1}, reward) given (h_t, a_t) depends on h_t only through b_t : indeed $\mathbb{E}[r(s_t, a_t) | h_t, a_t] = \bar{r}(b_t, a_t)$ and $b_{t+1} = \tau(b_t, a_t, o_{t+1})$ with $o_{t+1} \sim \sigma(\cdot | b_t, a_t)$. Thus any history-dependent POMDP policy induces on (b_t, a_t) a history-dependent policy for the belief MDP with the same expected reward, and a belief-MDP policy is implementable in the POMDP by computing b_t . Since the belief MDP has bounded rewards, the optimality theory of [Theorem 2.13](#) extends to it (with the measurability caveats discussed in the notes), and the optimal policy is a stationary function of b_t . ■

[Theorem 2.16](#), due in essence to Åström (1965), is the prototype of the internal-state construction of [Proposition 1.3](#): the belief is a compressed internal state that loses nothing relevant for decisions. It is also a warning. The belief space is a simplex of dimension $|\mathcal{S}|-1$, and computing optimal policies over it is intractable in general: finite-horizon POMDPs are PSPACE-complete, whereas finite MDPs are

(a) Markov decision process



(b) Partially observed: belief as state

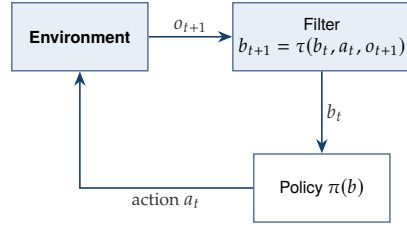


Figure 2.4. (a) In a Markov decision process the policy acts on the observed state. (b) In a partially observed process a filter compresses the history into the belief b_t , which becomes the state of an equivalent fully observed problem (Theorem 2.16).

P-complete (Papadimitriou and Tsitsiklis, 1987). For finite horizons the optimal value function is piecewise linear and convex in b (Smallwood and Sondik, 1973), which underlies exact and point-based solution methods (Kaelbling et al., 1998). Agents that cannot carry exact beliefs must compress further, and the consequences of that compression are a recurrent theme of Chapters 5 and 10. Figure 2.4 contrasts the two settings.

2.2.3 Epistemic uncertainty as partial observability

Suppose the agent observes the state but does not know the transition kernel, which belongs to a known family $\{P_\theta : \theta \in \Theta\}$ with Θ finite. Its uncertainty is epistemic, and it can be represented by a belief $\mu_t \in \Delta(\Theta)$ updated after each transition,

$$\mu_{t+1}(\theta) = \frac{\mu_t(\theta) P_\theta(s_{t+1} | s_t, a_t)}{\sum_{\theta'} \mu_t(\theta') P_{\theta'}(s_{t+1} | s_t, a_t)}. \quad (2.7)$$

Definition 2.17 (Bayes-adaptive MDP). The *Bayes-adaptive MDP* associated with $(\mathcal{S}, \mathcal{A}, \{P_\theta\}, r, \gamma)$ and prior μ_0 is the MDP with hyperstate $(s_t, \mu_t) \in \mathcal{S} \times \Delta(\Theta)$, reward $r(s_t, a_t)$, and transitions in which $s_{t+1} \sim \sum_{\theta} \mu_t(\theta) P_\theta(\cdot | s_t, a_t)$ and μ_{t+1} is given by (2.7). A policy is *Bayes optimal* if it is optimal in this MDP.

Proposition 2.18 (Epistemic uncertainty is partial observability). The *Bayes-adaptive MDP* is the belief MDP of the POMDP with hidden state $(s, \theta) \in \mathcal{S} \times \Theta$, static θ , transitions $P((s', \theta) | (s, \theta), a) = P_\theta(s' | s, a)$, and observation $o_{t+1} = s_{t+1}$. Consequently Bayes-optimal policies exist, can be taken to depend only on (s_t, μ_t) , and solve the exploration–exploitation trade-off exactly.

Proof. Since s_t is observed, the belief over (s_t, θ) is $\delta_{s_t} \otimes \mu_t$, and the update (2.6) for this POMDP reduces to (2.7). Apply Theorem 2.16. ■

Proposition 2.18 explains why exploration need not be an ad hoc addition to optimal control: a Bayes-optimal agent explores exactly as much as the value of

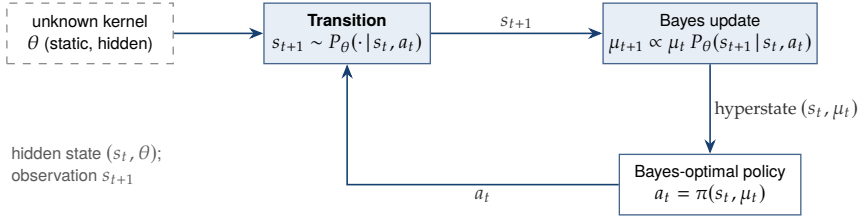


Figure 2.5. Epistemic uncertainty as partial observability. The unknown kernel θ is a hidden, static component of the state; a Bayes update maintains the posterior μ_t ; and the Bayes-optimal policy acts on the hyperstate (s_t, μ_t) .

the information it gains warrants, and the Gittins index solution of the discounted multi-armed bandit (Gittins, 1979) is the celebrated case in which this can be computed. It also marks the limits of the construction. The reduction treats θ as a hidden state and therefore requires the true kernel to be among the P_θ . It handles known unknowns and only known unknowns. The belief space is large even when $\mathcal{S}$ is small, and in practice agents approximate the Bayes-adaptive solution, as discussed in Chapter 3. Figure 2.5 depicts the construction.

Two faces of the belief

The belief b_t over hidden states and the belief μ_t over hidden parameters are the same object mathematically. The first summarizes what the agent does not observe about the present; the second summarizes what it does not know about the laws that generate the future. Both are sufficient statistics, and both are only as good as the model class over which they are defined.

2.3 Multi-Agent Decision Making: Games and Information Structures

When several decision makers interact, each is uncertain not only about nature but about the others, and the others' behavior depends on what they, in turn, believe. Game theory provides the concepts for this regress. We present four models of increasing generality and then the information structures that organize them. Figure 2.6 places them alongside the single-agent models of Section 2.2.

2.3.1 Normal-form games and Nash equilibrium

Definition 2.19 (Normal-form game). A *finite normal-form game* is a tuple $\mathcal{G} = (\mathcal{N}, (\mathcal{A}_i)_{i \in \mathcal{N}}, (u_i)_{i \in \mathcal{N}})$ with players $\mathcal{N} = \{1, \dots, N\}$, finite action sets $\mathcal{A}_i$, and utilities $u_i : \mathcal{A} \rightarrow \mathbb{R}$ on $\mathcal{A} := \prod_i \mathcal{A}_i$. A *mixed strategy* of player i is $\sigma_i \in$

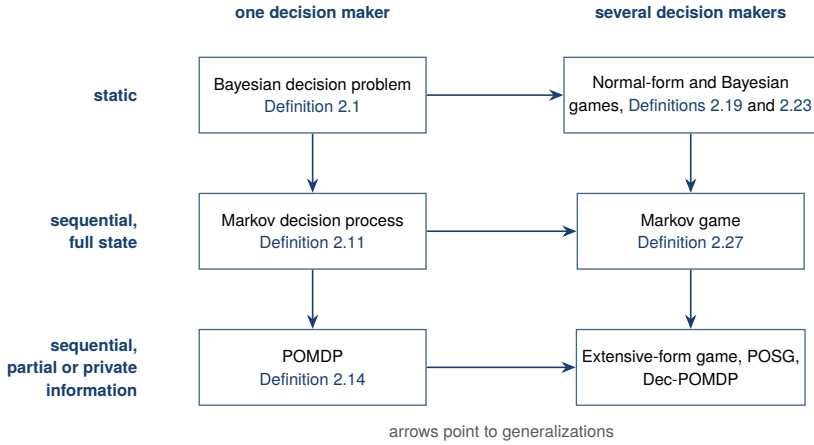


Figure 2.6. Decision models used in the book, organized by the number of decision makers and by what each observes. Arrows point from a model to its generalizations.

$\Sigma_i := \Delta(\mathcal{A}_i)$, and utilities extend multilinearly to profiles $\sigma = (\sigma_i, \sigma_{-i})$: $u_i(\sigma) := \sum_{a \in \mathcal{A}} \prod_j \sigma_j(a_j) u_i(a)$.

Definition 2.20 (Best response and Nash equilibrium). The *best-response correspondence* of player i is $\text{BR}_i(\sigma_{-i}) := \arg \max_{\sigma_i \in \Sigma_i} u_i(\sigma_i, \sigma_{-i})$. A profile σ^* is a *Nash equilibrium* if $\sigma_i^* \in \text{BR}_i(\sigma_{-i}^*)$ for every $i \in \mathcal{N}$.

Theorem 2.21 (Nash (1951, 1950)). *Every finite normal-form game has a Nash equilibrium in mixed strategies.*

Proof sketch. The set $\Sigma = \prod_i \Sigma_i$ is nonempty, compact, and convex. The correspondence $\sigma \mapsto \prod_i \text{BR}_i(\sigma_{-i})$ has nonempty convex values, because u_i is linear in σ_i , and a closed graph, by continuity of u_i and Berge's maximum theorem. Kakutani's fixed-point theorem (Theorem A.15) yields $\sigma^* \in \prod_i \text{BR}_i(\sigma_{-i}^*)$. ■

A mixed strategy is a best response if and only if every action in its support is a best response among pure actions. This *indifference principle* is the usual tool for computing mixed equilibria of small games.

Example 2.22 (Protecting two assets). A defender protects one of two assets, worth 2 and 1 to an attacker who attacks one of them. An attack on the protected asset yields 0; an attack on the unprotected asset yields its value, which the defender loses. The attacker's payoff matrix, with rows its targets and columns the defender's choices, is

	protect 1	protect 2
attack 1	0	2
attack 2	1	0

and the game is zero-sum. If the defender protects asset 1 with probability q , the attacker is indifferent when $2(1 - q) = q$, that is $q = 2/3$. If the attacker targets asset 1 with probability p , the defender is indifferent when $1 - p = 2p$, that is $p = 1/3$. The unique equilibrium has value $2/3$ to the attacker. Note that the attacker targets the more valuable asset *less* often, because the defender protects it more. ◀

2.3.2 Bayesian games

In many interactions players are uncertain about one another's payoffs or information. Harsanyi (1967) showed how to model this uncertainty by introducing *types*.

Definition 2.23 (Bayesian game). A *finite Bayesian game* is a tuple

$$(\mathcal{N}, (\mathcal{A}_i)_{i \in \mathcal{N}}, (\Theta_i)_{i \in \mathcal{N}}, \mu, (u_i)_{i \in \mathcal{N}}),$$

where Θ_i is player i 's finite set of types, $\mu \in \Delta(\Theta)$ with $\Theta = \prod_i \Theta_i$ is a *common prior* with $\mu(\theta_i) > 0$ for all θ_i , and $u_i : \mathcal{A} \times \Theta \rightarrow \mathbb{R}$. A (behavioral) strategy of player i is a map $\sigma_i : \Theta_i \rightarrow \Delta(\mathcal{A}_i)$.

The type θ_i is player i 's private information, and the common prior determines what each type believes about the others' types, $\mu(\theta_{-i} | \theta_i)$. The *interim* expected utility of type θ_i choosing $\alpha_i \in \Delta(\mathcal{A}_i)$ against strategies σ_{-i} is

$$U_i(\alpha_i, \sigma_{-i} | \theta_i) := \sum_{\theta_{-i}} \mu(\theta_{-i} | \theta_i) u_i(\alpha_i, \sigma_{-i}(\theta_{-i}); \theta_i, \theta_{-i}).$$

Definition 2.24 (Bayesian Nash equilibrium). A profile σ^* is a *Bayesian Nash equilibrium* if for every player i and every type θ_i ,

$$\sigma_i^*(\theta_i) \in \arg \max_{\alpha_i \in \Delta(\mathcal{A}_i)} U_i(\alpha_i, \sigma_{-i}^* | \theta_i).$$

A Bayesian Nash equilibrium is a Nash equilibrium of the normal-form game whose players are the types, so Theorem 2.21 guarantees existence. The modeling power of the construction comes with a demanding assumption: the prior μ is common, and it is common knowledge. Players may disagree about the world only because they have seen different information, not because they hold different models of how information is generated. Hierarchies of beliefs about beliefs can be represented without a common prior through universal type spaces (Mertens and Zamir, 1985), and the question of what agents can come to agree on is the subject of Aumann (1976); we return to both in Chapter 10.

2.3.3 Extensive-form games and information sets

Definition 2.25 (Extensive-form game, informal). A finite *extensive-form game* consists of a finite rooted tree whose non-terminal nodes are assigned either to a player in

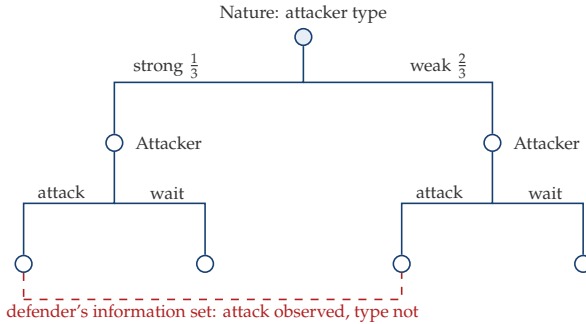


Figure 2.7. An extensive-form game with incomplete information. Nature draws the attacker's type, the attacker decides whether to attack, and the defender, who observes attacks but not types, cannot distinguish the two nodes joined by the dashed line. The defender's subsequent moves and the payoffs are omitted.

$\mathcal{N}$ or to chance; a set of actions at each node; fixed probabilities over actions at chance nodes; utilities at terminal nodes; and, for each player i , a partition of the nodes assigned to i into *information sets*, such that nodes in the same information set have the same available actions. A player cannot distinguish nodes in the same information set.

Figure 2.7 shows the defender of Chapter 1 facing an attacker whose type is drawn by nature. The defender observes whether an attack occurs but not the attacker's type, so the two nodes that follow an attack form a single information set of the defender.

A *pure strategy* selects an action at each information set; a *mixed strategy* randomizes over pure strategies; a *behavioral strategy* randomizes independently at each information set. A player has *perfect recall* if the player never forgets what it once knew or what it did, that is, if two nodes in the same information set are preceded by the same sequence of the player's own information sets and actions.

Theorem 2.26 (Kuhn (1953)). *In a finite extensive-form game in which player i has perfect recall, for every mixed strategy of i there is a behavioral strategy of i that induces the same distribution over terminal nodes against every strategy profile of the other players, and conversely.*

Perfect recall is thus what licenses the agent-level description of a player as a decision rule applied to its current information, the same license that Proposition 1.3 gives in the single-agent case. Without perfect recall, behavioral and mixed strategies differ, and the “agent” of a player at one information set cannot be treated as the same decision maker as at another, a phenomenon we meet again for agents with bounded memory. Refinements of Nash equilibrium for extensive-form games, including subgame perfection (Selten, 1975) and sequential

equilibrium (Kreps and Wilson, 1982), require optimality at information sets off the path of play, and so demand beliefs where experience provides none; this is the starting point of self-confirming equilibrium in Chapter 11.

2.3.4 Markov games and partially observed stochastic games

Definition 2.27 (Markov game). A (discounted) *Markov game* is a tuple

$$(\mathcal{N}, \mathcal{S}, (\mathcal{A}_i)_{i \in \mathcal{N}}, P, (r_i)_{i \in \mathcal{N}}, \gamma)$$

with finite $\mathcal{S}$ and $\mathcal{A}_i$, a transition kernel $P : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ on joint actions, rewards $r_i : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, and $\gamma \in [0, 1)$. A *stationary Markov strategy* of player i is $\pi_i : \mathcal{S} \rightarrow \Delta(\mathcal{A}_i)$, and $V_i^\pi(s) := \mathbb{E}_s^\pi[\sum_t \gamma^t r_i(s_t, a_t)]$. A profile π^* is a *Markov perfect equilibrium* if for each i and each s , $V_i^{\pi^*}(s) \geq V_i^{(\pi_i, \pi_{-i}^*)}(s)$ for every stationary π_i .

Markov games generalize MDPs to several agents and normal-form games to several stages. In the two-player zero-sum case, $r_1 = -r_2 =: r$, the dynamic programming theory of Theorem 2.13 carries over with the maximum replaced by the value of a matrix game. For a matrix $M \in \mathbb{R}^{\mathcal{A}_1 \times \mathcal{A}_2}$ let $\text{val}(M) := \max_{x \in \Delta(\mathcal{A}_1)} \min_{y \in \Delta(\mathcal{A}_2)} x^\top M y$.

Theorem 2.28 (Shapley (1953)). In a two-player zero-sum Markov game, the operator

$$(\mathcal{T}V)(s) := \text{val} \left[r(s, a_1, a_2) + \gamma \sum_{s'} P(s' | s, a_1, a_2) V(s') \right]_{a_1, a_2}$$

is a γ -contraction in the supremum norm. Its fixed point V^* is the value of the game from each initial state, and stationary strategies that play optimal strategies of the matrix games at V^* form an equilibrium.

Proof of the contraction. For matrices of equal size,

$$|\text{val}(M) - \text{val}(M')| \leq \max_{a_1, a_2} |M_{a_1 a_2} - M'_{a_1 a_2}|,$$

because $x^\top M y$ and $x^\top M' y$ differ by at most this amount for all mixed x, y , and the max-min preserves uniform bounds. The matrices defining $(\mathcal{T}V)(s)$ and $(\mathcal{T}W)(s)$ differ entrywise by at most $\gamma \|V - W\|_\infty$. The equilibrium claim is proved as in Theorem 2.13; see Fudenberg and Tirole (1991). ■

When players observe the state only partially and privately, the model becomes a *partially observed stochastic game*: each player i receives $o_{i,t+1} \sim O_i(\cdot | s_{t+1}, a_t)$. If all players share one reward, the model is a *decentralized POMDP*. The belief reduction of Theorem 2.16 fails here, because player i 's optimal action depends on what i believes the others believe, and so on. The finite-horizon decentralized POMDP is NEXP-complete even with two agents (Bernstein et al., 2002), and the tractable cases are those whose information structure allows the regress to be cut, as we now describe.

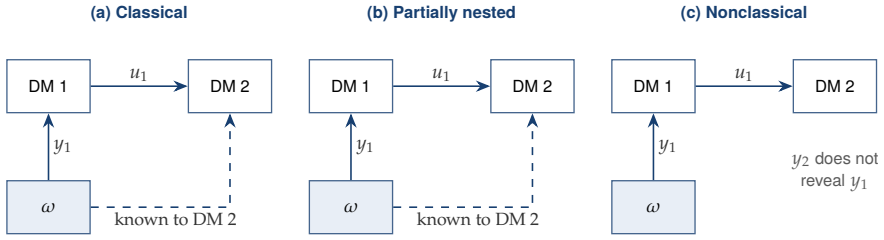


Figure 2.8. Information structures with two decisions, in which the first decision affects what the second observes. In (a) and (b) the second decision maker knows what the first knew; in (c) it does not, so the first decision can be used to signal.

2.3.5 Information structures

The *information structure* of a multi-agent problem specifies what each decision maker knows when it acts. Witsenhausen (1975) gave a formulation that makes the dependence of information on earlier decisions explicit (see also Witsenhausen, 1971), and it is the right one for our purposes because in agentic systems what one agent observes is typically the product of what others have done.

Definition 2.29 (Sequential team or game with an information structure). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space carrying the primitive randomness ω . There are K decisions, made in order $k = 1, \dots, K$ (by the same or different agents). Decision k observes

$$y_k = \eta_k(\omega, u_1, \dots, u_{k-1}) \in \mathcal{Y}_k$$

and chooses $u_k = g_k(y_k) \in \mathcal{U}_k$, where g_k is its decision rule. The *information structure* is the collection of maps $(\eta_k)_{k=1}^K$, and $\mathcal{I}_k := \sigma(y_k)$ is the information field of decision k (which depends on the rules $g_{1:k-1}$ in use).

Definition 2.30 (Classes of information structures). The information structure is

- (i) *static* if every η_k depends on ω alone, and *dynamic* otherwise;
- (ii) *classical* if for all $j < k$, y_j and u_j are functions of y_k (decision k knows everything decision j knew and did);
- (iii) *partially nested* if for all $j < k$ such that u_j affects y_k , the observation y_j is a function of y_k (whoever is influenced by a decision knows what that decision was based on);
- (iv) *nonclassical* otherwise.

Figure 2.8 contrasts the three classes in the simplest case of two decisions.

Classical structures are those of single-agent stochastic control with perfect recall, where dynamic programming applies. In linear-quadratic-Gaussian teams with partially nested information structures, linear decision rules are optimal (Ho and Chu, 1972). Beyond it, even the simplest problems can be hard.

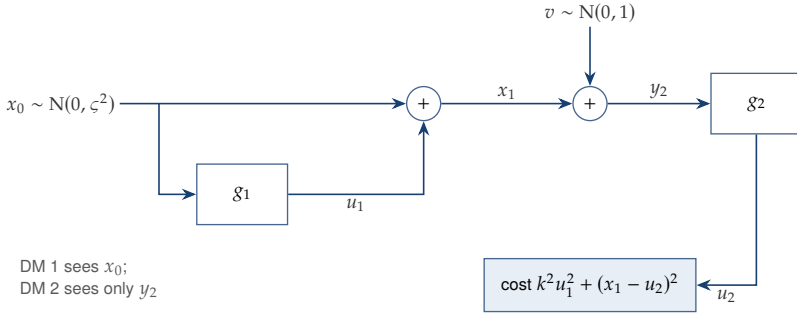


Figure 2.9. Witsenhausen’s counterexample. The first decision maker observes x_0 and shifts the state; the second observes the shifted state only through additive noise. Because u_1 affects y_2 while y_2 does not reveal x_0 , the information structure is nonclassical.

Example 2.31 (Witsenhausen’s counterexample). Let $x_0 \sim N(0, \varsigma^2)$ and $v \sim N(0, 1)$ be independent. The first decision maker observes $y_1 = x_0$ and applies $u_1 = g_1(x_0)$, producing $x_1 = x_0 + u_1$. The second observes $y_2 = x_1 + v$ and applies $u_2 = g_2(y_2)$. The team cost is

$$J(g_1, g_2) = \mathbb{E}[k^2 u_1^2 + (x_1 - u_2)^2], \quad k > 0.$$

All variables are Gaussian and the cost is quadratic, yet the information structure is nonclassical: u_1 affects y_2 , but y_2 does not reveal y_1 . Witsenhausen (1968) showed that for suitable (k, ς) , nonlinear rules strictly outperform the best linear ones. The first decision maker has an incentive to *signal*, shaping x_1 so that the second can estimate it, a dual role of control that linear rules cannot exploit. ◀

Figure 2.9 shows the structure of the counterexample.

The counterexample is the cleanest demonstration of a principle that governs this book: *when actions carry information, optimization and inference cannot be separated*. An agentic learner is in exactly this position with respect to itself, since its actions determine what it will learn, and a multi-agent learner is in this position with respect to others. Two constructive responses to nonclassical structures will be important later. The *common information approach* (Nayyar et al., 2013) identifies information shared by all agents and reduces the team problem to a centralized POMDP whose “actions” are prescriptions mapping private information to decisions. Information design, conversely, treats the information structure itself as a choice variable, which is the perspective of deception and zero-trust design in Chapter 13.

In a single-agent problem with a classical information structure, the belief is a sufficient statistic, optimal policies are functions of it, and learning has a well-defined target. As the information structure moves from classical to nonclassical, and from common priors to private models, each of these properties fails in turn, and with them the classical guarantees of learning. Much of Parts II to IV can be read as a sequence of responses to this failure.

2.4 Preference, Utility, Information, and Rationality

We have written utilities and priors as though they were given. This section explains in what sense they are, what it means for an agent to be rational with respect to them, and why more information helps a rational agent. Each answer rests on axioms, and knowing which axiom an agent violates is often the most informative diagnosis of its behavior.

2.4.1 Expected utility

Let X be a finite set of outcomes and $\mathcal{L} = \Delta(X)$ the set of *lotteries*. A *preference relation* $\succsim$ on $\mathcal{L}$ is a binary relation, with $>$ and $\sim$ its strict and indifference parts. For $p, q \in \mathcal{L}$ and $\alpha \in [0, 1]$, $\alpha p + (1 - \alpha)q$ denotes the compound lottery.

Definition 2.32 (von Neumann–Morgenstern axioms). (i) *Weak order*: $\succsim$ is complete and transitive.

(ii) *Continuity*: if $p > q > r$, there exist $\alpha, \beta \in (0, 1)$ with $\alpha p + (1 - \alpha)r > q > \beta p + (1 - \beta)r$.

(iii) *Independence*: for all p, q, r and $\alpha \in (0, 1]$, $p \succsim q$ if and only if $\alpha p + (1 - \alpha)r \succsim \alpha q + (1 - \alpha)r$.

Theorem 2.33 (von Neumann and Morgenstern (1944)). A preference relation $\succsim$ on $\mathcal{L}$ satisfies the axioms of Definition 2.32 if and only if there is $u : X \rightarrow \mathbb{R}$ such that

$$p \succsim q \iff \sum_x p(x)u(x) \geq \sum_x q(x)u(x).$$

Moreover u is unique up to positive affine transformations.

A proof can be found in Kreps (1988) or Mas-Colell et al. (1995). Theorem 2.33 concerns *objective* lotteries. Agents rarely face these; they face acts whose consequences depend on an unknown state. Savage (1954) showed that axioms on preferences over such acts yield both a utility and a subjective probability. The cleanest statement uses the framework of Anscombe and Aumann (1963), in which an *act* is a map $f : \Theta \rightarrow \mathcal{L}$ from states to lotteries.

Theorem 2.34 (Subjective expected utility, Anscombe and Aumann (1963)). Let Θ be finite and $\succsim$ a preference relation on acts. If $\succsim$ is a nontrivial weak order satisfying

continuity, independence (for mixtures of acts taken state by state), and state independence (the ranking of constant acts does not depend on the state in which they are evaluated), then there exist $u : X \rightarrow \mathbb{R}$, unique up to positive affine transformations, and a unique $\mu \in \Delta(\Theta)$ such that

$$f \succeq g \iff \sum_{\theta} \mu(\theta) \sum_x f(\theta)(x) u(x) \geq \sum_{\theta} \mu(\theta) \sum_x g(\theta)(x) u(x).$$

The importance of [Theorem 2.34](#) for agents is that beliefs are not postulated but *revealed*: an agent whose choices obey the axioms acts as if it held the prior μ and maximized (2.1). For an artificial agent, the theorem is best read in reverse. If the agent's choices cannot be represented this way, then attributing a belief to it, and interpreting its learning as belief revision, is a modeling assumption that must be justified by other means.

2.4.2 Rationality

Definition 2.35 (Subjective rationality). An action a is *rational* in decision problem $\mathcal{D}$ if it is a best response to *some* belief: there is $\nu \in \Delta(\Theta)$ with $a \in \arg \max_{a'} U(a', \nu)$. In a game, an action a_i is rational for player i if it is a best response to some belief $\nu_i \in \Delta(\mathcal{A}_{-i})$ about the others' actions.

Rationality in this sense constrains behavior only weakly, since beliefs are unrestricted. Rationality in a game is strengthened by requiring beliefs to be consistent with the rationality of others, and of others' beliefs about others, and so on.

Definition 2.36 (Rationalizability). Let $R_i^0 = \mathcal{A}_i$ and, for $n \geq 0$, let R_i^{n+1} be the set of $a_i \in R_i^n$ that are best responses to some belief $\nu_i \in \Delta(\prod_{j \neq i} R_j^n)$ (correlated beliefs allowed). The *rationalizable* actions of i are $R_i^\infty := \bigcap_n R_i^n$.

Proposition 2.37 (Bernheim (1984); Pearce (1984)). *In a finite game the sets R_i^n decrease and stabilize after finitely many steps at nonempty sets R_i^∞ . Every action in the support of a Nash equilibrium is rationalizable. Rationalizable actions are exactly those consistent with common knowledge of rationality.*

The gap between rationalizability and Nash equilibrium is a gap in *coordination of beliefs*: rationalizability asks that beliefs be consistent with others' rationality, while Nash equilibrium asks, in addition, that they be correct. Where correctness of beliefs comes from, if not from assumption, is the question that learning in games answers, and Chapter 6 takes it up.

2.4.3 Comparing information

[Theorem 2.9](#) says that observing an experiment never hurts a Bayesian decision maker. When is one experiment better than another *for every* decision problem?

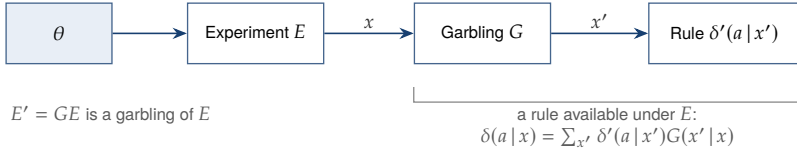


Figure 2.10. Garbling. Any decision rule that uses the garbled observation x' can be simulated by a rule that uses x and applies the garbling itself, which is why a garbled experiment is never more valuable (Theorem 2.39).

Definition 2.38 (Garbling). An experiment $E' : \Theta \rightarrow \Delta(\mathcal{X}')$ is a *garbling* of $E : \Theta \rightarrow \Delta(\mathcal{X})$ if there is a Markov kernel $G : \mathcal{X} \rightarrow \Delta(\mathcal{X}')$, independent of θ , with $E'(x' | \theta) = \sum_x G(x' | x) E(x | \theta)$ for all θ, x' .

Figure 2.10 illustrates the definition.

For an experiment E and a decision problem with utilities $\bar{u}$ and prior μ , let $W(E) := \max_{\delta} \mathcal{U}(\delta)$ be the value attainable with decision rules based on E .

Theorem 2.39 (Blackwell (1951, 1953)). E' is a garbling of E if and only if $W(E) \geq W(E')$ for every finite action set, every utility $\bar{u} : \mathcal{A} \times \Theta \rightarrow \mathbb{R}$, and every prior μ .

Proof of necessity. Suppose $E' = GE$ and let δ' be any decision rule for E' . The rule $\delta(a | x) := \sum_{x'} \delta'(a | x') G(x' | x)$ is available under E and induces, for every θ , the same distribution of actions as δ' under E' . Hence $\mathcal{U}(\delta) = \mathcal{U}(\delta')$, and maximizing over δ' gives $W(E) \geq W(E')$. Sufficiency is a separating hyperplane argument; see Blackwell (1953). ■

Remark 2.40 (Information can hurt in games). Theorems 2.9 and 2.39 fail when other decision makers are present. If it is common knowledge that a player is better informed, others adapt to that fact, and the player's equilibrium payoff may fall. A simple instance: a player who is known to observe an opponent's move before responding may lose the ability to commit, and hence the advantage of moving as a leader. The value of information in games depends on who knows that whom is informed, which is to say on the information structure, not on the experiment alone.

2.4.4 Where the foundations strain

The expected-utility agent of Theorem 2.34 has a single prior, revises it by Bayes' rule, and optimizes exactly. Each of these assumptions has well-documented limits, and each limit marks a place where later chapters must go beyond the framework of this chapter. Figure 2.11 summarizes the chain from preferences to equilibrium and the three points at which it strains.

Ambiguity. An urn contains 30 red balls and 60 balls that are black or yellow in unknown proportion. Many decision makers prefer a bet on red to a bet

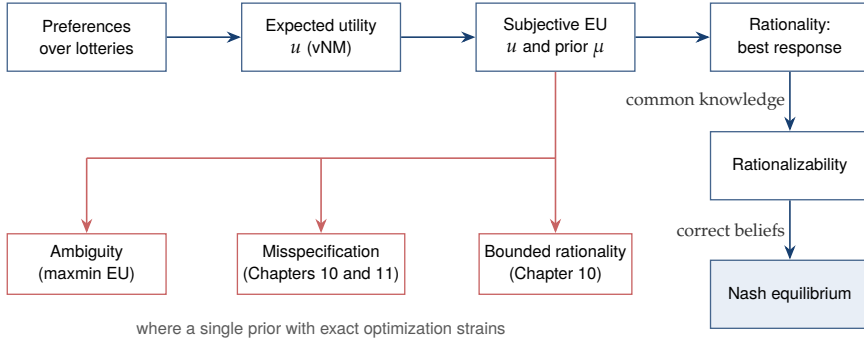


Figure 2.11. From preferences to solution concepts. Axioms on preferences over lotteries yield expected utility, and axioms on acts yield a utility together with a prior. Rationality, common knowledge of rationality, and correct beliefs lead to rationalizability and Nash equilibrium. Ambiguity, misspecification, and bounded rationality mark where a single prior with exact optimization ceases to be adequate.

on black, yet prefer a bet on “black or yellow” to a bet on “red or yellow” (Ellsberg, 1961). No probability μ rationalizes both choices: the first requires $\mu(\text{red}) > \mu(\text{black})$, the second $\mu(\text{black}) + \mu(\text{yellow}) > \mu(\text{red}) + \mu(\text{yellow})$. The pattern violates independence (Savage’s sure-thing principle) and reflects a preference for known over unknown probabilities. Gilboa and Schmeidler (1989) characterized preferences that weaken independence appropriately as those represented by

$$f \succsim g \iff \min_{\mu \in C} \sum_{\theta} \mu(\theta) u(f(\theta)) \geq \min_{\mu \in C} \sum_{\theta} \mu(\theta) u(g(\theta)) \quad (2.8)$$

for a closed convex set of priors C . Robust control and robust reinforcement learning optimize criteria of this form (Hansen and Sargent, 2008). They are an appropriate response when the agent can name the set of models it considers possible, that is, for a known family of unknowns.

Misspecification. Both (2.1) and (2.8) evaluate acts under models the agent can write down. When the true process is not among them, Bayesian updating converges to the best approximation within the model class (Proposition 1.16), the value of information computed by the agent is the value under a wrong likelihood, and the agent’s behavior may be self-confirming rather than optimal. Chapters 10 and 11 develop the theory of decision making under misspecification.

Bounded rationality. Exact Bayesian updating over $\Delta(\Theta)$ and exact optimization over policies are computationally intractable in the problems that interest us (Section 2.2.2). Simon (1955) argued that rationality should be understood procedurally, relative to the computational and informational resources of the

decision maker. Systematic departures from expected utility, such as the reference dependence and probability weighting documented by [Kahneman and Tversky \(1979\)](#), suggest that biological decision makers adopt such procedures. Artificial agents do so by design: every approximation of a belief, a value function, or a policy is a bounded-rationality assumption. Chapter 10 treats bounded rationality under limited computation, memory, and attention.

From one prior to many models

The framework of this chapter is complete for an agent with a correct model class, a single prior, and unlimited computation. The three strains identified above, ambiguity, misspecification, and bounded rationality, correspond to three departures from that ideal: several models without a prior among them, a model class that excludes the truth, and a model that cannot be used exactly. The cognitive level of agentic learning responds to the first and third by revising models and learning rules. The epistemic level responds to the second, which no revision within the model class can repair.

2.5 Risk, Robustness, and the Price of Caution

Section 2.4.4 identified ambiguity as the first place the single-prior framework strains, and (2.8) as the standard repair: evaluate an act by its worst expected utility over a set C of priors. That formula raises three questions it does not answer. What is the class of criteria it belongs to? Where does C come from? And what happens when the decision is not a single choice but a sequence? This section answers all three, because the answers are what Sections 9.4 and 13.2 later use, and because the third answer contains a surprise.

Throughout, Θ is finite, and we work with *losses* $X \in \mathbb{R}^\Theta$ rather than utilities, so that smaller is better; the translation to (2.8) is $X = -u \circ f$.

2.5.1 Coherent risk measures and their dual form

Definition 2.41 (Coherent risk measure). A functional $\rho : \mathbb{R}^\Theta \rightarrow \mathbb{R}$ is a *coherent risk measure* if for all $X, Y \in \mathbb{R}^\Theta$, $c \in \mathbb{R}$, and $t \geq 0$:

- (i) (monotonicity) $X \leq Y$ pointwise implies $\rho(X) \leq \rho(Y)$;
- (ii) (translation equivariance) $\rho(X + c\mathbf{1}) = \rho(X) + c$;
- (iii) (positive homogeneity) $\rho(tX) = t\rho(X)$;
- (iv) (subadditivity) $\rho(X + Y) \leq \rho(X) + \rho(Y)$.

The axioms are due to [Artzner et al. \(1999\)](#), who read them as requirements on a capital reserve: a larger loss needs a larger reserve, a certain extra loss of c needs exactly c more, the units do not matter, and merging two positions never requires more reserve than holding them apart. The fourth axiom is the one with

content, and it is what expectation and the worst case share but the quantile does not.

Example 2.42 (Conditional value at risk). For $\alpha \in (0, 1]$ and a distribution $\hat{\mu}$ on Θ ,

$$\text{CVaR}_\alpha(X) := \min_{s \in \mathbb{R}} \left\{ s + \frac{1}{\alpha} \mathbb{E}_{\hat{\mu}}[(X - s)_+] \right\}, \quad (2.9)$$

which for a continuous loss equals $\mathbb{E}[X \mid X \geq \text{VaR}_\alpha(X)]$, the mean of the worst α fraction of outcomes (Rockafellar and Uryasev, 2000). It is coherent; the quantile VaR_α itself satisfies the first three axioms and fails subadditivity, which is the technical reason risk management moved to CVaR. At $\alpha = 1$, $\text{CVaR}_1 = \mathbb{E}_{\hat{\mu}}$; as $\alpha \downarrow 0$ it tends to the worst case over the support of $\hat{\mu}$. ◀

The next theorem says that every coherent risk measure is a worst case over a set of probability distributions. Nothing more general is available, and nothing less general is needed.

Theorem 2.43 (Dual representation). ρ is a coherent risk measure on $\mathbb{R}^\Theta$ if and only if there is a nonempty closed convex $C \subseteq \Delta(\Theta)$ with

$$\rho(X) = \max_{\mu \in C} \mathbb{E}_\mu[X] \quad \text{for all } X \in \mathbb{R}^\Theta. \quad (2.10)$$

Proof. If ρ has the form (2.10), each axiom is immediate: a maximum of linear functionals with nonnegative coefficients summing to one is monotone, adds c when $c\mathbf{1}$ is added, is positively homogeneous, and is subadditive because the maximum of a sum is at most the sum of the maxima.

Conversely, positive homogeneity and subadditivity make ρ sublinear, so by Rockafellar (1970, Theorem 13.2) it is the support function of the nonempty closed convex set $C = \{\mu \in \mathbb{R}^\Theta : \langle \mu, X \rangle \leq \rho(X) \text{ for all } X\}$, that is, $\rho(X) = \max_{\mu \in C} \langle \mu, X \rangle$. It remains to place C in the simplex. Positive homogeneity gives $\rho(0) = 0$. Fix $\mu \in C$ and $\theta \in \Theta$, and suppose $\mu_\theta < 0$. For $t > 0$ the loss $X = -t\mathbf{e}_\theta$ satisfies $X \leq 0$, so $\rho(X) \leq \rho(0) = 0$ by monotonicity, whereas $\langle \mu, X \rangle = -t\mu_\theta \rightarrow +\infty$ as $t \rightarrow \infty$, contradicting $\langle \mu, X \rangle \leq \rho(X)$. Hence $\mu \geq 0$. Translation equivariance with $X = 0$ gives $\rho(c\mathbf{1}) = c$, so $\max_{\mu \in C} \langle \mu, \mathbf{1} \rangle = \rho(\mathbf{1}) = 1$ and $-\min_{\mu \in C} \langle \mu, \mathbf{1} \rangle = \rho(-\mathbf{1}) = -1$; the two together force $\langle \mu, \mathbf{1} \rangle = 1$ for every $\mu \in C$. Thus $C \subseteq \Delta(\Theta)$. ■

Corollary 2.44 (Ambiguity aversion is risk aversion in disguise). The maxmin expected-utility criterion (2.8) with prior set C ranks acts by $-\rho(-u \circ f)$, where ρ is the coherent risk measure represented by C through (2.10). Conversely every coherent risk measure arises this way.

So the two literatures of Section 2.4.4, ambiguity in decision theory and risk measurement in finance, are one literature with two sign conventions. What distinguishes applications is not the criterion but the set.

2.5.2 Where the ambiguity set comes from

An agent that must name C has essentially three constructions available, and they differ in what kind of surprise they anticipate. In each, $\hat{\mu}$ is the distribution the agent has estimated and $r \geq 0$ is a radius.

Divergence balls. Take $C = \{\mu : D_{\text{KL}}(\mu \parallel \hat{\mu}) \leq r\}$, or the same with another φ -divergence (Ben-Tal et al., 2013). The worst case has a closed form.

Proposition 2.45 (Worst case over a Kullback–Leibler ball). *For every $X \in \mathbb{R}^{\Theta}$ and $r > 0$,*

$$\sup_{\mu: D_{\text{KL}}(\mu \parallel \hat{\mu}) \leq r} \mathbb{E}_{\mu}[X] = \inf_{\beta > 0} \left\{ \beta r + \beta \log \mathbb{E}_{\hat{\mu}}[e^{X/\beta}] \right\}. \quad (2.11)$$

Proof. By the Donsker–Varadhan formula (Theorem A.7) applied to $g = X/\beta$,

$$\beta \log \mathbb{E}_{\hat{\mu}}[e^{X/\beta}] = \sup_{\mu \ll \hat{\mu}} \{ \mathbb{E}_{\mu}[X] - \beta D_{\text{KL}}(\mu \parallel \hat{\mu}) \},$$

so for every $\beta > 0$ the right side of (2.11) dominates $\sup_{\mu} \{ \mathbb{E}_{\mu}[X] - \beta(D_{\text{KL}}(\mu \parallel \hat{\mu}) - r) \}$, which in turn dominates the left side. The right side is therefore an upper bound, and it is the Lagrangian dual of the convex problem on the left, whose feasible set contains $\hat{\mu}$ with $D_{\text{KL}}(\hat{\mu} \parallel \hat{\mu}) = 0 < r$; Slater’s condition holds and the duality gap is zero. ■

Proposition 2.45 is the identity that makes robust training practical: the worst case over a whole ball of distributions costs one scalar minimization, and the minimizing β is a temperature that tilts the nominal distribution toward its expensive outcomes. Proposition 13.9 is this computation in a meta-learning loop.

Transport balls. Take $C = \{\mu : W_1(\mu, \hat{\mu}) \leq r\}$, the 1-Wasserstein ball (Blanchet and Murthy, 2019; Mohajerin Esfahani and Kuhn, 2018).

Proposition 2.46 (Worst case over a transport ball). *If X is L -Lipschitz on the metric space in which W_1 is computed, then*

$$\sup_{\mu: W_1(\mu, \hat{\mu}) \leq r} \mathbb{E}_{\mu}[X] \leq \mathbb{E}_{\hat{\mu}}[X] + Lr, \quad (2.12)$$

with equality on $\mathbb{R}$ when $X(\xi) = L\xi$.

Proof. Kantorovich–Rubinstein duality gives $\mathbb{E}_{\mu}[X] - \mathbb{E}_{\hat{\mu}}[X] \leq L W_1(\mu, \hat{\mu}) \leq Lr$ for every feasible μ . For the equality case, let μ be the law of $\xi + r$ under $\hat{\mu}$; then $W_1(\mu, \hat{\mu}) \leq r$ and $\mathbb{E}_{\mu}[X] = \mathbb{E}_{\hat{\mu}}[X] + Lr$. ■

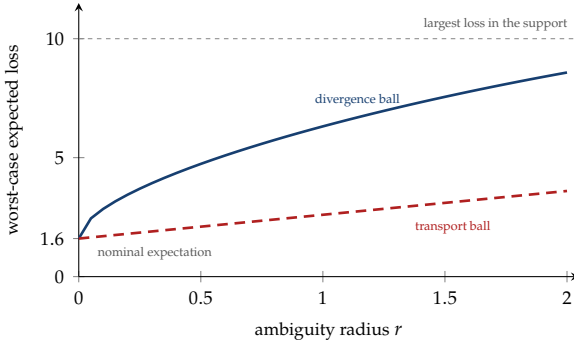


Figure 2.12. The price of caution, for a loss taking the values (0, 1, 2, 5, 10) with estimated probabilities (0.40, 0.30, 0.15, 0.10, 0.05) and nominal expectation 1.6. The worst case over a Kullback–Leibler ball, computed from Proposition 2.45, rises steeply and then saturates: no reweighting of the observed outcomes can exceed the largest of them. The worst case over a transport ball grows linearly and without bound, because moving mass creates outcomes the data never showed. The two radii are in different units, nats and distance, and each must be calibrated separately against the data.

Moment sets. Take $\mathcal{C}$ to be the distributions whose first two moments lie in confidence regions around the empirical ones (Delage and Ye, 2010). This is the weakest assumption of the three and gives the most conservative set.

The choice among them is not a matter of taste, and Figure 2.12 shows why. Every measure in a divergence ball is absolutely continuous with respect to $\hat{\mu}$, so $\sup_{\mu \in \mathcal{C}} \mathbb{E}_{\mu}[X] \leq \max\{X(\theta) : \hat{\mu}(\theta) > 0\}$ no matter how large r is: a divergence ball can reweight the outcomes the data have shown and cannot invent one they have not. A transport ball can, because moving mass a finite distance creates support where there was none, and this is exactly why its worst case grows without bound in r . In the vocabulary of Chapter 1, divergence balls are a model of known unknowns and transport balls reach a little way past them; neither reaches all the way, which is the subject of Chapter 12.

In every case the radius is a statistical object: it is chosen so that the ball covers the unknown true distribution with probability at least $1 - \alpha$, which makes the robust value a $1 - \alpha$ upper confidence bound on the true expected loss, and which makes r shrink as data accumulate. A robust agent is therefore not a permanently pessimistic one; it is an agent whose pessimism is indexed by its ignorance.

2.5.3 Risk in sequence: dynamic consistency

Applying a risk measure to the total loss of a multi-stage problem is the obvious thing to do and it is wrong, in a way worth seeing concretely.

Definition 2.47 (Dynamic consistency). A family of conditional preferences $\{\succsim_s\}$ at date one together with a preference $\succsim_0$ at date zero is *dynamically consistent* if, whenever $f \succsim_s g$ for every date-one state s with strict preference in at least one, $f \succ_0 g$.

Proposition 2.48 (Static CVaR is dynamically inconsistent). Let two date-one states U and D have probability $\frac{1}{2}$ each, and let losses be evaluated by $\text{CVaR}_{1/2}$, the mean of the worse half. Consider

$$\begin{aligned} \text{plan } A : & \text{ in } U, \text{ loss } 0 \text{ or } 1, \text{ each with probability } \frac{1}{2}; \\ & \text{ in } D, \text{ loss } 0 \text{ or } 4, \text{ each with probability } \frac{1}{2}; \\ \text{plan } B : & \text{ in } U, \text{ loss } 0; \quad \text{ in } D, \text{ loss } 3. \end{aligned}$$

Then $\text{CVaR}_{1/2}(A|U) = 1 > 0 = \text{CVaR}_{1/2}(B|U)$ and $\text{CVaR}_{1/2}(A|D) = 4 > 3 = \text{CVaR}_{1/2}(B|D)$, so B is strictly preferred in both states, while unconditionally $\text{CVaR}_{1/2}(A) = 2.5 < 3 = \text{CVaR}_{1/2}(B)$, so A is strictly preferred at date zero.

Proof. Use (2.9) with $\alpha = \frac{1}{2}$. Conditionally on U , plan A puts mass $\frac{1}{2}$ on each of 0 and 1, and $s = 0$ gives $0 + 2 \cdot \frac{1}{2} \cdot 1 = 1$, which is minimal; plan B is certain, so its CVaR is 0. Conditionally on D , the same computation gives 4 for A and 3 for B . Unconditionally, plan A has law $\mathbb{P}(0) = \frac{1}{2}, \mathbb{P}(1) = \frac{1}{4}, \mathbb{P}(4) = \frac{1}{4}$, and $s = 1$ gives $1 + 2 \cdot \frac{1}{4} \cdot 3 = 2.5$, which is minimal; plan B has law $\mathbb{P}(0) = \mathbb{P}(3) = \frac{1}{2}$ and CVaR equal to 3. ■

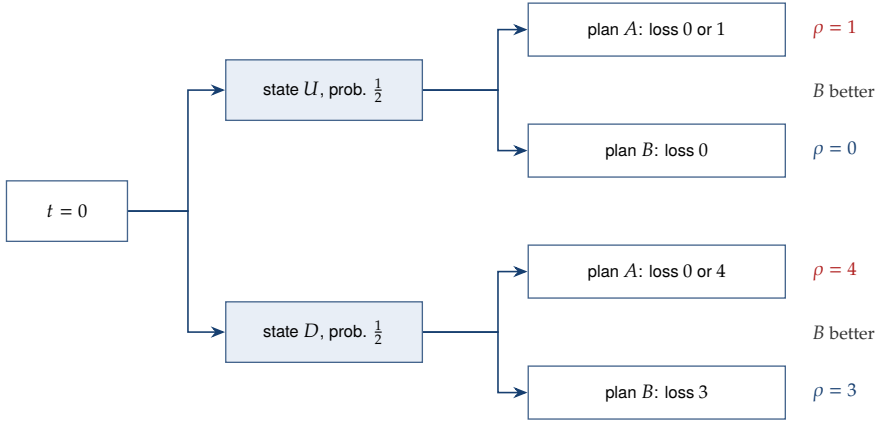
The mechanism is visible in the numbers. A static risk measure allocates a single tail budget across the whole tree, and a plan can concentrate its bad outcomes in one branch to use that budget efficiently; a conditional evaluation, which sees only its own branch, gives that branch a fresh budget and reaches the opposite conclusion. Nothing is paradoxical here, and nothing is repaired by choosing a different α : the same construction works for any $\alpha \in (0, 1)$. What is lost is the principle of optimality, and with it dynamic programming.

The repair is to build the criterion recursively rather than apply it once.

Definition 2.49 (Markov risk measure and the risk-averse Bellman operator). Let a controlled Markov chain have costs $c(s, a)$ and discount $\gamma \in [0, 1)$, and for each (s, a) let $\rho_{s,a}$ be a coherent risk measure on functions of the next state. The *risk-averse Bellman operator* is

$$(\mathcal{T}V)(s) = \min_a \{c(s, a) + \gamma \rho_{s,a}(V)\}. \quad (2.13)$$

Proposition 2.50 (Risk-averse dynamic programming). $\mathcal{T}$ is a γ -contraction in the supremum norm. Hence (2.13) has a unique fixed point V^* , value iteration converges to it geometrically, and a policy that is greedy for V^* is optimal for the nested criterion.



before the state is revealed: $\rho(A) = 2.5 < 3 = \rho(B)$, so A is better at $t = 0$

each branch fills its own half of the tail; pooling them does not

Figure 2.13. The reversal of Proposition 2.48. A static risk measure allots one tail budget to the whole tree. Plan A spends it in the branch where the bad outcome is worst and is cheap in the other, which the date-zero evaluation rewards; each date-one evaluation, seeing only its own branch, charges A the full tail and prefers B .

Proof. Let V, V' and $d = \|V - V'\|_\infty$. Then $V \leq V' + d\mathbf{1}$ pointwise, so monotonicity and translation equivariance give $\rho_{s,a}(V) \leq \rho_{s,a}(V') + d$, and by symmetry $|\rho_{s,a}(V) - \rho_{s,a}(V')| \leq d$. Since $|\min_a f(a) - \min_a g(a)| \leq \max_a |f(a) - g(a)|$,

$$|(\mathcal{T}V)(s) - (\mathcal{T}V')(s)| \leq \gamma \max_a |\rho_{s,a}(V) - \rho_{s,a}(V')| \leq \gamma d$$

for every s . Banach's theorem (Theorem A.14) gives the rest. $\blacksquare$

Only two of the four axioms were used, which is why the proposition covers more than risk aversion.

Corollary 2.51 (Robust Markov decision processes). *Let each pair (s, a) carry its own ambiguity set $\mathcal{P}(s, a)$ of transition distributions, an assumption called (s, a) -rectangularity, and let $\rho_{s,a}(V) = \max_{p \in \mathcal{P}(s,a)} \langle p, V \rangle$. Then $\rho_{s,a}$ is coherent and Proposition 2.50 applies, so the robust Bellman equation has a unique solution and robust value iteration converges.*

This is the result of Iyengar (2005) and Nilim and Ghaoui (2005), and the rectangularity hypothesis is not a technicality. It says that nature may choose the worst transition law at each state and action independently, which overstates nature's freedom when the same physical parameter governs several transitions, and it is what makes the problem tractable: without rectangularity the robust problem is NP-hard (Wiesemann et al., 2013).

2.5.4 The price of caution

Every device in this section buys a guarantee with a loss of value, and the exchange rate is worth naming. Over one stage, [Figure 2.12](#) is the rate: the gap between the robust and the nominal value is what the agent pays for a ball that covers the truth with confidence $1 - \alpha$. Over many stages, nesting compounds it, because a criterion that is pessimistic at each step is pessimistic about a trajectory of steps, and the worst case at stage t is evaluated as if nature will again choose adversarially at $t + 1$. For a robust MDP this is the price of rectangularity, paid once per stage.

Three consequences run through the rest of the book. Pessimism is a tool for *offline* decisions, where the agent cannot test its model against the world and must not be optimistic about what it has not seen; this is [Section 9.4](#), and [Proposition 9.27](#) is [Theorem 2.43](#) applied to a value function. Robustness is a tool for *training*, where the worst case over a ball of task distributions produces a policy that does not fail on the rare task; this is [Proposition 13.9](#), which is [Proposition 2.45](#) in a meta-learning loop. And the radius is a tool for *honesty*: an agent that reports which ball it optimized over has said something checkable about how much it does not know, which is more than a point estimate says.

Caution is a set, not an attitude

Every coherent way of being cautious is a worst case over a set of distributions, so the design question is never how cautious to be but which set to be cautious about. Divergence balls reweight what has been seen; transport balls reach past it; moment sets assume the least and cost the most. Applied once to a whole trajectory, a risk measure destroys the principle of optimality; applied recursively at each stage, it preserves it, at the price of a conservatism that compounds.

Notes and References

Statistical decision theory in the form of [Section 2.1.1](#) is treated comprehensively by [Berger \(1985\)](#); the equivalence of normal and extensive forms of Bayesian analysis goes back to [Raiffa and Schlaifer \(1961\)](#). The value of information was formalized in decision analysis by [Howard \(1966\)](#). The distinction between risk and uncertainty is due to [Knight \(1921\)](#). The aleatoric–epistemic terminology is discussed in engineering by [Der Kiureghian and Ditlevsen \(2009\)](#) and in machine learning by [Hüllermeier and Waegeman \(2021\)](#); the variance and entropy decompositions of [Proposition 2.4](#) are used for uncertainty estimation in deep learning by [Kendall and Gal \(2017\)](#) and [Depeweg et al. \(2018\)](#).

Markov decision processes are treated by [Puterman \(1994\)](#) and [Bertsekas \(2017\)](#), following [Bellman \(1957\)](#); the extension of [Theorem 2.13](#) to Borel state and

action spaces, which requires measurable selection arguments, is in Hernández-Lerma and Lasserre (1996). The belief reduction is due to Åström (1965); for the structure of optimal POMDP value functions see Smallwood and Sondik (1973), for algorithms Kaelbling et al. (1998), and for a modern treatment emphasizing controlled sensing Krishnamurthy (2016). Bayes-adaptive MDPs were developed by Duff (2002); Ghavamzadeh et al. (2015) survey Bayesian reinforcement learning.

For game theory we follow Fudenberg and Tirole (1991), Osborne and Rubinstein (1994), Myerson (1991), and Maschler et al. (2013); for dynamic games in the control tradition, Başar and Olsder (1999). Bayesian games are due to Harsanyi (1967), stochastic games to Shapley (1953), and the behavioral strategy theorem to Kuhn (1953). Information structures in the sense of Definition 2.29 are due to Witsenhausen (1968, 1971); partially nested structures to Ho and Chu (1972); the common information approach to Nayyar et al. (2013); and Yüksel and Başar (2013) give a comprehensive treatment of information structures in networked control. Oliehoek and Amato (2016) introduce decentralized POMDPs, and Bernstein et al. (2002) establish their complexity.

For expected utility see Fishburn (1970) and Kreps (1988); subjective expected utility is due to Savage (1954) and Anscombe and Aumann (1963). Rationalizability is due to Bernheim (1984) and Pearce (1984). The comparison of experiments is due to Blackwell (1951, 1953). The observation that more information may reduce a player's payoff in games goes back at least to Hirshleifer (1971). Ambiguity aversion originates with Ellsberg (1961) and maxmin expected utility with Gilboa and Schmeidler (1989); its connection to robust control is developed by Hansen and Sargent (2008). Bounded rationality is due to Simon (1955); see Rubinstein (1998) for formal models, and Kahneman and Tversky (1979) for prospect theory.

Coherent risk measures and Theorem 2.43 are due to Artzner et al. (1999); the optimization form of conditional value at risk in (2.9) is Rockafellar and Uryasev (2000), and Shapiro et al. (2021) give the definitive treatment of risk measures in stochastic programming, including the convex and law-invariant extensions omitted here. Distributionally robust optimization over φ -divergence balls is Ben-Tal et al. (2013), over Wasserstein balls Mohajerin Esfahani and Kuhn (2018) and Blanchet and Murthy (2019), and over moment sets Delage and Ye (2010). Dynamic consistency and the recursive construction of Definition 2.49 and Proposition 2.50 are Ruszczyński (2010); robust dynamic programming under rectangularity is Iyengar (2005) and Nilim and Ghaoui (2005), and the hardness of the non-rectangular problem is Wiesemann et al. (2013). The version of Proposition 2.48 given here, with its explicit two-state reversal, is formulated for this book; the phenomenon is standard.

Exercises

2.1 (Value at risk is not coherent) Let $\Theta = \{1, 2, 3, 4\}$ be equally likely and let X and Y be losses that are 10 on a single, different state each and 0 elsewhere. Compute $\text{VaR}_{0.3}(X)$, $\text{VaR}_{0.3}(Y)$, and $\text{VaR}_{0.3}(X + Y)$, and exhibit the failure of subadditivity. Then verify that $\text{CVaR}_{0.3}$ satisfies it on the same example, and identify the set C of [Theorem 2.43](#) for CVaR_α .

2.2 (Calibrating the radius) In [Proposition 2.45](#), suppose $\hat{\mu}$ is the empirical distribution of n independent draws on a finite Θ . Using the asymptotic $2nD_{\text{KL}}(\mu^\star \parallel \hat{\mu}) \Rightarrow \chi^2_{|\Theta|-1}$, choose r so that the ball covers $\mu^\star$ with probability $1 - \alpha$, and describe how the robust value of [Figure 2.12](#) moves as n grows. What does the same calculation say about the transport radius of [Proposition 2.46](#)?

2.3 (Nesting compounds) Take the two-state chain of [Proposition 2.48](#) and iterate it for T stages with discount γ , evaluating by the nested $\text{CVaR}_{1/2}$ of [Definition 2.49](#). Show that the nested value is bounded below by the T -fold composition of the one-step worst case and compare it with the static $\text{CVaR}_{1/2}$ of the total loss for $T = 2, 3, 4$. Which grows faster, and why?

2.4 (Decomposition in the Gaussian model) Let $y \mid \theta \sim N(\theta, \eta^2)$ and $\theta \sim N(m_0, \tau_0^2)$. After n conditionally independent observations, compute the aleatoric and epistemic terms of (2.3) for the predictive distribution of y_{n+1} , and verify that the epistemic term decays as $O(1/n)$ while the aleatoric term stays at η^2 .

2.5 (Value of information and cost) In [Example 2.10](#), let the detector have true positive rate β and false positive rate α . Characterize the region of (α, β) in which the detector has strictly positive value, and show that it is empty when the likelihood ratio β/α is below a threshold that depends on the prior and costs.

2.6 (Belief MDP for a two-state problem) A machine is working or broken, $S = \{w, b\}$. Each period a working machine breaks with probability 0.1; a broken machine stays broken until repaired. The agent may *inspect*, at cost 1, revealing the state, *repair*, at cost 5, which makes it working, or *wait*. A broken machine costs 3 per period. Write the belief MDP with state $b = \mathbb{P}(b) \in [0, 1]$, and argue that an optimal policy has a threshold structure in b .

2.7 (Bayes-adaptive bandit) Consider a one-armed bandit whose arm pays Bernoulli rewards with unknown mean $\theta \in \{\theta_L, \theta_H\}$, against a safe arm paying λ . Write the Bayes-adaptive MDP with state $\mu_t = \mathbb{P}(\theta = \theta_H)$, show that the optimal policy is a threshold in μ_t , and show that once the safe arm is chosen it is chosen forever. Relate this to the self-confirming trap of [Example 1.13](#).

2.8 (Classical versus nonclassical) For each of the following two-stage problems, determine whether the information structure of [Definition 2.29](#) is classical, partially nested, or nonclassical: (a) a single controller with perfect recall; (b)

two sensors that each observe a noisy copy of ω and report to a fusion center that knows both reports; (c) a language model agent that writes a summary of a document into memory, followed by a second agent that acts on the summary but not the document.

2.9 (Blackwell sufficiency) Let $\Theta = \{0, 1\}$ and consider experiments E_ϵ that report θ correctly with probability $1 - \epsilon$. Show that $E_{\epsilon'}$ is a garbling of E_ϵ if and only if $\epsilon \leq \epsilon' \leq 1/2$ (for $\epsilon, \epsilon' \leq 1/2$), by constructing the garbling kernel explicitly.

Computational Methods for Reasoning, Planning, and Learning

Chapter 2 specified what an agent ought to do; this chapter develops the computational methods by which it does so. Dynamic programming is the organizing principle. When the model is known, value iteration, policy iteration, and linear programming compute the fixed points characterized in Chapter 2. When the model can only be simulated, Monte Carlo estimation, rollout, and tree search replace expectations by samples and exact optimization by lookahead. When it can only be experienced, reinforcement learning turns the Bellman equation into a stochastic approximation scheme whose behavior is governed by an ordinary differential equation. When the agent is uncertain which model governs its environment, Bayesian methods plan over beliefs, and posterior sampling makes the exploration they require tractable. Throughout, we emphasize the guarantees each method carries and the assumptions on which those guarantees rest.

In this chapter

- Derive value iteration and policy iteration from the Bellman operators, with convergence rates, stopping rules, and the loss of acting greedily on an approximate value function.
- Solve the linear-quadratic regulator by the Riccati recursion, and bound the error of approximate dynamic programming.
- Analyze Monte Carlo estimation and importance sampling, prove the improvement property of rollout, and describe Monte Carlo tree search as lookahead at decision time.
- Present temporal-difference learning and Q-learning as stochastic approximation, prove the policy gradient theorem, and relate entropy regularization to mirror descent.
- Formulate Bayes-optimal and Bayes-adaptive planning, state the Gittins index theorem, and prove the information-ratio bound on the Bayesian regret of Thompson sampling.

An agent that closes the loop between action and observation must, at every step, convert what it knows into what it does. Chapter 2 characterized the right

	value-based	policy-based	actor-critic
known model	value and policy iteration, linear programming, LQR	policy search in a known model	approximate policy iteration
simulator	Monte Carlo evaluation, fitted value iteration	rollout, multistep lookahead	MCTS with learned value and policy
experience only	TD(0), Q-learning	REINFORCE, policy gradient	actor-critic, natural and mirror-descent updates
uncertain model	Bayes-adaptive planning, Gittins index	Thompson sampling, posterior sampling RL	information-directed sampling

weaker access to the model

Figure 3.1. The computational methods of this chapter, organized by the agent’s access to the environment (rows) and by what they represent (columns). Moving down the rows, expectations are replaced by samples and then by experience; the last row adds epistemic uncertainty about the model itself.

conversion for a Markov decision process: the optimal value function is the unique fixed point of the Bellman operator (Lemma 2.12 and Theorem 2.13), and a policy that is greedy with respect to it is optimal. It said nothing about how to compute the fixed point, and in agentic settings computation is the binding constraint. The methods of this chapter are the elementary instances of the learning rule L of Chapter 1, and they are ordered by how much access the agent has to the environment’s law. Dynamic programming (Section 3.1) assumes the transition kernel P and reward r are known. Monte Carlo methods and rollout (Section 3.2) assume only a simulator from which transitions can be drawn. Reinforcement learning (Section 3.3) assumes only the ability to act and observe. Bayesian methods (Section 3.4) assume that the kernel belongs to a known family indexed by $\theta \in \Theta$ over which the agent holds a prior. Figure 3.1 organizes the methods along this axis and along a second one, whether they represent values, policies, or both.

We keep the network defender as a running example. Its intrusion-response problem, introduced in Example 3.5, is small enough to be solved exactly and rich enough to illustrate each approximation. The chapter also prepares later material. The two-timescale arguments of Section 3.3.1 reappear when several agents learn at once (Chapters 6 and 7); the Gibbs variational principle of Section 3.3.5 underlies both multi-agent mirror descent (Chapter 7) and the KL-regularized

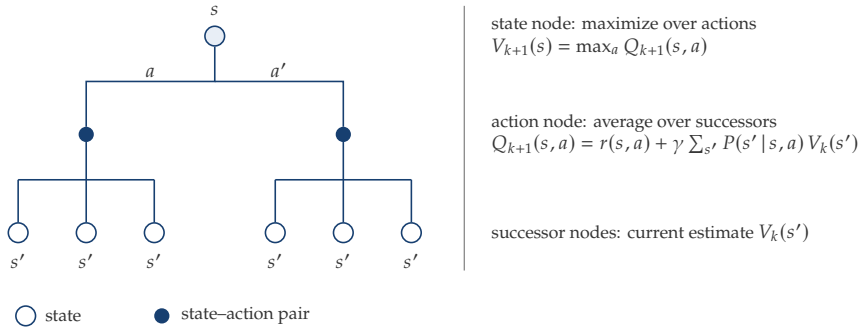


Figure 3.2. A Bellman backup. The value at a state is the maximum over actions of the expected one-step reward plus the discounted value of the successor states. Value iteration applies this operation at every state; sample-based methods replace the average over successors by samples, and search methods apply it recursively along a tree.

alignment of language models (Chapter 4); lookahead as test-time computation (Section 3.2.4) is the planning component of the architectures of Chapter 5; and the Bayes-optimal agent of Section 3.4 is the benchmark against which meta-learned agents are measured in Chapters 8 and 9.

3.1 Dynamic Programming and Optimal Control

Throughout this section $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma, \rho_0)$ is a finite discounted MDP as in Definition 2.11, with $|r(s, a)| \leq R_{\max}$. We use the operators $\mathcal{T}^\pi$ and $\mathcal{T}$ of Section 2.2.1, write

$$Q_V(s, a) := r(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s')$$

for the one-step lookahead value of $V \in \mathbb{R}^{\mathcal{S}}$, and call a deterministic policy π *greedy* with respect to V if $\pi(s) \in \arg \max_a Q_V(s, a)$ for every s , equivalently $\mathcal{T}^\pi V = \mathcal{T} V$. All norms on $\mathbb{R}^{\mathcal{S}}$ are supremum norms unless indicated otherwise. We use two consequences of Lemma 2.12 repeatedly: V^* is the unique fixed point of $\mathcal{T}$, and $\|V^*\|_\infty \leq R_{\max}/(1 - \gamma)$. Figure 3.2 depicts the elementary operation, the *backup*, from which all dynamic programming algorithms are assembled.

3.1.1 Finite-horizon problems and backward induction

Consider a horizon T , stage rewards $r_t : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, kernels P_t , and a terminal reward $r_T : \mathcal{S} \rightarrow \mathbb{R}$. The finite-horizon Bellman operators are $(\mathcal{T}_t V)(s) := \max_a [r_t(s, a) + \sum_{s'} P_t(s' | s, a) V(s')]$, and *backward induction* computes

$$V_T := r_T, \quad V_t := \mathcal{T}_t V_{t+1},$$

$$\pi_t(s) \in \arg \max_a \left[r_t(s, a) + \sum_{s'} P_t(s' | s, a) V_{t+1}(s') \right], \quad (3.1)$$

for $t = T - 1, \dots, 0$.

Proposition 3.1 (Backward induction). *The functions V_t of (3.1) are the optimal values-to-go, and the Markov deterministic policy $(\pi_0, \dots, \pi_{T-1})$ is optimal among all history-dependent randomized policies. Backward induction uses $O(T |\mathcal{S}|^2 |\mathcal{A}|)$ arithmetic operations.*

Proof. By induction on t downward. The claim holds at $t = T$. Suppose V_{t+1} is the optimal value-to-go from time $t + 1$ for every history, and that $\pi_{t+1:T-1}$ attains it. For any policy σ and history h_t ending in s , conditioning on (a_t, s_{t+1}) and using the induction hypothesis bounds the value-to-go of σ by $\sum_a \sigma_t(a | h_t) [r_t(s, a) + \sum_{s'} P_t(s' | s, a) V_{t+1}(s')] \leq V_t(s)$, with equality for π_t . Each stage performs one backup per state–action pair, each costing $O(|\mathcal{S}|)$ operations. ■

Backward induction is the finite-horizon version of the argument of [Theorem 2.13](#), and, applied to a game of perfect information, it is Zermelo’s algorithm, which computes a subgame-perfect equilibrium ([Selten, 1975](#)). Its cost is linear in the horizon and polynomial in $|\mathcal{S}|$, which is exactly the problem: in agentic settings $\mathcal{S}$ is typically combinatorial or continuous, and the rest of this chapter is concerned with circumventing the dependence on $|\mathcal{S}|$.

3.1.2 Value iteration

Value iteration starts from $V_0 \in \mathbb{R}^{\mathcal{S}}$ and sets $V_{k+1} := \mathcal{T} V_k$. By [Lemma 2.12](#) it converges to V^* geometrically. What an agent needs, however, is a policy, and a rule for when to stop. Both rest on the following bound, which quantifies how the quality of a value function transfers to the quality of the greedy policy it induces.

Proposition 3.2 (Loss of acting greedily). *Let $V \in \mathbb{R}^{\mathcal{S}}$ and let π be greedy with respect to V . Then*

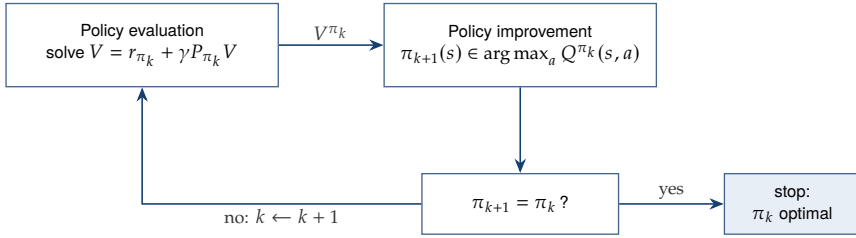
$$\|V^* - V^\pi\|_\infty \leq \frac{2\gamma}{1-\gamma} \|V - V^*\|_\infty.$$

Proof. Let $\varepsilon = \|V - V^*\|_\infty$. Since $\mathcal{T}^\pi V = \mathcal{T} V$,

$$\|V^* - \mathcal{T}^\pi V^*\|_\infty \leq \|\mathcal{T} V^* - \mathcal{T} V\|_\infty + \|\mathcal{T}^\pi V - \mathcal{T}^\pi V^*\|_\infty \leq 2\gamma \varepsilon.$$

Algorithm 3.1 Value iteration with an ϵ -optimal stopping rule**Require:** model (P, r) , discount γ , tolerance $\epsilon > 0$, initial V_0

- 1: $k \leftarrow 0$
- 2: **repeat**
- 3: $V_{k+1}(s) \leftarrow \max_a [r(s, a) + \gamma \sum_{s'} P(s' | s, a) V_k(s')]$ for all $s \in \mathcal{S}$
- 4: $k \leftarrow k + 1$
- 5: **until** $\|V_k - V_{k-1}\|_\infty < \epsilon(1 - \gamma)/(2\gamma)$
- 6: **return** V_k and a policy π greedy with respect to V_k



replacing exact evaluation by m applications of $\mathcal{T}^{\pi_k}$ gives modified policy iteration;
 $m = 1$ is value iteration, $m = \infty$ is policy iteration

Figure 3.3. Policy iteration alternates exact evaluation of the current policy with greedy improvement, and stops when the policy no longer changes. Interpolating the evaluation step between one backup and exact solution yields the family of modified policy iteration algorithms.

Using $V^\pi = \mathcal{T}^\pi V^\pi$,

$$\|V^\star - V^\pi\|_\infty \leq \|V^\star - \mathcal{T}^\pi V^\star\|_\infty + \|\mathcal{T}^\pi V^\star - \mathcal{T}^\pi V^\pi\|_\infty \leq 2\gamma\epsilon + \gamma\|V^\star - V^\pi\|_\infty,$$

and rearranging gives the claim. $\blacksquare$

The factor $1/(1 - \gamma)$ is the effective horizon: an error in the value estimate shifts the choice at every state, and the resulting losses accumulate over roughly $1/(1 - \gamma)$ steps. The bound is tight up to constants (Exercise 3.1). Value iteration with a stopping rule is Algorithm 3.1.

3.1.3 Policy iteration

Policy iteration (Howard, 1960) alternates two steps. *Policy evaluation* computes V^{π_k} by solving the linear system $(I - \gamma P_{\pi_k})V = r_{\pi_k}$, which is nonsingular because γP_{π_k} has spectral radius at most $\gamma < 1$. *Policy improvement* sets π_{k+1} greedy with respect to V^{π_k} , keeping $\pi_{k+1}(s) = \pi_k(s)$ whenever $\pi_k(s)$ is among the maximizers. The loop is shown in Figure 3.3.

Lemma 3.3 (Policy improvement). *Let π be a stationary policy and π' greedy with respect to V^π . Then $V^{\pi'} \geq \mathcal{T}V^\pi \geq V^\pi$. Moreover $V^{\pi'} = V^\pi$ if and only if $\mathcal{T}V^\pi = V^\pi$, in which case π and π' are optimal.*

Proof. Since π' is greedy, $\mathcal{T}^{\pi'}V^\pi = \mathcal{T}V^\pi \geq \mathcal{T}^\pi V^\pi = V^\pi$. Applying the monotone operator $\mathcal{T}^{\pi'}$ repeatedly, $(\mathcal{T}^{\pi'})^n V^\pi \geq \mathcal{T}^{\pi'}V^\pi$ for all $n \geq 1$, and the left side converges to $V^{\pi'}$ by Lemma 2.12. Hence $V^{\pi'} \geq \mathcal{T}V^\pi \geq V^\pi$. If $V^{\pi'} = V^\pi$, both inequalities are equalities, so V^π is the fixed point of $\mathcal{T}$, which is V^* by Theorem 2.13. Conversely, if $\mathcal{T}V^\pi = V^\pi$ then $V^\pi = V^*$ and $V^* \geq V^{\pi'} \geq V^\pi$ forces equality. ■

We can now state the convergence guarantees of the two basic algorithms together.

Theorem 3.4 (Convergence of value and policy iteration). *Let $\mathcal{M}$ be a finite discounted MDP.*

- (i) Value iteration. *For every V_0 and $k \geq 0$, $\|V_k - V^*\|_\infty \leq \gamma^k \|V_0 - V^*\|_\infty$ and $\|V_{k+1} - V^*\|_\infty \leq \frac{\gamma}{1-\gamma} \|V_{k+1} - V_k\|_\infty$. If $\|V_{k+1} - V_k\|_\infty < \epsilon(1-\gamma)/(2\gamma)$, then every policy greedy with respect to V_{k+1} is ϵ -optimal, $V^\pi \geq V^* - \epsilon$.*
- (ii) Policy iteration. *The iterates satisfy $V^{\pi_{k+1}} \geq \mathcal{T}V^{\pi_k} \geq V^{\pi_k}$ and*

$$\|V^* - V^{\pi_k}\|_\infty \leq \gamma^k \|V^* - V^{\pi_0}\|_\infty.$$

The algorithm terminates after at most $|\mathcal{A}|^{|\mathcal{S}|}$ improvement steps with a deterministic stationary policy π satisfying $V^\pi = V^$.*

In particular, both algorithms compute an optimal deterministic stationary policy, whose existence is guaranteed by Theorem 2.13.

Proof. (i) The first inequality is iteration of the contraction property. For the second, $\|V_{k+1} - V^*\| \leq \|\mathcal{T}V_k - \mathcal{T}V_{k+1}\| + \|\mathcal{T}V_{k+1} - \mathcal{T}V^*\| \leq \gamma \|V_k - V_{k+1}\| + \gamma \|V_{k+1} - V^*\|$; rearrange. Let $\delta = \|V_{k+1} - V_k\|$ and π greedy with respect to V_{k+1} . Then

$$\|V^\pi - V_{k+1}\| \leq \|\mathcal{T}^\pi V^\pi - \mathcal{T}^\pi V_{k+1}\| + \|\mathcal{T}V_{k+1} - \mathcal{T}V_k\| \leq \gamma \|V^\pi - V_{k+1}\| + \gamma \delta,$$

which gives $\|V^\pi - V_{k+1}\| \leq \gamma \delta / (1-\gamma)$. Combining this with the second inequality yields $\|V^\pi - V^*\| \leq 2\gamma \delta / (1-\gamma) < \epsilon$.

(ii) The first chain is Lemma 3.3. It gives $0 \leq V^* - V^{\pi_{k+1}} \leq \mathcal{T}V^* - \mathcal{T}V^{\pi_k}$, whose supremum norm is at most $\gamma \|V^* - V^{\pi_k}\|$; iterate. If $\pi_{k+1} \neq \pi_k$, the tie-breaking convention implies that $\mathcal{T}V^{\pi_k}(s) > V^{\pi_k}(s)$ at some state, so $V^{\pi_{k+1}} \neq V^{\pi_k}$ and, by monotonicity, no policy is visited twice. There are $|\mathcal{A}|^{|\mathcal{S}|}$ deterministic stationary policies. When $\pi_{k+1} = \pi_k$, we have $\mathcal{T}V^{\pi_k} = \mathcal{T}^{\pi_k}V^{\pi_k} = V^{\pi_k}$, and Lemma 3.3 shows that π_k is optimal. ■

The bound $|\mathcal{A}|^{|\mathcal{S}|}$ is very pessimistic. For a fixed discount factor, policy iteration is strongly polynomial: Ye (2011) showed that it terminates after $O\left(\frac{|\mathcal{S}||\mathcal{A}|}{1-\gamma} \log \frac{|\mathcal{S}|}{1-\gamma}\right)$

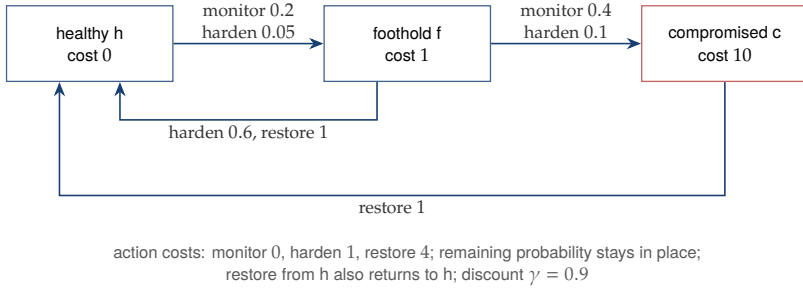


Figure 3.4. The intrusion-response MDP of Example 3.5. Arrows show the transitions that change the state, labeled by action and probability; probability not shown keeps the host in its current state.

iterations, a number independent of the rewards. Value iteration, by contrast, has no finite termination, but each of its iterations costs $O(|S|^2 |\mathcal{A}|)$ rather than the $O(|S|^3)$ of a linear solve. *Modified policy iteration* (Puterman, 1994) evaluates each policy by m applications of $\mathcal{T}^{\pi_k}$, interpolating between the two.

Example 3.5 (An intrusion-response MDP). A host protected by the defender is *healthy* (h), has an attacker *foothold* (f), or is *compromised* (c), with per-period costs 0, 1, and 10. The defender may *monitor* (cost 0), *harden* (cost 1), or *restore* the host from a clean image (cost 4). Under monitoring, a healthy host acquires a foothold with probability 0.2 and a foothold escalates to compromise with probability 0.4. Hardening reduces these probabilities to 0.05 and 0.1 and evicts a foothold with probability 0.6. Restoring returns the host to h. Compromise persists until restoration. Rewards are negative costs and $\gamma = 0.9$; Figure 3.4 shows the transitions.

Value iteration from $V_0 = 0$ gives

$$(V^*(h), V^*(f), V^*(c)) = (-6.34, -9.86, -19.70)$$

and the optimal policy *monitor when healthy, harden a foothold, restore when compromised*. The Q-values at V^* show how much each deviation costs: at a foothold, monitoring has value -13.42 , hardening -9.86 , and restoring -10.70 . Policy iteration started from *always monitor*, whose value is $(-51.7, -80.4, -100.0)$, first switches to *(harden, restore, restore)* with value $(-11.7, -15.6, -24.6)$ and reaches the optimal policy at the second improvement. Figure 3.5 compares the two algorithms. ◀

The example illustrates a general phenomenon: greedy policies typically stabilize long before the value iterates converge. Proposition 3.2 is a worst-case bound; in a finite MDP there is a gap $\Delta > 0$ between the optimal and second-best actions, and once $\|V_k - V^*\|_\infty < \Delta/(2\gamma)$ every greedy policy is optimal (Exercise 3.2).

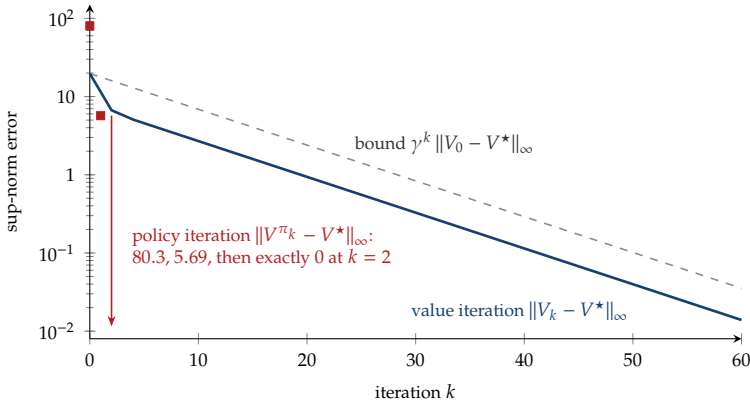


Figure 3.5. Value iteration and policy iteration on the MDP of Example 3.5. The error of value iteration decays at the geometric rate guaranteed by Theorem 3.4, though the greedy policy of V_k is already optimal from $k = 1$. Policy iteration system reaches V^* exactly after two improvements, each of which requires solving a linear system.

3.1.4 Linear programming

The Bellman equation also has a linear programming characterization (Manne, 1960). Let $\rho \in \Delta(\mathcal{S})$ have full support.

Proposition 3.6 (Linear programming formulation). *V^* is the unique optimal solution of*

$$\begin{aligned} \min_{V \in \mathbb{R}^{\mathcal{S}}} \quad & \sum_s \rho(s) V(s) \\ \text{subject to} \quad & V(s) \geq r(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \text{ for all } (s, a). \end{aligned} \quad (3.2)$$

Its dual is

$$\begin{aligned} \max_{d \geq 0} \quad & \sum_{s,a} d(s, a) r(s, a) \\ \text{subject to} \quad & \sum_a d(s', a) = \rho(s') + \gamma \sum_{s,a} P(s' | s, a) d(s, a) \text{ for all } s', \end{aligned} \quad (3.3)$$

whose feasible points are the unnormalized discounted state–action occupancy measures $d^\pi(s, a) = \sum_t \gamma^t \mathbf{P}_\rho^\pi(s_t = s, a_t = a)$ of stationary policies, with $\pi(a | s) = d(s, a) / \sum_{a'} d(s, a')$.

Proof of the primal claim. The constraints say $V \geq \mathcal{T}V$. By monotonicity of $\mathcal{T}$, $V \geq \mathcal{T}^n V \rightarrow V^*$, so every feasible V satisfies $V \geq V^*$, and V^* is feasible. Since ρ has full support, V^* is the unique minimizer. The dual statement follows from LP duality and the fact that the flow constraints are the stationarity equations of the discounted occupancy measure; see Puterman (1994, Section 6.9). ■

The dual is the more consequential formulation for this book. It turns sequential decision making into optimization over a convex polytope of occupancy measures, on which constraints, such as a budget on disruptive defensive actions, are linear. It is also the object that regularized and game-theoretic formulations act on: entropy-regularized dynamic programming (Section 3.3.5) and learning in Markov games (Chapter 6) are naturally expressed in occupancy-measure coordinates.

3.1.5 Linear–quadratic control

The one continuous-state problem that dynamic programming solves in closed form is the linear–quadratic regulator. The state $s_t \in \mathbb{R}^n$ and control $a_t \in \mathbb{R}^m$ evolve as

$$s_{t+1} = As_t + Ba_t + \xi_t, \quad (3.4)$$

where the ξ_t are independent with mean zero and covariance Σ , and the agent minimizes

$$\mathbb{E} \left[\sum_{t=0}^{T-1} (s_t^\top C s_t + a_t^\top D a_t) + s_T^\top C_T s_T \right],$$

with $C, C_T \geq 0$ and $D > 0$. (We minimize costs here, following control convention.)

Theorem 3.7 (Riccati recursion). Define $\Lambda_T := C_T$ and, for $t = T - 1, \dots, 0$,

$$\begin{aligned} K_t &:= (D + B^\top \Lambda_{t+1} B)^{-1} B^\top \Lambda_{t+1} A, \\ \Lambda_t &:= C + A^\top \Lambda_{t+1} A - A^\top \Lambda_{t+1} B (D + B^\top \Lambda_{t+1} B)^{-1} B^\top \Lambda_{t+1} A. \end{aligned} \quad (3.5)$$

Then the optimal cost-to-go is $V_t(s) = s^\top \Lambda_t s + \kappa_t$ with $\kappa_T = 0$ and $\kappa_t = \kappa_{t+1} + \text{tr}(\Lambda_{t+1} \Sigma)$, and the optimal policy is the linear feedback $a_t = -K_t s_t$. The gains K_t do not depend on Σ .

Proof. By backward induction (Proposition 3.1, with minimization). Suppose $V_{t+1}(s) = s^\top \Lambda_{t+1} s + \kappa_{t+1}$ with $\Lambda_{t+1} \geq 0$. Since ξ_t has mean zero and is independent of (s_t, a_t) ,

$$\mathbb{E} V_{t+1}(As + Ba + \xi_t) = (As + Ba)^\top \Lambda_{t+1} (As + Ba) + \text{tr}(\Lambda_{t+1} \Sigma) + \kappa_{t+1}.$$

The function $a \mapsto a^\top D a + (As + Ba)^\top \Lambda_{t+1} (As + Ba)$ is strictly convex because $D + B^\top \Lambda_{t+1} B > 0$; setting its gradient to zero gives $a = -K_t s$, and substituting gives $s^\top \Lambda_t s$ with Λ_t as in (3.5), which is positive semidefinite as the minimum of positive semidefinite quadratic forms. ■

The independence of K_t from Σ is *certainty equivalence*: the optimal controller for the stochastic problem is the optimal controller for the deterministic problem obtained by replacing ξ_t with its mean. It holds because the noise is additive

and independent of the controls, and it fails as soon as the control affects what is learned, for instance when (A, B) are unknown. That failure is the *dual control* problem of Section 3.4 and of Chapter 9. For $T \rightarrow \infty$, if (A, B) is stabilizable and $(A, C^{1/2})$ is detectable, Λ_t converges to the unique stabilizing solution of the algebraic Riccati equation obtained by setting $\Lambda_t = \Lambda_{t+1}$ in (3.5), and the stationary gain minimizes the average cost (Anderson and Moore, 1990; Kalman, 1960).

3.1.6 Approximate dynamic programming

When $\mathcal{S}$ is large, V cannot be stored as a table, and the backup cannot be applied exactly. *Fitted value iteration* represents values in a class $\mathcal{F} \subseteq \mathbb{R}^{\mathcal{S}}$, for instance $V_w(s) = \phi(s)^\top w$ or a neural network, and sets $V_{k+1} := \text{Fit}(\mathcal{T}V_k)$, where Fit regresses sampled backups $\hat{y}_i = \max_a [r(s_i, a) + \gamma \frac{1}{m} \sum_j V_k(s'_{ij})]$ at sampled states s_i onto $\mathcal{F}$. Whatever the source of error, the following bound shows how it propagates.

Proposition 3.8 (Approximate value iteration). *Suppose that $\|V_{k+1} - \mathcal{T}V_k\|_\infty \leq \varepsilon$ for all k , and let π_k be greedy with respect to V_k . Then*

$$\limsup_{k \rightarrow \infty} \|V_k - V^\star\|_\infty \leq \frac{\varepsilon}{1 - \gamma}, \quad \limsup_{k \rightarrow \infty} \|V^\star - V^{\pi_k}\|_\infty \leq \frac{2\gamma\varepsilon}{(1 - \gamma)^2}.$$

Proof. $\|V_{k+1} - V^\star\| \leq \|V_{k+1} - \mathcal{T}V_k\| + \|\mathcal{T}V_k - \mathcal{T}V^\star\| \leq \varepsilon + \gamma\|V_k - V^\star\|$. Unrolling, $\|V_k - V^\star\| \leq \gamma^k \|V_0 - V^\star\| + \varepsilon \sum_{j < k} \gamma^j$, which gives the first bound; the second follows from Proposition 3.2. ■

For approximate policy iteration, in which the evaluation of π_k has error at most δ and the improvement step is ε -greedy in value, the analogous bound is $\limsup_k \|V^\star - V^{\pi_k}\|_\infty \leq (\varepsilon + 2\gamma\delta)/(1 - \gamma)^2$ (Bertsekas and Tsitsiklis, 1996, Section 6.2). Two caveats temper these results. First, the factor $1/(1 - \gamma)^2$ is unavoidable in the worst case, so long effective horizons amplify small regression errors. Second, the assumption of a uniform sup-norm error is rarely verifiable, because regression controls errors in a weighted L_2 norm under the sampling distribution, and the composition of a projection with $\mathcal{T}$ need not be a contraction in any norm. Bounds in L_p norms, which involve concentrability coefficients measuring how far the state distributions induced by policies can drift from the sampling distribution, are developed by Munos and Szepesvári (2008). The same mismatch between the distribution on which values are fit and the distribution on which they are used reappears as distribution shift in learned world models (Chapter 9).

3.2 Monte Carlo Methods and Rollout

Dynamic programming requires the transition probabilities. In many agentic settings the agent instead has a *simulator*: a digital twin of a network, a physics engine, or a generative model that can be queried for a sample $s' \sim P(\cdot | s, a)$ from any state. Monte Carlo methods replace expectations by sample averages, which removes the dependence of each backup on $|\mathcal{S}|$ at the price of statistical error. Rollout and tree search use the simulator at decision time, spending computation on the states the agent actually encounters.

3.2.1 Monte Carlo estimation and importance sampling

Fix a policy π and a state s . The *truncated return* of a trajectory started at s is $G_H := \sum_{t=0}^{H-1} \gamma^t r(s_t, a_t)$, and the *Monte Carlo estimator* $\hat{V}_n(s)$ averages G_H over n independent trajectories generated by π .

Proposition 3.9 (Monte Carlo evaluation). *The bias of the estimator is at most $\gamma^H R_{\max}/(1 - \gamma)$ in absolute value, and for every $\epsilon > 0$,*

$$\mathbb{P}\left(\left|\hat{V}_n(s) - \mathbb{E} \hat{V}_n(s)\right| \geq \epsilon\right) \leq 2 \exp\left(-\frac{n\epsilon^2(1 - \gamma)^2}{2R_{\max}^2}\right).$$

Proof. The bias is the expected discounted reward after time H , bounded by $\sum_{t \geq H} \gamma^t R_{\max}$. Each G_H lies in an interval of length at most $2R_{\max}/(1 - \gamma)$, and Hoeffding's inequality (Hoeffding, 1963) gives the tail bound. ■

The sample size needed for accuracy ϵ is $O(R_{\max}^2/((1 - \gamma)^2\epsilon^2))$, independent of $|\mathcal{S}|$. This independence is the reason simulation scales where dynamic programming does not, and Kearns et al. (2002) exploited it to show that near-optimal actions in an MDP can be computed with a number of simulator calls that does not depend on the size of the state space, though it grows exponentially in the effective horizon.

Often the trajectories available were generated by a *behavior policy* β other than the *target policy* π one wishes to evaluate: the defender may wish to assess a new response policy from logs of the current one. *Importance sampling* reweights each trajectory $\tau = (s_0, a_0, \dots, s_H)$ by the likelihood ratio

$$\varrho_{0:H-1} := \prod_{t=0}^{H-1} \frac{\pi(a_t | s_t)}{\beta(a_t | s_t)},$$

in which the unknown transition probabilities cancel.

Proposition 3.10 (Importance sampling). *Suppose $\pi(a | s) > 0$ implies $\beta(a | s) > 0$. Then $\mathbb{E}_{\beta}[\varrho_{0:H-1} G_H] = \mathbb{E}_{\pi}[G_H]$, so the importance-sampling estimator is unbiased. Its*

second moment is $\mathbb{E}_\beta[\varrho_{0:H-1}^2 G_H^2] = \mathbb{E}_\pi[\varrho_{0:H-1} G_H^2]$. In particular, if $\mathcal{S}$ is a single state and π, β are fixed distributions on $\mathcal{A}$, then

$$\mathbb{E}_\beta[\varrho_{0:H-1}^2] = (1 + \chi^2(\pi \parallel \beta))^H, \quad \chi^2(\pi \parallel \beta) := \sum_a \frac{\pi(a)^2}{\beta(a)} - 1.$$

Proof. The probability of τ under π is $\rho_0(s_0) \prod_t \pi(a_t | s_t) P(s_{t+1} | s_t, a_t)$, and the ratio of this to its counterpart under β is $\varrho_{0:H-1}$ wherever the latter is positive. Hence $\mathbb{E}_\beta[\varrho f(\tau)] = \mathbb{E}_\pi[f(\tau)]$ for every f ; take $f = G_H$ and $f = \varrho G_H^2$. In the single-state case the ratios are independent across t under β , each with second moment $\sum_a \pi(a)^2 / \beta(a)$. ■

Unbiasedness is cheap; the variance is the difficulty. Unless $\pi = \beta$, the second moment of the weights grows exponentially with the horizon, so ordinary importance sampling is useful only for short horizons or nearly identical policies. Per-decision importance sampling, which weights the reward at time t by $\varrho_{0:t}$ only, and weighted importance sampling, which normalizes the weights at the cost of a vanishing bias, reduce the variance (Precup et al., 2000). Off-policy evaluation of this kind reappears when meta-learned agents must evaluate candidate policies from data collected by earlier ones (Chapter 8).

3.2.2 Rollout and approximate policy improvement

Rollout performs one step of policy iteration online. Starting from a *base policy* π_b , which may be a heuristic, a hand-written playbook, or a previously learned policy, the agent at state s_t evaluates each first action by simulating π_b thereafter, and takes the best (Figure 3.6).

Definition 3.11 (Rollout policy). Let π_b be a stationary base policy. The *exact rollout policy* is $\pi_{ro}(s) \in \arg \max_a Q^{\pi_b}(s, a)$, where $Q^{\pi_b}(s, a) = r(s, a) + \gamma \sum_{s'} P(s' | s, a) V^{\pi_b}(s')$. A *sampled rollout policy* $\hat{\pi}_{ro}$ selects $\arg \max_a \hat{Q}(s, a)$, where $\hat{Q}(s, a)$ is a Monte Carlo estimate of $Q^{\pi_b}(s, a)$ from simulated trajectories that take a and then follow π_b , possibly truncated at horizon H with a terminal approximation $\gamma^H \tilde{V}(s_H)$.

Proposition 3.12 (Rollout improves on the base policy). Let π_b , π_{ro} , and $\hat{\pi}_{ro}$ be as in Definition 3.11.

- (i) The exact rollout policy satisfies $V^{\pi_{ro}} \geq \mathcal{T} V^{\pi_b} \geq V^{\pi_b}$.
- (ii) If $|\hat{Q}(s, a) - Q^{\pi_b}(s, a)| \leq \delta$ at all states and actions at which $\hat{\pi}_{ro}$ is evaluated, then $V^{\hat{\pi}_{ro}} \geq V^{\pi_b} - 2\delta/(1 - \gamma)$.

Proof. (i) is Lemma 3.3 with $\pi = \pi_b$. (ii) Let $\hat{a} = \hat{\pi}_{ro}(s)$. Since $V^{\pi_b}(s) = \sum_a \pi_b(a | s) Q^{\pi_b}(s, a) \leq \max_a Q^{\pi_b}(s, a)$,

$$Q^{\pi_b}(s, \hat{a}) \geq \hat{Q}(s, \hat{a}) - \delta = \max_a \hat{Q}(s, a) - \delta \geq \max_a Q^{\pi_b}(s, a) - 2\delta \geq V^{\pi_b}(s) - 2\delta.$$

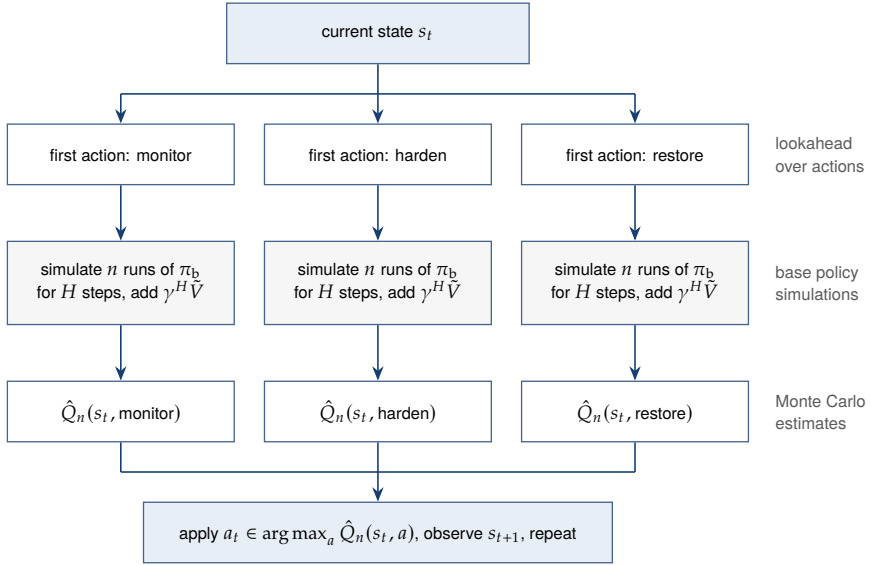


Figure 3.6. Rollout at the current state. Each candidate first action is evaluated by simulating the base policy, optionally truncated with a terminal value approximation, and the action with the best estimate is applied. The procedure is repeated at the next state.

Thus $\mathcal{T}^{\hat{\pi}_{\text{ro}}} V^{\pi_b} \geq V^{\pi_b} - 2\delta \mathbf{1}$. Applying the monotone operator $\mathcal{T}^{\hat{\pi}_{\text{ro}}}$, which satisfies $\mathcal{T}^{\hat{\pi}_{\text{ro}}}(V - c\mathbf{1}) = \mathcal{T}^{\hat{\pi}_{\text{ro}}} V - \gamma c\mathbf{1}$, n times gives $(\mathcal{T}^{\hat{\pi}_{\text{ro}}})^n V^{\pi_b} \geq V^{\pi_b} - 2\delta \sum_{j < n} \gamma^j \mathbf{1}$; let $n \rightarrow \infty$. ■

Part (i) is the *cost improvement property*. It is what makes rollout attractive for agents: the guarantee holds for any base policy, requires no training, and is realized only at visited states. The price is online computation, roughly $|\mathcal{A}| \cdot n \cdot H$ simulator calls per decision. Combined with [Proposition 3.9](#) and a union bound over actions, $n = O(R_{\max}^2 \log(|\mathcal{A}|/\eta)/((1-\gamma)^2\delta^2))$ trajectories per action suffice for the hypothesis of part (ii) to hold at a given state with probability $1 - \eta$, provided H is chosen so that the truncation bias is below $\delta/2$. Rollout was used by [Tesauro and Galperin \(1997\)](#) to improve a backgammon program at play time, and it was developed as a general method for combinatorial and stochastic optimization by [Bertsekas et al. \(1997\)](#) and [Bertsekas \(2020\)](#).

Example 3.13 (Rollout for intrusion response). In [Example 3.5](#), take as base policy the playbook *monitor*, and *restore only when compromised*, with value $V^{\pi_b} = (-15.43, -24.00, -27.88)$. The Q-values of the base policy at a foothold are -24.00 for monitoring, -19.32 for hardening, and -18.88 for restoring; at a healthy host they are -15.43 , -15.27 , and -17.88 . The rollout policy therefore hardens healthy hosts and restores both footholds and compromised hosts.

Its value $(-11.72, -15.55, -24.55)$ improves on the playbook at every state, as [Proposition 3.12](#) guarantees, although it is not optimal. Rolling out from the rollout policy, which amounts to a second step of policy iteration, recovers the optimal policy. $\blacktriangleleft$

Rollout extends naturally to deeper lookahead. The ℓ -step lookahead policy with terminal approximation $\tilde{V}$ chooses at s the first action of an optimal ℓ -stage plan with terminal reward $\gamma^\ell \tilde{V}$, that is, it is greedy with respect to $\mathcal{T}^{\ell-1} \tilde{V}$.

Proposition 3.14 (Multistep lookahead). *If π_ℓ is greedy with respect to $\mathcal{T}^{\ell-1} \tilde{V}$, then*

$$\|V^\star - V^{\pi_\ell}\|_\infty \leq \frac{2\gamma^\ell}{1-\gamma} \|\tilde{V} - V^\star\|_\infty.$$

Proof. $\|\mathcal{T}^{\ell-1} \tilde{V} - V^\star\|_\infty \leq \gamma^{\ell-1} \|\tilde{V} - V^\star\|_\infty$ by the contraction property; apply [Proposition 3.2](#). $\blacksquare$

Each additional step of lookahead thus contracts the effect of errors in the terminal approximation by a factor γ , at a computational cost that grows as $(|\mathcal{A}| \cdot b)^\ell$ for branching factor b of the simulator. The next two subsections concern how to spend that budget selectively and what the trade-off means for agents.

3.2.3 Monte Carlo tree search

Monte Carlo tree search (MCTS) builds a lookahead tree incrementally, allocating simulations to the branches that appear most promising while continuing to sample the others. Its selection rule comes from the multi-armed bandit problem. Consider K arms with rewards in $[0, 1]$ and means $\mu_1, \dots, \mu_K$, let $\mu^\star = \max_i \mu_i$ and $\Delta_i = \mu^\star - \mu_i$. The UCB1 rule plays each arm once and then, at round n , the arm maximizing $\bar{x}_i + \sqrt{2 \log n / n_i}$, where $\bar{x}_i$ is the empirical mean of the n_i rewards obtained from arm i .

Theorem 3.15 (Auer et al. (2002)). *After n rounds, the expected regret of UCB1 is at most*

$$8 \sum_{i:\Delta_i>0} \frac{\log n}{\Delta_i} + \left(1 + \frac{\pi^2}{3}\right) \sum_{i=1}^K \Delta_i.$$

The bonus $\sqrt{2 \log n / n_i}$ is an upper confidence bound: it is large for arms that have been sampled rarely, so an arm whose empirical mean is low is still tried occasionally, at a rate that decreases as evidence accumulates. *Upper confidence bounds applied to trees* (UCT) (Kocsis and Szepesvári, 2006) treats each internal node of the search tree as a separate bandit. Each iteration of the algorithm has four phases ([Figure 3.7](#)): *selection* descends from the root, choosing at node s the action maximizing

$$\hat{Q}(s, a) + c \sqrt{\frac{\log N(s)}{N(s, a)}}, \quad (3.6)$$

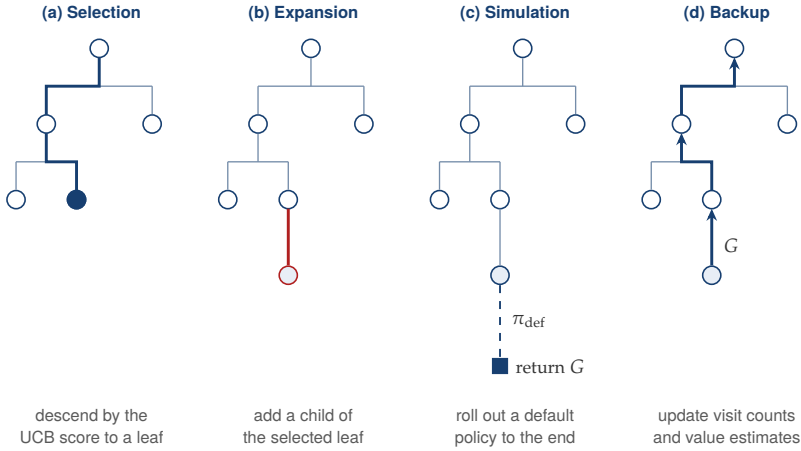


Figure 3.7. One iteration of Monte Carlo tree search. (a) The tree policy selects actions by an upper confidence rule down to a leaf. (b) A new node is added. (c) A default policy is simulated from the new node to estimate its value. (d) The return is propagated back along the path, updating visit counts and value estimates.

where $N(s)$ and $N(s, a)$ are visit counts; *expansion* adds a child to the tree when a leaf is reached; *simulation* estimates the value of the new node by a rollout of a default policy; and *backup* propagates the return along the selected path, incrementing counts and updating the averages $\hat{Q}$. MCTS in this form is due to Coulom (2007) and Kocsis and Szepesvári (2006); Browne et al. (2012) survey its variants.

Theorem 3.16 (Consistency of UCT; Kocsis and Szepesvári (2006)). *Consider a finite-horizon problem represented by a finite tree with rewards in $[0, 1]$, and run UCT with exploration terms of the form $2C_p \sqrt{\log N(s)/N(s, a)}$ for a suitable constant C_p . Then, as the number of iterations n grows, the bias of the value estimate at the root is $O((\log n)/n)$, and the probability of selecting a suboptimal action at the root converges to zero at a polynomial rate.*

The guarantee is asymptotic, and the constants can be very poor: Coquelin and Munos (2007) exhibit trees on which the number of iterations before UCT identifies the optimal action grows as an iterated exponential in the depth, because the logarithmic bonus explores deep suboptimal branches too rarely. In practice, the effectiveness of MCTS derives from domain knowledge injected in two places. A learned *value function* replaces or complements the random rollouts at the leaves, and a learned *policy prior* $p(a | s)$ biases the selection rule, for example through a bonus proportional to $p(a | s) \sqrt{N(s)} / (1 + N(s, a))$, a variant of the predictor-augmented upper confidence rule of Rosin (2011). This combination was the core of AlphaGo (Silver et al., 2016) and of AlphaGo Zero (Silver et al.,

2017), in which the visit distribution produced by search serves as the training target for the policy network and the search result as the target for the value network.

3.2.4 Lookahead as test-time computation

The methods of this section invite a distinction that is central to modern agents. An agent may invest computation *before* deployment, to learn a value function $\hat{V}$ or a policy $\hat{\pi}$ that is cheap to evaluate, or *at decision time*, to improve on them by lookahead from the current state. [Proposition 3.14](#) makes the trade-off quantitative: an ℓ -step search multiplies the effect of an offline error $\|\hat{V} - V^*\|_\infty$ by $\gamma^{\ell-1}$, at a cost exponential in ℓ for exhaustive search and much less for selective search.

AlphaGo Zero is the canonical demonstration that the two investments are complementary rather than competing. Search is a *policy improvement operator*: given a network, MCTS produces a stronger policy at the current state. Training is a *distillation operator*: it fits the network to the improved policy. Alternating the two is approximate policy iteration ([Section 3.1.6](#)) in which evaluation is amortized into a network and improvement is performed by search. The same structure organizes reasoning in large language models. Sampling several chains of thought and selecting among them with a verifier, or searching over partial reasoning steps with a learned value, spends inference compute to improve on the base policy defined by the model, and fine-tuning on the selected outputs distills the improvement; Chapter 4 develops these methods. The analysis of this section shows what such procedures can and cannot guarantee: improvement over the base policy when the evaluator is accurate ([Proposition 3.12](#)), and no improvement when it is not, since a biased evaluator can lead the lookahead to exploit its errors.

Planning is policy improvement at the current state

Rollout, multistep lookahead, and tree search all compute, for the state the agent is in, an approximation to one or more steps of policy improvement with respect to a base policy or value function. Their guarantees are inherited from the monotonicity of the Bellman operator, and their errors from the evaluator. How much to search, and where, is itself a decision, which becomes the metareasoning problem of Chapter 5.

3.3 Reinforcement Learning

Reinforcement learning dispenses with the simulator. The agent observes only the trajectory $s_0, a_0, r_0, s_1, a_1, r_1, \dots$ that its own actions generate, and must estimate values or improve policies from it. The algorithms are incremental: each

transition nudges an estimate in a direction whose expectation is the correction that dynamic programming would apply. The mathematical tool that explains when such schemes converge is stochastic approximation, which interprets a noisy recursion as the discretization of an ordinary differential equation.

3.3.1 Stochastic approximation and the ODE method

Consider the recursion in $\mathbb{R}^d$

$$x_{n+1} = x_n + \alpha_n [h(x_n) + M_{n+1}], \quad (3.7)$$

where $h : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a mean field, $\alpha_n > 0$ are step sizes, and M_{n+1} is noise. Robbins and Monro (1951) introduced such schemes to find a root of a function observed with noise. Let $\mathcal{F}_n := \sigma(x_0, M_1, \dots, M_n)$.

Assumption 3.17 (Stochastic approximation). (i) h is Lipschitz continuous.

(ii) The step sizes satisfy $\sum_n \alpha_n = \infty$ and $\sum_n \alpha_n^2 < \infty$.

(iii) (M_n) is a martingale difference sequence with respect to $(\mathcal{F}_n)$, and $\mathbb{E}[\|M_{n+1}\|^2 | \mathcal{F}_n] \leq K(1 + \|x_n\|^2)$ for some $K < \infty$.

(iv) The iterates are bounded: $\sup_n \|x_n\| < \infty$ almost surely.

The first condition in (ii) ensures that the iterates can travel any distance; the second ensures that the accumulated noise $\sum_n \alpha_n M_{n+1}$ converges. Define the ODE time $\tau_n := \sum_{m < n} \alpha_m$ and let $\bar{x}(\cdot)$ be the piecewise linear interpolation with $\bar{x}(\tau_n) = x_n$.

Theorem 3.18 (ODE method; Borkar (2008)). Under Assumption 3.17, almost surely the sequence (x_n) converges to a compact connected internally chain transitive invariant set of the ODE

$$\dot{x}(\tau) = h(x(\tau)). \quad (3.8)$$

In particular, if (3.8) has a globally asymptotically stable equilibrium x^* , then $x_n \rightarrow x^*$ almost surely.

Idea of the proof. Write $\bar{x}(\tau_{n+k}) = x_n + \sum_{m=n}^{n+k-1} \alpha_m h(x_m) + \sum_{m=n}^{n+k-1} \alpha_m M_{m+1}$. By (ii) and (iii), $\sum_m \alpha_m M_{m+1}$ is a martingale with summable conditional variances on the event that the iterates are bounded, hence convergent, so its tail sums vanish. The middle term is a Riemann sum for $\int h(\bar{x}(\tau)) d\tau$ with mesh $\alpha_m \rightarrow 0$. Gronwall's inequality then shows that, on every window of fixed ODE-time length T , the interpolated trajectory started at τ_n stays within a distance of the solution of (3.8) started at x_n that vanishes as $n \rightarrow \infty$. A process with this property is an *asymptotic pseudotrajectory* of the flow, and its limit sets are internally chain transitive (Benaïm, 1999). ■

Theorem 3.18 separates the analysis of a learning algorithm into two parts: the deterministic dynamics (3.8), which determine where the algorithm can

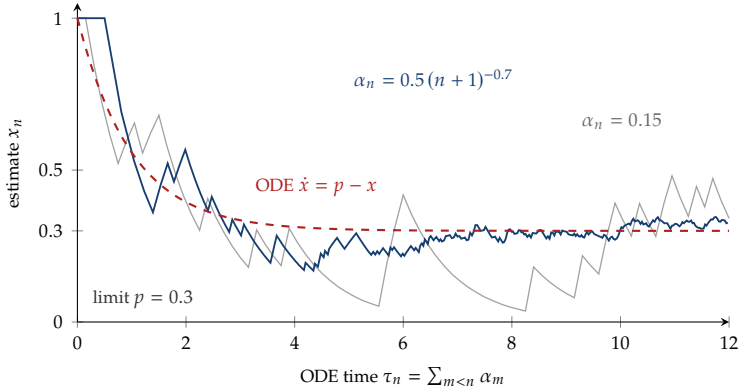


Figure 3.8. Stochastic approximation in [Example 3.19](#), plotted against ODE time. With decreasing step sizes the iterate follows the ODE and converges to $p = 0.3$; with a constant step size it follows the ODE at first and then fluctuates around the equilibrium indefinitely.

go, and the conditions of [Assumption 3.17](#), which guarantee that the noise does not prevent it from getting there. Boundedness (iv) is often the hardest condition to verify. [Borkar and Meyn \(2000\)](#) give a practical criterion: if the limit $h_\infty(x) := \lim_{c \rightarrow \infty} h(cx)/c$ exists and the origin is a globally asymptotically stable equilibrium of $\dot{x} = h_\infty(x)$, then (iv) holds under (i)–(iii). The approach originates with [Ljung \(1977\)](#); see [Kushner and Yin \(2003\)](#) for weaker noise conditions.

Example 3.19 (Estimating an alarm rate). The defender estimates the probability p that a detector fires on a flow. With alarms $y_n \in \{0, 1\}$ it updates $x_{n+1} = x_n + \alpha_n(y_{n+1} - x_n)$, which is (3.7) with $h(x) = p - x$ and $M_{n+1} = y_{n+1} - p$. The ODE $\dot{x} = p - x$ has the globally asymptotically stable equilibrium p , and $x_n \in [0, 1]$ whenever $x_0 \in [0, 1]$ and $\alpha_n \leq 1$, so [Theorem 3.18](#) gives $x_n \rightarrow p$ almost surely when $\alpha_n = 0.5(n+1)^{-0.7}$. With the constant step size $\alpha_n = 0.15$, condition (ii) fails; the iterate tracks the ODE initially but then fluctuates around p with a spread of order $\sqrt{\alpha}$, which is the price paid for the ability to track a rate that changes over time ([Figure 3.8](#)). With $\alpha_n = 1/(n+1)$, x_n is the sample mean. ◀

Several learning processes in this book couple recursions that run at different speeds. Consider

$$x_{n+1} = x_n + \alpha_n [h(x_n, y_n) + M_{n+1}^{(1)}], \quad y_{n+1} = y_n + \beta_n [g(x_n, y_n) + M_{n+1}^{(2)}], \quad (3.9)$$

with $\beta_n/\alpha_n \rightarrow 0$. On the time scale of x , the slow iterate y is nearly constant; on the time scale of y , the fast iterate has already equilibrated.

Theorem 3.20 (Two time scales; [Borkar \(1997\)](#)). Suppose h and g are Lipschitz, both step-size sequences satisfy [Assumption 3.17\(ii\)](#), $\beta_n/\alpha_n \rightarrow 0$, the noise sequences satisfy

Assumption 3.17(iii), and the iterates are bounded almost surely. Suppose that for each y the ODE $\dot{x} = h(x, y)$ has a globally asymptotically stable equilibrium $\lambda(y)$, with λ Lipschitz, and that $\dot{y} = g(\lambda(y), y)$ has a globally asymptotically stable equilibrium y^ . Then $(x_n, y_n) \rightarrow (\lambda(y^*), y^*)$ almost surely.*

The theorem is the single-agent prototype of a pattern that recurs throughout the book: a fast learner tracking the equilibrium of a slowly changing environment. Actor–critic methods (Section 3.3.4) are one instance; learning in games in which players adapt at different rates (Chapters 6 and 7), and meta-learning in which an outer loop revises the learning rule of an inner loop (Chapter 8), are others. When the fast dynamics have no globally stable equilibrium, as when learners cycle, the conclusion fails, and Chapter 7 develops what can be said instead.

3.3.2 Temporal-difference learning and Q-learning

Temporal-difference (TD) learning (Sutton, 1988) estimates V^π from a single trajectory generated by π . After each transition (s_t, a_t, r_t, s_{t+1}) , TD(0) computes the TD error and updates the visited entry,

$$\delta_t := r_t + \gamma V_t(s_{t+1}) - V_t(s_t), \quad V_{t+1}(s_t) = V_t(s_t) + \alpha_t \delta_t. \quad (3.10)$$

Given the past, the expected TD error is $(\mathcal{T}^\pi V_t)(s_t) - V_t(s_t)$, so TD(0) is an asynchronous stochastic approximation of the fixed-point iteration for $\mathcal{T}^\pi$: it uses a single sampled successor instead of the full expectation, and it bootstraps from its own current estimate instead of the full return. In the tabular case its convergence to V^π follows from the same asynchronous argument as for Q-learning below.

For large state spaces, *linear* TD(0) uses $V_w(s) = \phi(s)^\top w$ with features $\phi : \mathcal{S} \rightarrow \mathbb{R}^d$ and updates $w_{t+1} = w_t + \alpha_t \delta_t \phi(s_t)$. Let the Markov chain induced by π be irreducible and aperiodic with stationary distribution ν , let $\Phi \in \mathbb{R}^{\mathcal{S} \times d}$ have linearly independent columns, and let Π_ν denote the projection onto $\{\Phi w\}$ in the norm $\|V\|_\nu^2 = \sum_s \nu(s) V(s)^2$. Tsitsiklis and Van Roy (1997) proved that, with Robbins–Monro step sizes (Assumption A.17) and on-policy sampling, w_t converges almost surely to the unique solution w^* of the *projected Bellman equation* $\Phi w = \Pi_\nu \mathcal{T}^\pi \Phi w$, and that

$$\|\Phi w^* - V^\pi\|_\nu \leq \frac{1}{1 - \gamma} \|\Pi_\nu V^\pi - V^\pi\|_\nu.$$

The limiting ODE is $\dot{w} = \Phi^\top N(\gamma P_\pi - I)\Phi w + \Phi^\top N r_\pi$, with $N = \text{diag}(\nu)$, and its stability rests on the fact that P_π is nonexpansive in $\|\cdot\|_\nu$, which holds precisely because states are sampled from the stationary distribution of the policy being evaluated. If states are sampled from another distribution (off-policy learning), the combination of function approximation, bootstrapping, and off-policy sampling can make the iterates diverge. This combination, called the *deadly triad* by Sutton

and Barto (2018), is the reason stable deep reinforcement learning requires additional devices, such as target networks and experience replay (Mnih et al., 2015).

Q-learning (Watkins, 1989; Watkins and Dayan, 1992) learns Q^* directly, without a model and independently of the policy generating the data. After observing (s_t, a_t, r_t, s_{t+1}) it updates

$$Q_{t+1}(s_t, a_t) = Q_t(s_t, a_t) + \alpha_t(s_t, a_t) \left[r_t + \gamma \max_{a'} Q_t(s_{t+1}, a') - Q_t(s_t, a_t) \right], \quad (3.11)$$

leaving other entries unchanged, where $\alpha_t(s, a) = 0$ unless $(s, a) = (s_t, a_t)$.

Theorem 3.21 (Convergence of Q-learning). *Let $\mathcal{M}$ be a finite discounted MDP in which the reward r_t observed at (s_t, a_t) is random, conditionally independent of the past given (s_t, a_t) , with mean $r(s_t, a_t)$ and bounded variance, and $s_{t+1} \sim P(\cdot | s_t, a_t)$. Let the actions a_t be chosen by any rule that depends on the past, and suppose that the step sizes $\alpha_t(s, a) \in [0, 1]$ are adapted to the past and satisfy, for every (s, a) ,*

$$\sum_{t \geq 0} \alpha_t(s, a) = \infty, \quad \sum_{t \geq 0} \alpha_t(s, a)^2 < \infty \quad \text{almost surely.}$$

Then $Q_t \rightarrow Q^$ almost surely.*

Proof sketch. The operator

$$(HQ)(s, a) := r(s, a) + \gamma \sum_{s'} P(s' | s, a) \max_{a'} Q(s', a')$$

is a γ -contraction in the supremum norm whose fixed point is Q^* (Section 2.2.1). The update (3.11) is

$$Q_{t+1}(s, a) = Q_t(s, a) + \alpha_t(s, a) \left[(HQ_t)(s, a) - Q_t(s, a) + w_t(s, a) \right],$$

where the noise $w_t(s_t, a_t) = r_t + \gamma \max_{a'} Q_t(s_{t+1}, a') - (HQ_t)(s_t, a_t)$ has conditional mean zero and conditional variance at most $K(1 + \|Q_t\|_\infty^2)$. This is an asynchronous stochastic approximation driven by a sup-norm contraction. Tsitsiklis (1994, Theorems 1 and 3) shows that for such schemes the iterates remain bounded and converge to the fixed point under the stated step-size conditions; Jaakkola et al. (1994) give an independent proof. ■

The step-size condition implicitly requires every state–action pair to be updated infinitely often, so the theorem is really a statement about exploration: the data must cover $\mathcal{S} \times \mathcal{A}$. An ϵ -greedy behavior policy, which takes a uniformly random action with probability ϵ , ensures this in any MDP in which every state is reachable from every other under some policy. The theorem is asymptotic; finite-time rates depend sharply on the step-size schedule and on the mixing of the behavior policy (Even-Dar and Mansour, 2003). It also assumes a stationary environment. When other agents learn simultaneously, the transition kernel faced by each agent drifts, H changes over time, and the guarantee is lost (Chapter 7).

3.3.3 Policy gradient methods

Value-based methods derive a policy from a value function. Policy gradient methods instead parameterize a stochastic policy π_w , $w \in \mathbb{R}^d$, and ascend the objective $J(w) := \mathbb{E}_{s_0 \sim \rho} [V^{\pi_w}(s_0)]$ directly. This is natural when actions are continuous, when the optimal policy is stochastic, as in games, or when the policy is a large pretrained model, as in the fine-tuning of language models (Chapter 4). Let

$$d_\rho^\pi(s) := (1 - \gamma) \sum_{t \geq 0} \gamma^t \mathbf{P}_\rho^\pi(s_t = s)$$

be the normalized discounted state occupancy, and $A^\pi(s, a) := Q^\pi(s, a) - V^\pi(s)$ the *advantage*. We begin with an identity that compares any two policies.

Lemma 3.22 (Performance difference; Kakade and Langford (2002)). *For stationary policies π, π' and $\rho \in \Delta(\mathcal{S})$,*

$$\mathbb{E}_\rho [V^{\pi'}(s_0)] - \mathbb{E}_\rho [V^\pi(s_0)] = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\rho^{\pi'}} \mathbb{E}_{a \sim \pi'(\cdot | s)} [A^\pi(s, a)].$$

Proof. Under π' , $V^{\pi'}(s_0) = \mathbb{E}^{\pi'} [\sum_t \gamma^t r(s_t, a_t)]$, and, by telescoping and boundedness of V^π , $-V^\pi(s_0) = \mathbb{E}^{\pi'} [\sum_t \gamma^t (\gamma V^\pi(s_{t+1}) - V^\pi(s_t))]$. Adding,

$$V^{\pi'}(s_0) - V^\pi(s_0) = \mathbb{E}^{\pi'} \left[\sum_t \gamma^t (r(s_t, a_t) + \gamma V^\pi(s_{t+1}) - V^\pi(s_t)) \right],$$

and conditioning on (s_t, a_t) turns each term into $\gamma^t A^\pi(s_t, a_t)$. Averaging over $s_0 \sim \rho$ and using the definition of $d_\rho^{\pi'}$ gives the claim. ■

Theorem 3.23 (Policy gradient theorem). *Let $\mathcal{S}, \mathcal{A}$ be finite and $w \mapsto \pi_w(a | s)$ be differentiable for every (s, a) . Then*

$$\nabla_w J(w) = \frac{1}{1 - \gamma} \mathbb{E}_{s \sim d_\rho^{\pi_w}} \mathbb{E}_{a \sim \pi_w(\cdot | s)} \left[\nabla_w \log \pi_w(a | s) Q^{\pi_w}(s, a) \right],$$

and the same identity holds with $Q^{\pi_w}(s, a)$ replaced by $Q^{\pi_w}(s, a) - b(s)$ for any baseline $b : \mathcal{S} \rightarrow \mathbb{R}$, in particular by $A^{\pi_w}(s, a)$. (Where $\pi_w(a | s) = 0$, the integrand is read as $\nabla_w \pi_w(a | s) Q^{\pi_w}(s, a) / \pi_w(a | s)$ weighted by $\pi_w(a | s)$, that is, as $\nabla_w \pi_w(a | s) Q^{\pi_w}(s, a)$.)

Proof. Write $\pi = \pi_w$ and fix one coordinate of w , denoting its partial derivative by a prime. The value of a finite MDP is a rational function of the policy with nonvanishing denominator, $V^\pi = (I - \gamma P_\pi)^{-1} r_\pi$, so it is differentiable. Differentiating $V^\pi(s) = \sum_a \pi(a | s) Q^\pi(s, a)$ and $Q^\pi(s, a) = r(s, a) + \gamma \sum_{s'} P(s' | s, a) V^\pi(s')$ gives

$$V^{\pi'}(s) = \underbrace{\sum_a \pi'(a | s) Q^\pi(s, a)}_{f(s)} + \gamma \sum_{s'} P_\pi(s' | s) V^{\pi'}(s'),$$

that is, $V^{\pi'} = f + \gamma P_{\pi} V^{\pi'}$, whence $V^{\pi'} = (I - \gamma P_{\pi})^{-1} f = \sum_t \gamma^t P_{\pi}^t f$. Averaging over ρ , $J' = \sum_t \gamma^t \rho^{\top} P_{\pi}^t f = \frac{1}{1-\gamma} \sum_s d_{\rho}^{\pi}(s) f(s)$. Finally $\pi'(a|s) = \pi(a|s)(\log \pi(a|s))'$. For the baseline, $\sum_a \pi'(a|s)b(s) = b(s)(\sum_a \pi(a|s))' = 0$. ■

The theorem, due to Sutton et al. (2000), has two features that make it the basis of practical algorithms. The gradient does not involve the derivative of the state distribution, which depends on the unknown dynamics, and it is an expectation under the policy's own behavior, so it can be estimated from trajectories. The REINFORCE estimator (Williams, 1992) replaces $Q^{\pi_w}(s_t, a_t)$ by the observed return from time t , which is unbiased but has high variance; subtracting a learned baseline $b(s_t) \approx V^{\pi_w}(s_t)$ reduces the variance without introducing bias. Sutton et al. (2000) also show that Q^{π_w} can be replaced by a linear approximation in the compatible features $\nabla_w \log \pi_w(a|s)$, fit by least squares under $d_{\rho}^{\pi_w} \otimes \pi_w$, without biasing the gradient.

J is not concave in w , so gradient ascent is guaranteed in general only to approach stationary points. For the tabular softmax parameterization $\pi_w(a|s) \propto \exp(w_{s,a})$, however, every stationary point is globally optimal when ρ has full support, and projected and softmax gradient ascent converge to V^* ; the rates depend on distribution mismatch coefficients measuring how well ρ covers the occupancy of an optimal policy (Agarwal et al., 2021). The performance difference lemma is the key to such results: it expresses global suboptimality as an average of advantages, which gradient steps control locally.

3.3.4 Actor–critic methods

An *actor–critic* method combines the two families. A *critic* with parameters v estimates the value of the current policy by TD learning, and an *actor* updates the policy parameters w in the direction of the policy gradient estimated with the critic's TD error, which is an estimate of the advantage (Figure 3.9):

$$v_{t+1} = v_t + \alpha_t \delta_t \nabla_v V_{v_t}(s_t), \quad w_{t+1} = w_t + \beta_t \delta_t \nabla_w \log \pi_{w_t}(a_t|s_t), \quad (3.12)$$

with $\delta_t = r_t + \gamma V_{v_t}(s_{t+1}) - V_{v_t}(s_t)$. Since $\mathbb{E}[\delta_t | s_t, a_t] = A^{\pi_w}(s_t, a_t)$ when the critic is exact, the actor's expected update is the policy gradient of Theorem 3.23, up to the weighting of states.

Equation (3.12) is an instance of (3.9), and choosing $\beta_t/\alpha_t \rightarrow 0$ makes the critic the fast process. Theorem 3.20 then reduces the analysis to two simpler questions: whether TD learning converges for a fixed policy, which is the result of Tsitsiklis and Van Roy (1997), and whether the actor's ODE, driven by the converged critic, ascends J . Konda and Tsitsiklis (2003) carried out this program for linear critics using compatible features in the average-reward setting, showing that $\liminf_t \|\nabla J(w_t)\|$ is zero, or small when the critic uses TD(λ) with $\lambda < 1$. Bhatnagar et al. (2009) extended the analysis to natural-gradient actors. Soft

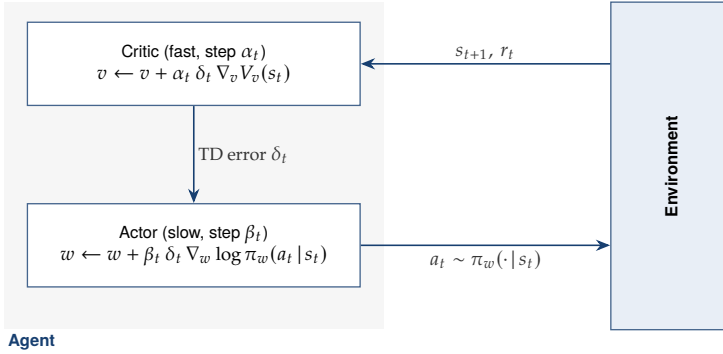


Figure 3.9. Actor–critic as a two-timescale stochastic approximation. The critic evaluates the current policy by TD learning with large steps; the actor follows the policy gradient estimated from the TD error with smaller steps. [Theorem 3.20](#) justifies analyzing each as if the other were frozen or converged.

actor–critic ([Haarnoja et al., 2018](#)), a widely used deep variant, combines the two-timescale structure with the entropy regularization of the next subsection.

3.3.5 Entropy regularization and policy mirror descent

Many modern algorithms regularize policies toward high entropy or toward a reference policy. The common mathematical core is the following variational identity.

Lemma 3.24 (Gibbs variational principle). *Let $\mathcal{A}$ be finite, $q \in \mathbb{R}^{\mathcal{A}}$, $p_0 \in \Delta(\mathcal{A})$ with full support, and $\tau > 0$. Then*

$$\max_{p \in \Delta(\mathcal{A})} \left\{ \langle q, p \rangle - \tau D_{\text{KL}}(p \| p_0) \right\} = \tau \log \sum_a p_0(a) e^{q(a)/\tau},$$

and the maximum is attained uniquely at $p^*(a) = p_0(a) e^{q(a)/\tau} / \sum_{a'} p_0(a') e^{q(a')/\tau}$.

Proof. Let $Z = \sum_a p_0(a) e^{q(a)/\tau}$. For any p , a direct computation gives $\langle q, p \rangle - \tau D_{\text{KL}}(p \| p_0) = \tau \log Z - \tau D_{\text{KL}}(p \| p^*)$. The divergence is nonnegative and vanishes only at $p = p^*$. ■

With p_0 uniform, the regularizer is entropy up to a constant, and the lemma yields the *soft Bellman operator*

$$(\mathcal{T}_\tau V)(s) := \tau \log \sum_a \exp \left(\frac{1}{\tau} \left[r(s, a) + \gamma \sum_{s'} P(s' | s, a) V(s') \right] \right),$$

which is again a monotone γ -contraction (the log-sum-exp function is monotone and nonexpansive under constant shifts). Its fixed point $V_\tau^\star$ is the optimal value of the entropy-regularized objective $\mathbb{E}[\sum_t \gamma^t (r(s_t, a_t) + \tau H(\pi(\cdot | s_t)))]$, the optimal policy is the softmax $\pi_\tau^\star(\cdot | s) = \text{softmax}(Q_\tau^\star(s, \cdot)/\tau)$, and $V^\star \leq V_\tau^\star \leq V^\star + \tau \log |\mathcal{A}| / (1 - \gamma)$ because $\max_a q(a) \leq \tau \log \sum_a e^{q(a)/\tau} \leq \max_a q(a) + \tau \log |\mathcal{A}|$. Regularized MDPs of this kind are studied systematically by Geist et al. (2019) and Neu et al. (2017).

With p_0 the current policy, the lemma yields *policy mirror descent*: at each iteration and each state,

$$\pi_{k+1}(\cdot | s) = \arg \max_{p \in \Delta(\mathcal{A})} \left\{ \eta \langle Q^{\pi_k}(s, \cdot), p \rangle - D_{\text{KL}}(p \| \pi_k(\cdot | s)) \right\}, \quad (3.13)$$

that is, $\pi_{k+1}(a | s) \propto \pi_k(a | s) e^{\eta Q^{\pi_k}(s, a)}$.

This is mirror descent on the simplex at each state, with the negative entropy as mirror map and the action values as linear payoffs, and it coincides with the natural policy gradient (Kakade, 2002) for the tabular softmax parameterization.

Theorem 3.25 (Policy mirror descent; Agarwal et al. (2021)). *Let rewards lie in $[0, 1]$, let π_0 be uniform and $\eta > 0$, and let*

$$\pi_{k+1}(a | s) \propto \pi_k(a | s) \exp(\eta A^{\pi_k}(s, a)/(1 - \gamma)) \quad \text{for all } s, a.$$

Then for every $\rho \in \Delta(\mathcal{S})$ and $T \geq 1$,

$$\mathbb{E}_\rho[V^\star(s_0)] - \mathbb{E}_\rho[V^{\pi_T}(s_0)] \leq \frac{\log |\mathcal{A}|}{\eta T} + \frac{1}{(1 - \gamma)^2 T}.$$

Since replacing Q^{π_k} by A^{π_k} changes the update only by a state-dependent normalization, this is (3.13) with step size $\eta/(1 - \gamma)$. The proof combines the performance difference lemma with the standard three-point analysis of mirror descent, and the rate does not depend on $|\mathcal{S}|$ or on distribution mismatch. With entropy regularization the convergence becomes linear: Cen et al. (2022) show that entropy-regularized natural policy gradient converges to $Q_\tau^\star$ at a geometric rate, and Lan (2023) obtains linear rates for policy mirror descent with general convex regularizers and increasing step sizes.

One update, three settings

The update $\pi_{k+1} \propto \pi_k \exp(\eta Q)$ appears three times in this book. Here it is policy mirror descent for a single agent against a fixed environment. In Chapter 7 each player of a game applies it to payoffs that depend on the others' play, which yields multi-agent mirror descent and its nonconvergent dynamics. In Chapter 4, the solution $\pi^\star \propto \pi_{\text{ref}} \exp(r/\beta)$ of the KL-regularized objective $\mathbb{E}_\pi[r] - \beta D_{\text{KL}}(\pi \| \pi_{\text{ref}})$ used to align language

models is [Lemma 3.24](#) with p_0 the reference model. The KL term plays the same role in all three: it keeps each update close to a trusted distribution.

3.4 Bayesian Decision Making under Uncertainty

The methods above estimate values and policies for a single, fixed environment and treat exploration as a device for collecting data. An agent that is uncertain about the parameter $\theta \in \Theta$ governing its environment can do better: it can reason about how its actions affect what it will learn and value that information exactly. [Proposition 2.18](#) showed that this reasoning is planning in the belief MDP whose state includes the posterior. This section develops the computational side: how beliefs are updated, what Bayes-optimal policies are, why computing them is hard, and which tractable procedures retain strong guarantees.

3.4.1 Posterior updating and conjugacy

For a family of observation laws $\{Q_\theta\}_{\theta \in \Theta}$ and prior μ_0 , the posterior after observations $y_1, \dots, y_n$ is $\mu_n(d\theta) \propto \prod_{i \leq n} Q_\theta(y_i) \mu_0(d\theta)$. For general Θ this requires integration, and in practice it is approximated by sampling or variational methods. A family of priors is *conjugate* to the likelihood if the posterior remains in the family; updating then reduces to updating finitely many parameters, which are sufficient statistics of the data.

- Proposition 3.26** (Standard conjugate updates). (i) Beta–Bernoulli. If $\theta \sim \text{Beta}(\alpha, \beta)$ and $y_i \in \{0, 1\}$ are conditionally i.i.d. Bernoulli(θ), then $\theta \mid y_{1:n} \sim \text{Beta}(\alpha + \sum_i y_i, \beta + n - \sum_i y_i)$.
- (ii) Dirichlet–categorical. If the row $P(\cdot \mid s, a) \sim \text{Dir}(c(s, a, \cdot))$ independently across (s, a) , and transitions are observed, then the posterior is $\text{Dir}(c(s, a, \cdot) + N(s, a, \cdot))$ independently across (s, a) , where $N(s, a, s')$ counts observed transitions from (s, a) to s' .
- (iii) Normal–normal. If $\theta \sim N(m_0, \tau_0^2)$ and $y_i \mid \theta \sim N(\theta, \eta^2)$, then $\theta \mid y_{1:n} \sim N(m_n, \tau_n^2)$ with $\tau_n^{-2} = \tau_0^{-2} + n\eta^{-2}$ and $m_n = \tau_n^2(\tau_0^{-2}m_0 + \eta^{-2} \sum_i y_i)$.

Proof. In each case the product of the prior density and the likelihood is, as a function of θ , proportional to a density of the stated form: $\theta^{\alpha-1+k}(1-\theta)^{\beta-1+n-k}$ in (i), $\prod_{s'} P(s' \mid s, a)^{c(s, a, s')-1+N(s, a, s')}$ in (ii), and the exponential of a quadratic in θ with the stated coefficients in (iii). ■

Conjugacy makes the hyperstate (s_t, μ_t) of [Definition 2.17](#) finite-dimensional: for a Dirichlet prior over transition kernels it is the state together with the transition counts. It does not make the belief space small. By [Proposition 2.5](#), the posterior concentrates as data accumulate, but only along the directions that the agent’s actions explore, which is the source of the exploration problem.

3.4.2 Bayes-optimal policies and Bayes-adaptive planning

Definition 3.27 (Bayes-optimal policy). Let $\{\mathcal{M}_\theta = (\mathcal{S}, \mathcal{A}, P_\theta, r, \gamma)\}_{\theta \in \Theta}$ be a family of MDPs sharing states, actions, and rewards, and let $\mu_0 \in \Delta(\Theta)$ be a prior. For a history-dependent policy σ and initial state s , the *Bayesian value* is

$$J(\sigma; s, \mu_0) := \int_{\Theta} \mathbb{E}_{\theta, s}^{\sigma} \left[\sum_{t \geq 0} \gamma^t r(s_t, a_t) \right] \mu_0(d\theta),$$

and σ^B is *Bayes optimal* if $J(\sigma^B; s, \mu_0) = \sup_{\sigma} J(\sigma; s, \mu_0)$ for every s .

The Bayesian value averages over the environment before the agent acts; it does not ask the agent to be optimal for each θ , which is impossible without knowing θ . By Proposition 2.18, the Bayes-optimal policies are the optimal policies of the Bayes-adaptive MDP of Definition 2.17, so they can be taken to be stationary functions of the hyperstate, and the Bayesian value satisfies a Bellman equation on hyperstates.

Proposition 3.28 (Bayes-adaptive Bellman equation). Let Θ be finite, let the predictive kernel be $\bar{P}_\mu(\cdot | s, a) := \sum_{\theta} \mu(\theta) P_\theta(\cdot | s, a)$, and $\mu^{s,a,s'}$ the posterior (2.7). The optimal Bayesian value $V^B(s, \mu) := \sup_{\sigma} J(\sigma; s, \mu)$ is the unique bounded solution of

$$V^B(s, \mu) = \max_{a \in \mathcal{A}} \left[r(s, a) + \gamma \sum_{s'} \bar{P}_\mu(s' | s, a) V^B(s', \mu^{s,a,s'}) \right], \quad (3.14)$$

and any policy selecting a maximizing action at each hyperstate is Bayes optimal.

Proof. The right side defines an operator on bounded functions on $\mathcal{S} \times \Delta(\Theta)$ that is a monotone γ -contraction by the argument of Lemma 2.12, which does not use finiteness of the state space. The optimality statements follow as in Theorem 2.13 applied to the Bayes-adaptive MDP, whose hyperstates reachable from a given (s, μ) form a countable set, so no measurability issues arise. ■

Equation (3.14) makes the value of information explicit. The action a matters both through its reward and through the posterior $\mu^{s,a,s'}$ it induces, and the convexity of values in beliefs, the property behind Theorem 2.9, makes informative actions more valuable than their immediate rewards suggest. A *certainty-equivalent* agent, which acts optimally for the posterior mean model $\bar{P}_{\mu_t}$ as if it were known, ignores this term, and can settle into the self-confirming trap of Example 1.13.

Figure 3.10 shows the lookahead tree of (3.14) for a two-armed Bernoulli bandit with independent uniform priors. Even there, the number of distinct hyperstates reachable in d steps grows polynomially in d for each arm but the tree of action–outcome sequences grows as $(2K)^d$, and exact solution is feasible only for short horizons or special structure. In general, finite-horizon Bayes-adaptive planning is a POMDP problem, subject to the intractability results cited in Section 2.2.2. Practical methods approximate (3.14). BEETLE exploits the piecewise polynomial

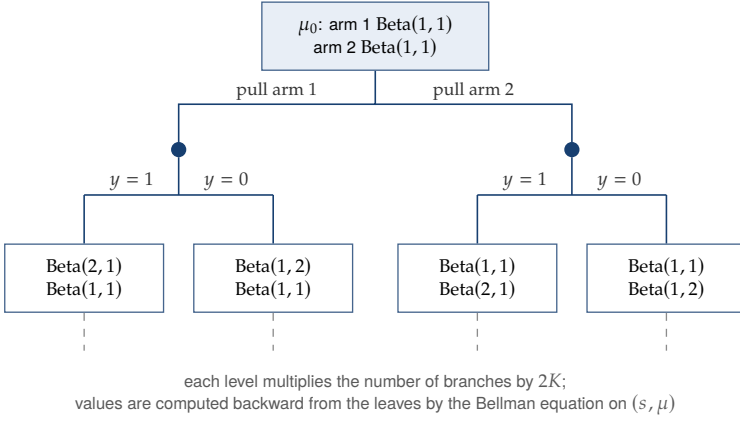


Figure 3.10. Hyperstate tree of a Bayes-adaptive two-armed Bernoulli bandit with independent Beta(1, 1) priors. Each action leads to an outcome drawn from the predictive distribution (here both outcomes have probability $\frac{1}{2}$) and to an updated posterior. Bayes-optimal values are computed backward through this tree, whose size grows exponentially with depth.

structure of the value in Dirichlet parameters (Poupart et al., 2006). BAMCP applies Monte Carlo tree search to the hyperstate tree, sampling a model from the posterior at the root of each simulation so that no posterior needs to be updated inside the tree (Guez et al., 2012), which is Section 3.2.3 applied to Definition 2.17. Optimistic methods add an exploration bonus that decays with visit counts and are near-Bayes-optimal in polynomial time in a sense made precise by Kolter and Ng (2009).

3.4.3 The Gittins index

In one important case the Bayes-adaptive problem decomposes. In a *discounted multi-armed bandit*, arm $k \in \{1, \dots, K\}$ has a state x_k evolving as a Markov chain only when the arm is played, playing it yields reward $R_k(x_k)$, and the states of the other arms are frozen. A Bernoulli bandit with independent Beta priors is of this form, with $x_k = (\alpha_k, \beta_k)$ the posterior parameters and the chain given by Proposition 3.26(i).

Theorem 3.29 (Gittins index theorem; Gittins (1979)). *For arm k in state x define the index*

$$G_k(x) := \sup_{\tau \geq 1} \frac{\mathbb{E} \left[\sum_{t=0}^{\tau-1} \gamma^t R_k(x_k(t)) \mid x_k(0) = x \right]}{\mathbb{E} \left[\sum_{t=0}^{\tau-1} \gamma^t \mid x_k(0) = x \right]},$$

Algorithm 3.2 Thompson sampling for a Bernoulli bandit

Require: prior parameters (α_k, β_k) for $k = 1, \dots, K$

- 1: **for** $t = 1, 2, \dots$ **do**
- 2: sample $\hat{\theta}_{t,k} \sim \text{Beta}(\alpha_k, \beta_k)$ independently for each k
- 3: play $a_t \in \arg \max_k \hat{\theta}_{t,k}$ and observe $y_t \in \{0, 1\}$
- 4: $\alpha_{a_t} \leftarrow \alpha_{a_t} + y_t, \quad \beta_{a_t} \leftarrow \beta_{a_t} + 1 - y_t$
- 5: **end for**

the supremum being over stopping times of the arm's own process. The policy that at each time plays an arm with the largest current index is optimal for the discounted multi-armed bandit.

The index of an arm depends only on that arm, so a problem whose state space is a product of K spaces is solved by K one-dimensional computations. The index is the constant reward rate of a standard arm that would make the agent indifferent between it and arm k , and for a Bayesian arm it exceeds the posterior mean by an amount that reflects the value of information about that arm. Short proofs are given by Whittle (1980) and Weber (1992); see Gittins et al. (2011). The decomposition is fragile: it fails when arms' states evolve while not played, when there are switching costs, when rewards are undiscounted, and when playing one arm is informative about another. The last of these is the typical case for agents whose options share structure, and it motivates the approximate methods below.

3.4.4 Thompson sampling

Thompson sampling (Thompson, 1933) replaces planning over beliefs by a single posterior sample. At each round it draws $\hat{\theta}_t \sim \mu_t$ and acts optimally for $\hat{\theta}_t$. Equivalently, it selects each action with the posterior probability that the action is optimal (Algorithm 3.2). The rule explores automatically: an action that might be optimal is tried with probability equal to the agent's belief that it is, and that probability shrinks as evidence against it accumulates. In an MDP, *posterior sampling for reinforcement learning* draws a model at the start of each episode and follows its optimal policy for the episode (Osband et al., 2013).

To analyze it, consider a general bandit setting. A random parameter $\theta \sim \mu_0$ determines, for each action $a \in \mathcal{A}$, the law of an outcome $Y_{t,a}$ and a reward $R(Y_{t,a}) \in [0, 1]$. Let $A^* \in \arg \max_a \mathbb{E}[R(Y_{t,a}) | \theta]$ be an optimal action (a random variable, since θ is), let $H_t = (A_1, Y_{1,A_1}, \dots, A_{t-1}, Y_{t-1,A_{t-1}})$, and write $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot | H_t]$, $\mathbb{P}_t$ and I_t for conditional expectation, probability, and mutual information given H_t . The *Bayesian regret* of a policy over T rounds is

$$\text{BR}(T) := \mathbb{E} \left[\sum_{t=1}^T (R(Y_{t,A^*}) - R(Y_{t,A_t})) \right],$$

where the expectation is over θ , the outcomes, and the policy's randomization. The analysis of Russo and Van Roy (2016) rests on the *information ratio*

$$\Gamma_t := \frac{(\mathbb{E}_t[R(Y_{t,A^\star}) - R(Y_{t,A_t})])^2}{I_t(A^\star; (A_t, Y_{t,A_t}))}, \quad (3.15)$$

which compares the square of the expected regret incurred in round t with the information gained in that round about the identity of the optimal action.

Theorem 3.30 (Information-ratio bound; Russo and Van Roy (2016)). *In the bandit setting above, the following hold.*

(i) *If the information ratio of a policy satisfies $\Gamma_t \leq \bar{\Gamma}$ almost surely for all t , then*

$$\text{BR}(T) \leq \sqrt{\bar{\Gamma} H(A^\star) T}.$$

(ii) *For Thompson sampling with a finite action set, $\Gamma_t \leq |\mathcal{A}|/2$. Consequently,*

$$\text{BR}(T) \leq \sqrt{\frac{1}{2} |\mathcal{A}| T H(A^\star)} \leq \sqrt{\frac{1}{2} |\mathcal{A}| T \log |\mathcal{A}|}.$$

Proof of (i). Let $\Delta_t = \mathbb{E}_t[R(Y_{t,A^\star}) - R(Y_{t,A_t})]$ and $g_t = I_t(A^\star; (A_t, Y_{t,A_t}))$, so that $\Delta_t \leq \sqrt{\bar{\Gamma} g_t}$. By the tower property and the Cauchy–Schwarz inequality,

$$\text{BR}(T) = \mathbb{E} \sum_{t=1}^T \Delta_t \leq \sqrt{\bar{\Gamma}} \mathbb{E} \sum_{t=1}^T \sqrt{g_t} \leq \sqrt{\bar{\Gamma} T \mathbb{E} \sum_{t=1}^T g_t}.$$

Since $H_{t+1} = (H_t, A_t, Y_{t,A_t})$, the chain rule for mutual information gives $\mathbb{E} \sum_{t \leq T} g_t = I(A^\star; H_{T+1}) \leq H(A^\star)$. ■

The proof of (ii), which bounds Γ_t by relating the regret of Thompson sampling to Kullback–Leibler divergences between the conditional outcome distributions given $A^\star = a$ and their mixture and then applying Pinsker's inequality, is given in Russo and Van Roy (2016); the same paper shows $\Gamma_t \leq d/2$ for linear bandits in dimension d and $\Gamma_t \leq 1/2$ under full information. The bound in (ii) does not depend on how close the arms are, in contrast to the instance-dependent bound of Theorem 3.15. The entropy $H(A^\star)$ quantifies prior knowledge: a prior that already concentrates on one action has little to learn and incurs little regret.

Example 3.31 (Selecting a detector configuration). The defender can deploy one of $K = 3$ detector configurations per time window and observes whether the deployed configuration catches a planted test intrusion. The catch rates θ_k are unknown, with independent uniform priors. Figure 3.11 shows the Bayesian regret, averaged over 3000 draws of θ from the prior, of Thompson sampling and of the greedy rule that deploys the configuration with the highest posterior mean. Greedy selection exploits early, and with positive probability it never again

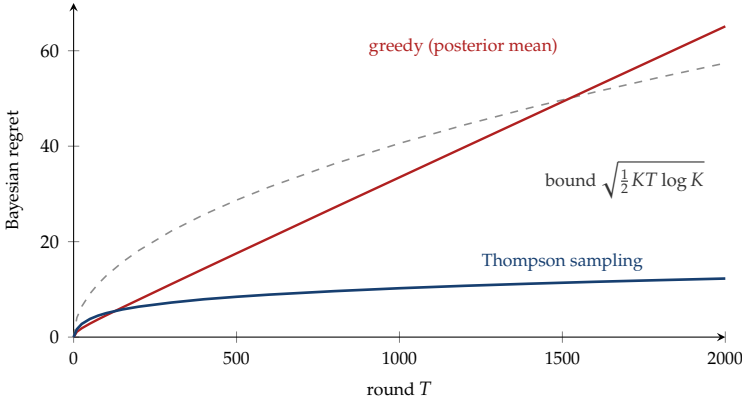


Figure 3.11. Bayesian regret in [Example 3.31](#), averaged over 3000 problem instances drawn from the prior. Greedy selection with respect to the posterior mean incurs linear regret; Thompson sampling incurs sublinear regret below the information-ratio bound of [Theorem 3.30](#).

deploys the best configuration after a few unlucky observations, so its regret grows linearly and eventually exceeds the bound of [Theorem 3.30](#). Thompson sampling stays well below the bound, with regret 12.3 after 2000 windows, against a bound of 57.4. $\triangleleft$

3.4.5 Information-directed exploration

[Theorem 3.30\(i\)](#) holds for any policy, which suggests choosing actions to make the information ratio small. *Information-directed sampling* ([Russo and Van Roy, 2018](#)) selects at each round the action distribution $v \in \Delta(\mathcal{A})$ minimizing

$$\frac{\Delta_t(v)^2}{g_t(v)}, \quad \begin{aligned} \Delta_t(v) &:= \sum_a v(a) \mathbb{E}_t[R(Y_{t,A^*}) - R(Y_{t,a})], \\ g_t(v) &:= \sum_a v(a) \mathbb{I}_t(A^*; Y_{t,a}). \end{aligned}$$

Its information ratio is at most that of Thompson sampling, so it inherits the bounds of [Theorem 3.30](#), and it can be much better when some actions are informative but certainly suboptimal: a diagnostic probe that the defender knows will not stop an attack but that reveals which detector is best is never chosen by Thompson sampling, which only plays actions that might be optimal, but may be chosen by information-directed sampling.

The objective makes the connection to [Chapter 2](#) explicit. The denominator is the expected reduction in the entropy of the posterior over the optimal action, a decision-relevant form of the epistemic uncertainty of [Proposition 2.4](#), and the

procedure pays for information in the currency of regret. The same trade-off arises whenever an agent can take actions whose only effect is informational, such as querying a threat-intelligence service or calling a retrieval tool; Chapter 5 treats tool calls in these terms.

Remark 3.32 (Known unknowns). Every result of this section presupposes that the prior is well specified: the true parameter θ^* lies in the support of μ_0 , and the likelihoods P_θ are the correct ones. Bayes optimality is optimality *on average under the prior*, the regret of [Theorem 3.30](#) is averaged over the prior, and the posterior concentrates on the truth only if the truth is among the hypotheses. This is the regime of known unknowns, which the cognitive chapters (Chapters 8 and 9) take as their starting point when the prior itself is learned from related tasks. When $\theta^* \notin \Theta$, the posterior concentrates instead on the models closest to the truth in Kullback–Leibler divergence ([Proposition 1.16](#)), exploration driven by posterior uncertainty can stop while the model is still wrong, and the guarantees become statements about a subjective model. The epistemic chapters of Part IV, beginning with Chapter 10, develop the theory for that case.

Notes and References

Dynamic programming originates with [Bellman \(1957\)](#), and policy iteration with [Howard \(1960\)](#). For discounted MDPs we follow [Puterman \(1994\)](#) and [Bertsekas \(2017\)](#); measurable-selection issues on Borel spaces are treated by [Bertsekas and Shreve \(1978\)](#) and [Hernández-Lerma and Lasserre \(1996\)](#). The stopping rule of [Theorem 3.4](#) is standard ([Puterman, 1994, Section 6.3](#)), and the greedy loss bound of [Proposition 3.2](#) is due to [Singh and Yee \(1994\)](#). Strong polynomiality of policy iteration for a fixed discount factor is due to [Ye \(2011\)](#). The linear programming formulation goes back to [Manne \(1960\)](#). The linear–quadratic regulator and the Riccati equation are due to [Kalman \(1960\)](#); [Anderson and Moore \(1990\)](#) give a thorough treatment. Approximate dynamic programming and its error bounds are developed in [Bertsekas and Tsitsiklis \(1996\)](#), with L_p analyses of fitted value iteration in [Munos and Szepesvári \(2008\)](#).

Rollout as online policy improvement was introduced by [Tesauro and Galperin \(1997\)](#) and [Bertsekas et al. \(1997\)](#); [Bertsekas \(2020\)](#) develops its connections to policy iteration and multi-agent problems. The sparse sampling algorithm of [Kearns et al. \(2002\)](#) established that planning with a simulator need not depend on the size of the state space. For importance sampling in reinforcement learning, see [Precup et al. \(2000\)](#). UCB1 is due to [Auer et al. \(2002\)](#), UCT to [Kocsis and Szepesvári \(2006\)](#), and MCTS in its modern form to [Coulom \(2007\)](#); [Coquelin and Munos \(2007\)](#) analyze its worst case, and [Browne et al. \(2012\)](#) survey the field. The combination of search with learned policies and values follows [Silver et al. \(2016, 2017\)](#).

For reinforcement learning, Sutton and Barto (2018) is the standard introduction, Bertsekas and Tsitsiklis (1996) the classical theoretical treatment, and Szepesvári (2010) a concise mathematical survey. Stochastic approximation originates with Robbins and Monro (1951); the ODE method with Ljung (1977); and the dynamical-systems viewpoint used here with Benaïm (1999) and Borkar (2008), with stability criteria in Borkar and Meyn (2000), two-timescale schemes in Borkar (1997), and weaker conditions in Kushner and Yin (2003). Temporal-difference learning is due to Sutton (1988) and its analysis with linear function approximation to Tsitsiklis and Van Roy (1997). Q-learning was introduced by Watkins (1989), with a first convergence proof in Watkins and Dayan (1992) and the stochastic approximation proofs of Tsitsiklis (1994) and Jaakkola et al. (1994). Deep Q-learning is due to Mnih et al. (2015). Policy gradient methods go back to Williams (1992) and Sutton et al. (2000); actor–critic algorithms were analyzed by Konda and Tsitsiklis (2000, 2003) and Bhatnagar et al. (2009). The natural policy gradient is due to Kakade (2002) and the performance difference lemma to Kakade and Langford (2002). The global convergence theory of policy gradient methods is developed by Agarwal et al. (2021), Cen et al. (2022), and Lan (2023); entropy-regularized MDPs are unified by Neu et al. (2017) and Geist et al. (2019), and soft actor–critic is due to Haarnoja et al. (2018).

Bayes-adaptive MDPs were developed by Duff (2002); Ghavamzadeh et al. (2015) survey Bayesian reinforcement learning. BEETLE is due to Poupart et al. (2006), BAMCP to Guez et al. (2012), and near-Bayesian exploration bounds to Kolter and Ng (2009). The index theorem is due to Gittins (1979), with alternative proofs by Whittle (1980) and Weber (1992); Gittins et al. (2011) is the standard monograph. Thompson sampling was proposed by Thompson (1933); the information-theoretic analysis is due to Russo and Van Roy (2016) and information-directed sampling to Russo and Van Roy (2018); posterior sampling for reinforcement learning is analyzed by Osband et al. (2013). Lattimore and Szepesvári (2020) give a comprehensive treatment of bandit algorithms, including frequentist analyses of Thompson sampling and upper confidence bounds.

Exercises

3.1 (Tightness of the greedy loss bound) Construct, for each $\gamma \in (0, 1)$ and $\varepsilon > 0$, an MDP with two actions and a function V with $\|V - V^*\|_\infty = \varepsilon$ such that a policy greedy with respect to V satisfies $\|V^* - V^\pi\|_\infty \geq c \gamma \varepsilon / (1 - \gamma)$ for a constant $c > 0$ independent of γ and ε . (Hint: use a chain in which a small misranking of two actions is repeated at every step.)

3.2 (Action gap) In a finite MDP, let

$$\Delta := \min_s \left(\max_a Q^*(s, a) - \max_{a \notin \arg \max Q^*(s, \cdot)} Q^*(s, a) \right),$$

with the convention that the inner maximum over an empty set is $-\infty$. Show that if $\|V - V^*\|_\infty < \Delta/(2\gamma)$, every policy greedy with respect to V is optimal. Deduce that value iteration identifies an optimal policy after finitely many iterations, and give a bound on their number. Compute Δ for [Example 3.5](#).

3.3 (Infinite-horizon LQR) For the scalar system $s_{t+1} = as_t + ba_t + \xi_t$ with cost $cs_t^2 + da_t^2$, $b \neq 0$, $c, d > 0$, show that the Riccati recursion (3.5) converges to the positive root of a quadratic equation, compute the stationary gain, and verify that $|a - bK| < 1$ even when $|a| > 1$. Explain why the gain is the same whether ξ_t has variance 0 or σ^2 , and how the answer changes if a is unknown and must be estimated from the closed-loop trajectory.

3.4 (Variance of per-decision importance sampling) Let $\mathcal{S}$ be a single state, $|\mathcal{A}| = 2$, β uniform, $\pi(a_1) = 1 - \pi(a_2) = p$, and rewards $r(a_1) = 1$, $r(a_2) = 0$ with $\gamma = 1$ and horizon H . Compute the variances of the ordinary importance-sampling estimator of $\mathbb{E}_\pi[G_H]$ and of the per-decision estimator $\sum_{t < H} \varrho_{0:t} r(a_t)$. Show that both grow exponentially in H at the rate $1 + \chi^2(\pi \| \beta)$ when $p \neq 1/2$, compare their leading constants, and explain why weighted importance sampling avoids this growth at the price of a bias.

3.5 (Softmax policy gradient) For the tabular softmax parameterization, in which $\pi_w(a | s) \propto \exp(w_{s,a})$, use [Theorem 3.23](#) to show that

$$\frac{\partial J}{\partial w_{s,a}} = \frac{1}{1 - \gamma} d_{\rho}^{\pi_w}(s) \pi_w(a | s) A^{\pi_w}(s, a).$$

Explain why gradient ascent can be slow when an optimal action has small probability under π_w , and why the natural gradient update (3.13) does not suffer from this problem.

3.6 (Bayes-optimal versus Thompson sampling) A one-armed Bernoulli bandit has an unknown arm with mean $\theta \in \{0.3, 0.7\}$, prior $\mathbb{P}(\theta = 0.7) = q$, and a known arm paying 0.5. With $\gamma = 0.9$, solve the Bayes-adaptive Bellman equation (3.14) numerically on a grid of beliefs, show that the Bayes-optimal policy is a threshold in the posterior, and compare its Bayesian value with those of Thompson sampling and of the certainty-equivalent policy. For which q does the certainty-equivalent policy never try the unknown arm, and what is the resulting loss?

3.7 (Thompson sampling under misspecification) Repeat [Example 3.31](#) with a prior that assigns zero probability to catch rates above 0.5, while the true rate of one configuration is 0.8. Describe the posterior of that configuration, determine whether Thompson sampling still identifies it, and relate the outcome to [Remark 3.32](#).

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